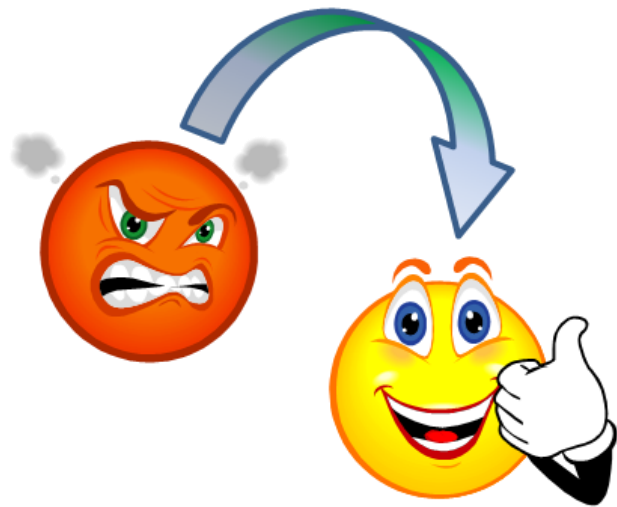


بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Training Program for Service Technicians

2025



Customer Service Department

Today's Learning



➤ Session 1

- PEL Products' Details with Unique Selling Points
- Troubleshooting / Major Error Codes of Inverter Air Conditioner
- Split AC Installation SOP for Installer
- Introduction and Troubleshooting of LED Television

0900 ~ 1730 hrs

➤ Session 2

- Hands-On Training to PEL Products



Session 1



- ☐ PEL Products' Details with Unique Selling Points.
- ☐ Troubleshooting of Major Error Codes
- ☐ SOP of Split AC Installation



Direct Cool Refrigerators

REFRIGERATOR LINEUP SUMMARY 2024

S. NO	Model	Capacity (LTR/CFT)	Single Door Series	Life Pro Series (VCM Door)	Inverter Series (VCM Door)	Non-Inverter Series (Flat GD)	Non-Inverter (Curved GD)	Inverter Series (Flat GD)	ULTRA Inverter Series (Curved GD)	DIGITRON Inverter Series (Flat GD)
1	1100 VCM	109 / 3.8	✓	–	–	–	–	–	–	–
2	1400 VCM	135 / 5.0	✓	–	–	–	–	–	–	–
3	1400 GD	135 / 5.0	✓	–	–	–	–	–	–	–
4	2000	168 / 6.0	–	✓	–	✓	–	–	–	–
5	2200	194 / 7.0	–	✓	–	✓	–	–	–	–
6	2350	233 / 8.0	–	✓	✓	✓	–	✓	–	–
7	2550	255 / 9.0	–	✓	✓	✓	✓	✓	✓	–
8	6360	310 / 10.9	–	✓	✓	✓	✓	✓	✓	✓
9	6460	334 / 11.8	–	✓	✓	✓	✓	–	✓	–
10	21860	354 / 12.5	–	✓	✓	✓	–	✓	✓	–
11	21960	382 / 13.5	–	–	–	✓	✓	–	–	✓
12	22260	408 / 14.5	–	✓	–	✓	✓	✓	✓	✓



PEL SINGLE DOOR Series (Non-Inverter)



PEL

SINGLE DOOR SERIES
NON INVERTER

1100 VCM

1400 VCM

1400 GD

1400 GD

BRUSHED METTALIC GRAY

BRUSHED METTALIC GRAY

RED BLAZE

PMI

INSTA COOL

25 MIN FAST FREEZE

99.9% PUREST COPPER

FOOD FRIENDLY CABINET

PEL SINGLE DOOR Series (Non-Inverter)

Technical Specifications

DESCRIPTION		Units	PRLP 1100 VCM	PRLP 1400 VCM / PRGD 1400 GD
INTERNAL Capacity	Total	Liters	109	135
	Freezer	Liters	-	-
	Refrigerator	Liters	109	135
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	100	110
	Current Consumption	Ampere	0.75	0.80
	Star Rating (Cooling)	-	4 Star	4 Star
	Climate Class	-	T	T
GENERAL FEATURES	Defrosting	Freezer	Manual	Manual
		Refrigerator	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes
	Humidity Controller	-	-	-
	Refrigerant	-	R134a	R134a
	Evaporator	-	Roll Bond	Roll Bond
	Condenser Type	-	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	Available
	Shelves Type	-	Wire Type	Wire Type
DIMENSIONS (only Cabinet without Door)	Height	mm	802	947
	Width	mm	490	490
	Depth	mm	430	430
WEIGHT	Net Weight	Kg	27	29
	Gross Weight	Kg	30	33



PEL LIFE PRO SERIES (VCM DOOR) NON-INVERTER



LIFE PRO SERIES

VCM DOOR – NON INVERTER

Unique Features

- Clean and Wide Look
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant



GRACIOUS
GRAPHITE



BRUSHED
METALLIC GRAY



METALLIC
GOLDEN BROWN



METALLIC
TEXTURE GRAY

PEL Range – Refrigerator



PEL LIFE PRO SERIES (VCM DOOR) NON-INVERTER



Technical Specifications

									JUMBO SERIES	
DESCRIPTION		Units	PRLP 2000	PRLP 2200	PRLP 2350	PRLP 2550	PRLP 6360	PRLP 6460	PRLP 21860	PRLP 22260
INTERNAL Capacity	Total	Liters	168	194	233	255	310	334	354	408
	Freezer	Liters	59	59	85	85	90	90	113	141
	Refrigerator	Liters	109	135	148	170	220	244	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	115	115	130	140	132	147	147	178
	Current Consumption	Ampere	0.85	0.85	0.90	1.10	0.78	0.80	0.80	1.20
	Start Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T	T	T	T	T	T
GENERAL FEATURES	Defrosting Freezer	-	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual
	Defrosting Refrigerator	-	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	-	-	-	-	Yes	Yes	No	No
	Refrigerant	-	R134a	R134a	R134a	R134a	R600a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Roll Bond	Roll Bond	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	Available	Available	Available	Available	Available	-	-
DIMENSIONS (only Cabinet without Door)	Shelves Type	-	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type
	Height	mm	1255	1400	1440	1540	1570	1660	1610	1814
	Width	mm	490	490	540	540	600	600	662	662
	Depth	mm	430	430	471	471	540	540	540	540
WEIGHT	Net Weight	Kg	39	41	46	49	53	55	64	68
	Gross Weight	Kg	44	46	54	57	60	63	73	78

*Unique features in models 6360, 6460, 21860 and 22260 only

PEL INVERTER SERIES (VCM DOOR)



INVERTER SERIES

VCM DOOR

Unique Features

- Clean and Wide Look
- Upto 40% Energy Saving
- Jet Cool 360°
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant



Inverteron
TECHNOLOGY



SILVER
MOONLIGHT



GOLD
SILK

PEL INVERTER SERIES (VCM DOOR)

Technical Specifications



							JUMBO SERIES
DESCRIPTION		Units	PRINVOVCM 2350	PRINVOVCM 2550	PRINVOVCM 6360	PRINVOVCM 6460	PRINVOVCM 21860
INTERNAL Capacity	Total	Liters	233	255	310	334	354
	Freezer	Liters	85	85	90	90	113
	Refrigerator	Liters	148	170	220	244	241
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	130	130	82	108	108
	Current Consumption	Ampere	0.80	0.80	0.65	0.88	0.88
	Start Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T	T	T
GENERAL FEATURES	Defrosting Freezer	-	Manual	Manual	Manual	Manual	Manual
	Defrosting Refrigerator	-	Manual	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	-	-	Yes	Yes	No
	Refrigerant	-	R134a	R134a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	Available	-	-	-
DIMENSIONS (Cabinet without Door)	Shelves Type	-	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type
	Height	mm	1440	1540	1570	1660	1610
	Width	mm	540	540	600	600	662
	Depth	mm	471	471	540	540	540
WEIGHT	Net Weight	Kg	46	49	53	55	64
	Gross Weight	Kg	54	57	60	63	73

*Unique features in models 6360, 6460 and 21860 only

*Upto 40% Energy Saving is applicable for all Inverter models w.r.t Non-Inverter R134a refrigerator



PEL NON-INVERTER SERIES (FLAT GLASS DOOR)



NON INVERTER SERIES FLAT GLASS DOOR

Unique Features

- Clean and Wide Look
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant



RED BLAZE



PURPLE BLAZE



PMI - RED

PEL NON-INVERTER SERIES (FLAT GLASS DOOR)



Technical Specifications

									JUMBO SERIES		
	DESCRIPTION	Units	PRGD 2000	PRGD 2200	PRGD 2350	PRGD 2550	PRGD 6360	PRGD 6460	PRGD 21860	PRGD 21960	PRGD 22260
INTERNAL Capacity	Total	Liters	168	194	233	255	310	334	354	382	408
	Freezer	Liters	59	59	85	85	90	90	113	141	141
	Refrigerator	Liters	109	135	148	170	220	244	241	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	115	115	130	140	132	147	147	178	178
	Current Consumption	Ampere	0.85	0.9	0.9	1.1	0.78	0.80	0.80	1.2	1.2
	Start Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star	4 Star
GENERAL FEATURES	Climate Class	-	T	T	T	T	T	T	T	T	T
	Defrosting Freezer	-	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual
	Defrosting Refrigerator	-	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	-	-	-	-	Yes	Yes	No	No	No
	Refrigerant	-	R134a	R134a	R134a	R134a	R600a	R600a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Roll Bond	Roll Bond	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
DIMENSIONS (only Cabinet without Door)	Child Lock with Key	-	Available	Available	Available	Available	-	-	-	-	-
	Shelves Type	-	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type
	Height	mm	1255	1400	1440	1540	1570	1660	1610	1724	1814
WEIGHT	Width	mm	490	490	540	540	600	600	662	662	662
	Depth	mm	430	430	471	471	540	540	540	540	540
	Net Weight	Kg	39	41	46	49	57	59	68	70	72
	Gross Weight	Kg	44	46	54	57	64	67	77	80	82

*Unique features in models 6360, 6460, 21860, 21960 and 22260 only



PEL NON-INVERTER SERIES (CURVED GLASS DOOR)



NON INVERTER SERIES CURVED GLASS DOOR

Unique Features

- Clean and Wide Look
- Jet Cool 360°
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant



RED BLAZE



MAROON BLAZE



GREY BLAZE

PEL Range – Refrigerator



PEL NON-INVERTER SERIES (CURVED GLASS DOOR)



Technical Specifications

						JUMBO SERIES	
	DESCRIPTION	Units	PRCGD 2550	PRCGD 6360	PRCGD 6460	PRCGD 21960	PRCGD 22260
INTERNAL Capacity	Total	Liters	255	310	334	382	408
	Freezer	Liters	85	90	90	141	141
	Refrigerator	Liters	170	220	244	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	140	132	147	178	178
	Current Consumption	Ampere	1.1	0.78	0.80	1.2	1.2
	Start Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T	T	T
GENERAL FEATURES	Defrosting	Freezer	-	Manual	Manual	Manual	Manual
		Refrigerator	-	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	No	Yes	Yes	No	No
	Refrigerant	-	R134A	R600a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	-	-	-	-
	Shelves Type	-	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type
DIMENSIONS (only Cabinet without Door)	Height	mm	1540	1570	1660	1724	1814
	Width	mm	540	600	600	662	662
	Depth	mm	471	540	540	540	540
WEIGHT	Net Weight	Kg	49	57	59	70	72
	Gross Weight	Kg	57	64	67	80	82

*Unique features in models 6360, 6460, 21960 and 22260 only

PEL INVERTER SERIES (FLAT GLASS DOOR)



INVERTER SERIES FLAT GLASS DOOR

Unique Features

- Clean and Wide Look
- Upto 40% Energy Saving
- Jet Cool 360°
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant



Inverteron
TECHNOLOGY



MAROONBLAZE



GREY BLAZE

PEL INVERTER SERIES (FLAT GLASS DOOR)

Technical Specifications



							JUMBO SERIES
DESCRIPTION		Units	PRINVOFGD 2350	PRINVOFGD 2550	PRINVOFGD 6360	PRINVOFGD 21860	PRINVOFGD 22260
INTERNAL Capacity	Total	Liters	233	255	310	354	408
	Freezer	Liters	85	85	90	113	141
	Refrigerator	Liters	148	170	220	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	130	130	82	108	108
	Current Consumption	Ampere	0.8	0.8	0.65	0.88	0.88
	Start Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T	T	T
GENERAL FEATURES	Defrosting Freezer	-	Manual	Manual	Manual	Manual	Manual
	Refrigerator	-	Manual	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	-	-	Yes	No	No
	Refrigerant	-	R134a	R134a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	Available	-	-	-
DIMENSIONS (only Cabinet without Door)	Shelves Type	-	Wire Type	Wire Type	Wire Type	Wire Type	Wire Type
	Height	mm	1440	1540	1570	1610	1814
	Width	mm	540	540	600	662	662
	Depth	mm	471	471	540	540	540
WEIGHT	Net Weight	Kg	46	49	57	68	72
	Gross Weight	Kg	54	57	64	77	82

*Unique features in models 6360, 21860 and 22260 only

*Upto 40% Energy Saving is applicable for all Inverter models w.r.t Non-Inverter R134a refrigerant

PEL ULTRA INVERTER SERIES (CURVED GLASS DOOR)



ULTRA INVERTER SERIES

CURVED GLASS DOOR

Unique Features

- Clean and Wide Look
- Upto 40% Energy Saving
- Jet Cool 360°
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Eco Friendly R600a Refrigerant
- Toughened Glass Shelves (Can bear load of upto 135kg)
- Deodorization & Sterilization (Eliminates 99.99% bacteria and bad odour)



RED BLAZE



PURPLE BLAZE

PEL ULTRA INVERTER SERIES (CURVED GLASS DOOR)



Technical Specifications

			JUMBO SERIES				
DESCRIPTION		Units	PRUINVOCGD 2550	PRUINVOCGD 6360	PRUINVOCGD 6460	PRUINVOCGD 21860	PRUINVOCGD 22260
INTERNAL Capacity	Total	Liters	255	310	334	354	408
	Freezer	Liters	85	90	90	113	141
	Refrigerator	Liters	170	220	244	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	130	82	108	108	108
	Current Consumption	Ampere	0.8	0.65	0.88	0.88	0.88
	Star Rating (Cooling)	-	4 Star	4 Star	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T	T	T
GENERAL FEATURES	Defrosting Freezer	-	Manual	Manual	Manual	Manual	Manual
	Defrosting Refrigerator	-	Manual	Manual	Manual	Manual	Manual
	Temperature Control (Adjustable)	-	Mechanical	Mechanical	Mechanical	Mechanical	Mechanical
	Crispo Tray	-	Yes	Yes	Yes	Yes	Yes
	Humidity Controller	-	-	Yes	Yes	No	No
	Refrigerant	-	R134a	R600a	R600a	R600a	R600a
	Evaporator	-	Roll Bond	Tube on Sheet	Tube on Sheet	Tube on Sheet	Tube on Sheet
	Condenser Type	-	Natural Draft	Natural Draft	Natural Draft	Natural Draft	Natural Draft
	Child Lock with Key	-	Available	-	-	-	-
DIMENSIONS (only Cabinet without Door)	Shelves Type	-	Glass	Glass	Glass	Glass	Glass
	Height	mm	1540	1570	1660	1610	1814
	Width	mm	540	600	600	662	662
	Depth	mm	471	540	540	540	540
WEIGHT	Net Weight	Kg	49	57	59	68	72
	Gross Weight	Kg	57	64	67	77	82

*Unique features in models 6360, 6460, 21860 and 22260 only except Deodorization and Sterilization which is common in all models of this category

*Upto 40% Energy Saving is applicable for all Inverter models w.r.t Non-Inverter R134a refrigerant

PEL DIGITRON INVERTER SERIES (FLAT GLASS DOOR)



DIGITRON INVERTER SERIES

FLAT GLASS DOOR

Unique Features

- Clean and Wide Look
- Digital Smart Display
- Upto 40% Energy Saving
- Jet Cool 360°
- Complete Light Zone (Elegant LED's in both Ref and Frz sections)
- Intelligent 4 Modes Operation (Holiday, Insta Cool, Energy Saver, Fast Freeze)
- Delay Start Safety Feature
- Eco Friendly R600a Refrigerant
- Toughened Glass Shelves (Can bear load of upto 135kg)



RED BLAZE



Inverter on
TECHNOLOGY

PEL Range – Refrigerator



PEL DIGITRON INVERTER SERIES (FLAT GLASS DOOR)



Technical Specifications

			JUMBO SERIES		
	DESCRIPTION	Units	PRINVOGDDT 6360	PRINVOGDDT 21960	PRINVOGDDT 22260
INTERNAL Capacity	Total	Liters	310	382	408
	Freezer	Liters	90	141	141
	Refrigerator	Liters	220	241	267
PERFORMANCE	Voltage/Frequency	Volts/Hz	220V / 50 Hz	220V / 50 Hz	220V / 50 Hz
	Power Consumption	Watt	108	108	108
	Current Consumption	Ampere	0.88	0.88	0.88
	Star Rating (Cooling)	-	4 Star	4 Star	4 Star
	Climate Class	-	T	T	T
GENERAL FEATURES	Defrosting	Freezer	-	Manual	Manual
		Refrigerator	-	Manual	Manual
	Temperature Control (Adjustable)		-	Digital	Digital
	Crispo Tray		-	Yes	Yes
	Humidity Controller		-	Yes	No
	Refrigerant		-	R600a	R600a
	Evaporator		-	Tube on Sheet	Tube on Sheet
	Condenser Type		-	Natural Draft	Natural Draft
	Child Lock with Key		-	-	-
	Shelves Type		-	Glass	Glass
DIMENSIONS (only Cabinet without Door)	Height	mm	1570	1724	1814
	Width	mm	600	662	662
	Depth	mm	540	540	540
WEIGHT	Net Weight	Kg	57	70	72
	Gross Weight	Kg	64	80	82

*Unique features in models 6360, 21960 and 22260 only

*Upto 40% Energy Saving is applicable for all Inverter models w.r.t Non-Inverter R134a refrigerant

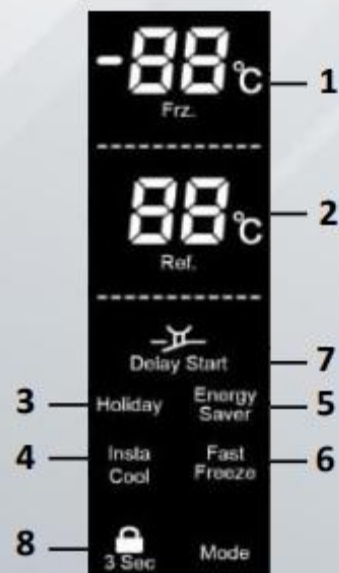
PEL DIGITRON INVERTER SERIES (FLAT GLASS DOOR)



DIGITRON INVERTER OPERATION GUIDE

Important Instructions:

- When power supply is plugged, compressor will start after 3 minutes
- The control panel gets LOCK and lights will turn OFF automatically after 1 minute
 - To activate the control panel, touch **MODE** or **LOCK** button once
- To change the **MODE**, first unlock the panel by pressing **LOCK** button for 3 seconds and then choose your desired **MODE**.



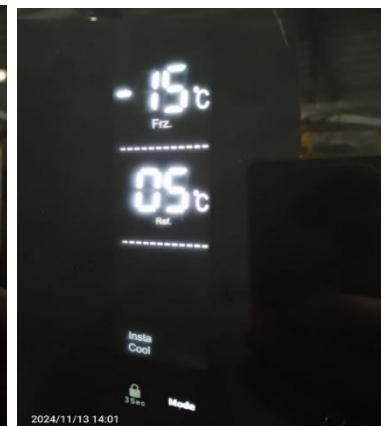
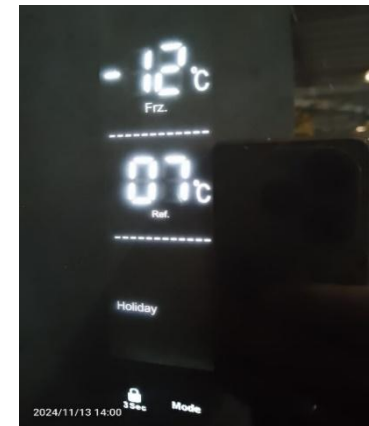
Sr. #	Description	Function
1	Frz Temp	Freezer section temperature
2	Ref Temp	Refrigerator section temperature
3	Holiday	Ref: +7°C and Frz: -12°C
4	Insta Cool	Ref: +5°C and Frz: -15°C
5	Energy Saver	Ref: +3°C and Frz: -18°C
6	Fast Freeze	Ref: +0°C and Frz: -22°C
7	Delay Start	When compressor is in delay/rest mode
8	Child Lock	Press for 3 seconds to Lock/Unlock the control panel

*Available in 6360, 21960 and 22260 models in inverter only

PEL DIGITRON INVERTER SERIES (Display Setting Demo)

- New Models of Digitron series are operated by Display panel.
- Display having 4 cooling zones option as **Holiday – Insta Cool – Energy saver – Fast Freeze**.(As per below table)
- A Lock option is also available to lock and unlock cooling zone by pressing it 3 sec at respective requirement.
- Cooling zones as per below table. ↓

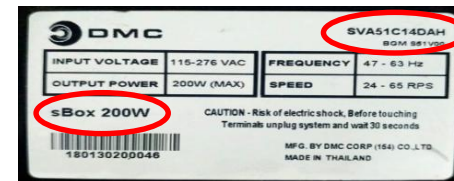
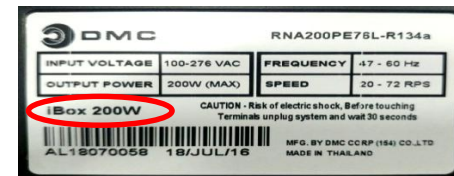
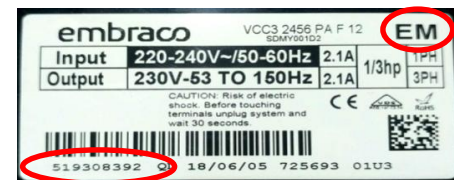
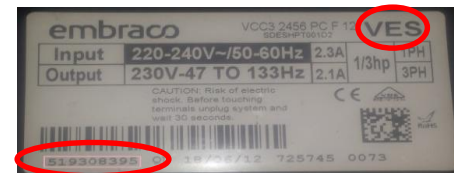
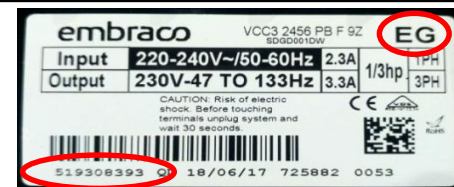
Sr. No	Function	Frz. Temp (°C)	Ref Temp (°C)
1	Holiday	-12	7
2	Insta cool	-15	5
3	Energy Saver	-18	3
4	Fast freeze	-22	0



PEL Range – Refrigerator



Inverter Compressor & PCB Kit For Inverter Refrigerators (OLD DOM 2023 & Before)

REF Model	Compressor	Inverter Kit	Inverter Code	Inverter Kit Label
2350 2550	Panasonic SVA51C14DAH (1/5hp)	DMC sBox - 200 W	SVA51C14DAH	
6350 6450 21850 21950 22250	Panasonic EEI76C13DCH (1/3hp)	DMC iBox - 200 W	IBOX	
2350 2550 6350 6450	Embraco VEMY 6H (1/5hp)	Embraco	EM 519308392	
2350 2550 6450	Embraco VESA 7H (1/4hp)	Embraco	VES 519308395	
21850 21950 22250	Embraco VEGC 8H (1/3hp)	Embraco	EG 519308393	

PEL Range – Refrigerator



Inverter Compressor & PCB Kit For Inverter Refrigerators (DOM 2024 & Onwards)

 INVERTER COMPRESSOR & KIT INFORMATION FOR REF & DF				
Dated: 23-12-2024				
REF Model	Compressor	Inverter Kit	Inverter Code	Inverter Kit Label
6350 6450 21850 22250	Wanbao VATZ72L	HUAGUANG Wanbao	WB 51.16.18	
2000 2200 2350 2550	Wanbao VATZ60	HUAGUANG Wanbao	WB 51.16.16	
6360 PDF-100 PDFT-155	Embraco FMXA9CE	FMX VES 2456 DJ F 00	519306719	
6360 6460 21860 21960 22260 All DF Models	Wanbao VETZ110L	HUAGUANG Wanbao	WB 52.16.11	

Inverter Compressor & PCB Kit For Inverter Refrigerators (Error Codes)

Inverter Kit WANBAO VATZ72L (WB-51.16.18)

故障代码 Fault code:

序号 No.	相应故障 Corresponding fault	故障代码 Code
1	过电流保护 Over current	LED 闪烁 1 次 LED flashes 1 times
2	缺相或未接电机（三相全断）保护 Phase failure or no compressor is connected at all.	LED 闪烁 2 次 LED flashes 2 times
3	欠电压 Low voltage	LED 闪烁 3 次 LED flashes 3 times
4	硬件触发过电流保护 Hardware triggered over current detected	LED 闪烁 4 次 LED flashes 4 times
5	启动失败、堵转或无法维持最低转速 Start failure、rotor locked or fail to maintain min speed.	LED 闪烁 5 次 LED flashes 5 times
6	过速，出现预料之外的过高转速 Unexpected high speed detected.	LED 闪烁 6 次 LED flashes 6 times
7	过电压 Over voltage	LED 闪烁 7 次 LED flashes 7 times
8	功率保护（转速降低至指定最低转速时输出功率仍高于规定保护功率，或输出功率触及设定的停机功率限值） Over power	LED 闪烁 8 次 LED flashes 8 times

注：故障代码根据实际板子软件故障模式添加。

Note: Fault code is added according to the actual software failure mode.



PEL Range – Refrigerator



Inverter Compressor & PCB Kit For Inverter Refrigerators (Error Codes)

Inverter Kit Embraco FMXA9CE (FMX VES 2456 DJF 00)

LED Pattern	Period	Status Description
1 flash	15 seconds	Normal operation
2 flashes	5 seconds	Control communication problem
3 flashes	5 seconds	Inverter problem
4 flashes	5 seconds	Compressor problem
1 flash	0.5 seconds	Temperature protection active

This function helps services technicians to diagnose possible fault components by blinking a green LED inside the box.

PEL Range – Refrigerator



Inverter Compressor & PCB Kit For Inverter Refrigerators (Error Codes)

Inverter Kit for WANBAO VETZ110L & VATZ60 (WB-52.16.11) & (WB-51.16.16)

No.	Corresponding fault	Code
1	Over power	LED flashes 3 times
2	Start failure	LED flashes 4 times
3	Rotor locked	LED flashes 5 times
4	Low voltage	LED flashes 6 times
5	Over voltage	LED flashes 7 times
6	Over current protection in Software	LED flashes 8 times
7	Phase failure or no compressor is connected at all	LED flashes 9 times
8	Hardware triggered over current detected	LED flashes 10 times
9	Over-temperature protection	LED flashes 11 times
10	AD self-checking failure	LED flashes 12 times
11	Software over current at startup	LED flashes 13 times

PEL Range – Refrigerator



Inverter Compressor & PCB Kit For Inverter Refrigerators

Solar Panel;

- | | |
|---------------------------|-------------|
| 1. Solar Panel | 600 Watt |
| 2. Inverter/controller | 1~3 KVA |
| 3. Batteries (for backup) | 75~165 Amps |

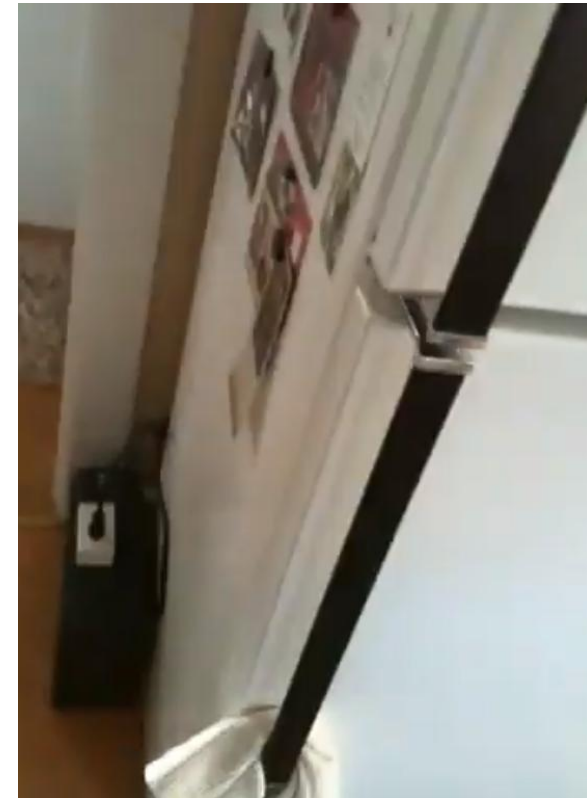
Note: use more solar panels, if you are consuming household load.



Uninterruptible Power Supply;

- | | |
|---------------------------|-------------|
| 1. UPS | 1KVA |
| 2. Batteries (for backup) | 75~165 Amps |

Note: use high capacity UPS, if you are consuming household load.



❑ PEL Deep Freezer



PEL Range – Deep Freezer



NON-INVERTER SERIES



Description	UOM	PDF 100 Arctic Pro Freeze	PDF 130 Arctic Pro Freeze	PDF 150 Arctic Pro Freeze	PDF 135 Arctic Pro Freeze	PDF 155 Arctic Pro Freeze
Capacity	Liter	275	370	410	370	410
Door Type	-	Single	Single	Single	Double	Double
Voltage / Frequency	V/Hz	220/50	220/50	220/50	220/50	220/50
Power Consumption	W	152	183	183	152	183
Current Consumption	A	1.21	1.5	1.5	1.3	1.5
Temperature Control	-	Mechanical (EBD077B1719 Danfoss No short circuit/no spark)	Mechanical	Mechanical	Mechanical	Mechanical
Refrigerant	-	R600a (60g)	R600a (65g)	R600a (65g)	R600a (65g)	R600a (65g)

PEL Range – Deep Freezer



INVERTER SERIES



Description	UOM	PDINT 100 PEL Artic Inverter on	PDINT 130 PEL Artic Inverter on	PDINT 150 PEL Artic Inverter on	PDINT 135 PEL Artic Inverter on	PDINT 155 PEL Artic Inverter on
Capacity	Liter	275	370	410	370	410
Door Type	-	Single	Single	Single	Double	Double
Voltage / Frequency	V/Hz	220/50	220/50	220/50	220/50	220/50
Power Consumption	W	152	183	183	152	183
Current Consumption	A	1.21	1.5	1.5	1.3	1.5
Temperature Control	-	Mechanical (EBD077B1719 Danfoss No short circuit/no spark)	Mechanical	Mechanical	Mechanical	Mechanical
Refrigerant	-	R600a (60g)	R600a (65g)	R600a (65g)	R600a (65g)	R600a (65g)

PEL Range – Deep Freezer



Direct Cool Deep Freezer

Arctic InverterOn

Model	Single Door	Double Door
	Capacity (Liter/Cft)	
PDIN70-100	275/10	-
PDIN70-130	370/13	-
PDIN70-150	410/15	-
PDINT70-135	-	370/13.5
PDINT70-155	-	410/15.5

Only sBox Kit used

Arctic Profreeze/ Profreeze Extended

Model	Single Door	Double Door
	Capacity (Liter/Cft)	
PDF70-100	275/10.0	-
PDF70-130	370/13.0	-
PDF70-150	410/15.0	-
PDFT70-135	-	370/13.5
PDFT70-155	-	410/15.5

Single Door

Double Door

Single Door

Double Door



Direct Cool Deep Freezer

Vertical Deep Freezer

Front View



Model	Single Door
	Capacity (Liter/Cft)
PVF-6D	239/8.5
PVF-7D	281/10.0

INSIDE VIEW





PEL Range – Deep Freezer



R-600a Deep Freezer

1. Units Circuitry

PDF - 100

- Gas weight: 60g [R600a]
- Capillary 0.031" Length : 10' 5"
- Flow Rate : 4.25 l/min
- Compressor: HXK95AT with Capacitor 4uF
- Return Line 3 ft
- Evaporator: 6 Turns
- Condenser : Medium WTC
- Dryer: Same as existing 12 g
- Fan Blade: Suction Type
- Harness Same as Existing
- Thermostat Model: 077B1719 EBD

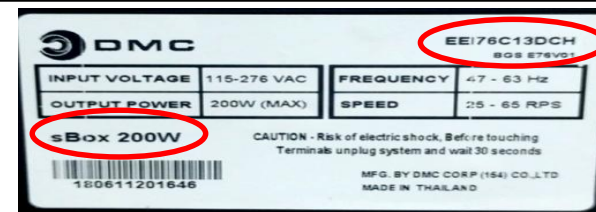
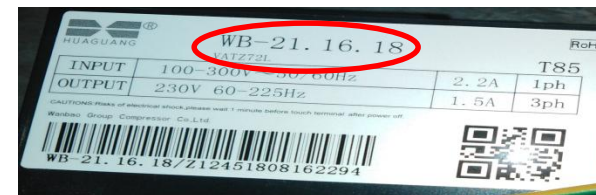
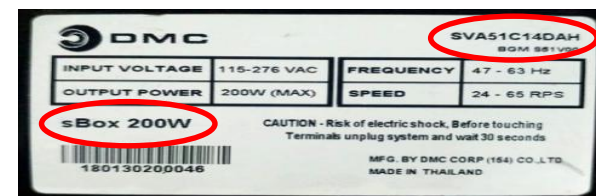
PDF - 135

- Gas weight: 65g [R600a]
- Capillary 0.036" Length : 13' 5"
- Flow Rate : 5.5 l/min
- Compressor: HXK12AT with Capacitor 4uF
- Return Line 4 ft
- Evaporator : 6 Turn
- Condenser : Medium WTC
- Dryer: Same as existing 12 g
- Fan Blade: Suction Type
- Harness Same as Existing
- Thermostat Model: EBD 095X1129LA

PEL Range – Deep Freezer



Inverter Compressor & PCB Kit For Inverter Deep Freezer (Old DOM 2023 & Before)

DF Model	Compressor	Inverter Kit	Inverter Code	Inverter Kit Label
PDF-370T(135) PDF-370S(135)	1. Panasonic EEI76C13DCH	1. DMC sBox - 200 W	1. EEI76C13DCH	
New PDF-410S (155) PDF-410T(155)	Wanbao VATZ72L	HUAGUANG – T85	WB- 21. 16. 18 VATZ72L	
PDF-275(100) PDF-370T(135)	Panasonic SVA51C14DAH	DMC sBox - 200 W	SVA51C14DAH	
PDF-275(100) PDF-370T(135)	Embraco VEMY 6H	Embraco	EM 519308392	

PEL Range – Deep Freezer



Inverter Compressor & PCB Kit For Inverter Deep Freezer (DOM 2024 & Onward)

 INVERTER COMPRESSOR & KIT INFORMATION FOR REF & DF				
Dated: 23-12-2024				
REF Model	Compressor	Inverter Kit	Inverter Code	Inverter Kit Label
6360 PDINT – 100 / 155	Embraco FMXA9CE	FMX VES 2456 DJ F 00	519306719	
6360 6460 21860 21960 22260 All DF Models	Wanbao VETZ110L	HUAGUANG Wanbao	WB 52.16.11	



PDINT – 100 / 130 / 135 /150 /155

DOM 2024 Production
100/155 Inverter
Models we used FMX
Comp and Kit

PEL Range – Deep Freezer



Inverter Compressor & PCB Kit For Inverter Refrigerators (Error Codes)

Inverter Kit Embraco FMXA9CE (FMX VES 2456 DJF 00)

LED Pattern	Period	Status Description
1 flash	15 seconds	Normal operation
2 flashes	5 seconds	Control communication problem
3 flashes	5 seconds	Inverter problem
4 flashes	5 seconds	Compressor problem
1 flash	0.5 seconds	Temperature protection active

This function helps services technicians to diagnose possible fault components by blinking a green LED inside the box.

Inverter Compressor & PCB Kit For Inverter Refrigerators (Error Codes)

Inverter Kit for Wanbao VETZ110L (WB-52.16.11)

No.	Corresponding fault	Code
1	Over power	LED flashes 3 times
2	Start failure	LED flashes 4 times
3	Rotor locked	LED flashes 5 times
4	Low voltage	LED flashes 6 times
5	Over voltage	LED flashes 7 times
6	Over current protection in Software	LED flashes 8 times
7	Phase failure or no compressor is connected at all	LED flashes 9 times
8	Hardware triggered over current detected	LED flashes 10 times
9	Over-temperature protection	LED flashes 11 times
10	AD self-checking failure	LED flashes 12 times
11	Software over current at startup	LED flashes 13 times

PEL Range – Deep Freezer



Inverter Deep Freezer run on Solar Panel and UPS

Solar Panel;

- | | |
|------------------------------|-------------|
| 1. Solar Panel | 600 Watt |
| 2. Inverter / Controller Box | 1~3 KVA |
| 3. Batteries (for backup) | 75~165 Amps |

Note: use more solar panels, if you are consuming household load.



Uninterruptible Power Supply;

- | | |
|---------------------------|-------------|
| 1. UPS | 1KVA |
| 2. Batteries (for backup) | 75~165 Amps |

Note: use high capacity UPS, if you are consuming household load.



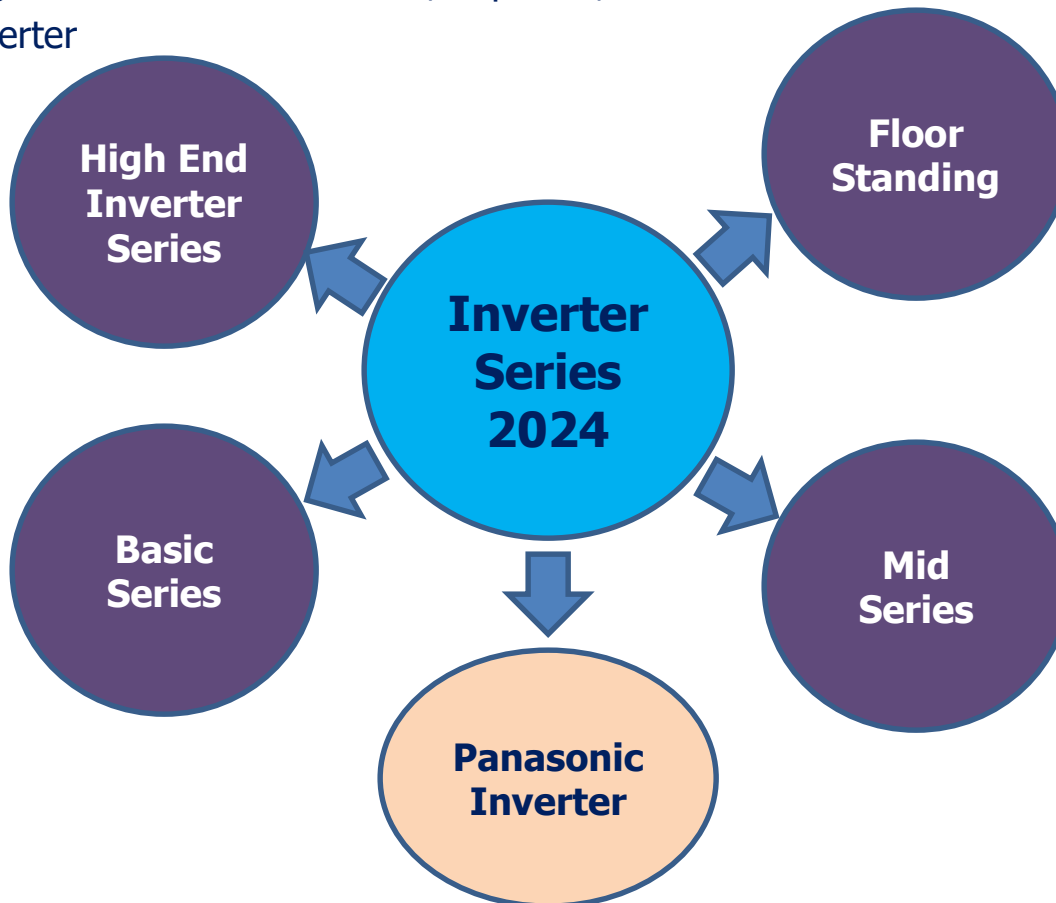
❑ PEL Air Conditioner





Inverter Range 2024 : Subcategories

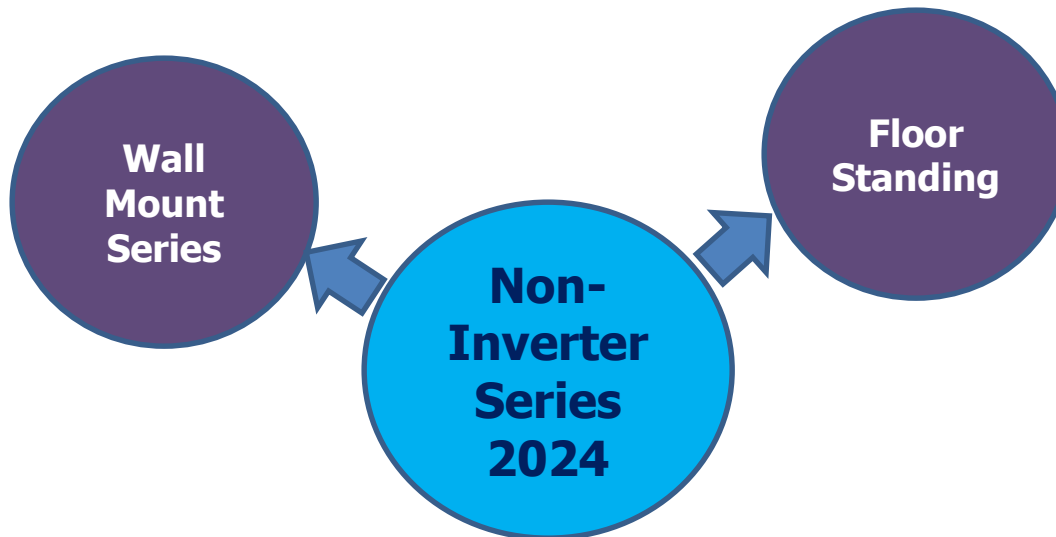
1. Basic Series : ALPHA, AERO Extend
2. Mid Series : Sublime T3, Jumbo Black
3. High End Inverter Series : Turbo DC Ultimate T3, Jumbo DC-Classic, Jumbo DC Prime T3
4. Floor Standing Series : Bold+, Supreme, Ultimate
5. Panasonic Inverter





Non-Inverter Range 2024 : Subcategories

1. Wall mount : MAJESTIC T3, MAJESTIC GLORY
2. Floor Standing : BOLD, REGAL



Inverter (Variable Speed) Air Conditioners

1. Basic Series

Inverteron
TECHNOLOGY



ALPHA Basic Inverter

Inverteron
TECHNOLOGY

- EER 3.0
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- R32 Refrigerant

Capacity	IDU Size (mm)
12K	870
18K	977
24K	1148



Superior Performance & Energy Saving



*Operates on as Low as 2.5A



*Minimal Start-Up Voltage at 140 Volts



Automatically Cleans the AC & Eliminates Moisture



Strong & Durable Copper Coils for Best Cooling

*Testing under controlled conditions

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Double Row Condenser (18k Only)
- LVS @ 140 Volts
- Gold Fin Evaporator and Condenser

Inverter (Variable Speed) Air Conditioners

1. Basic Series Specification:

PEL Model Name			12K Alpha	18K Alpha	24K Alpha
Function			H&C	H&C	H&C
Capacity			12K	18K	22K
Rated Capacity	Cooling	W	3500	5200	6450
	Heating	W	3700	5200	6740
Rated Power Consumption	Cooling	W	1160	1705	2160
	Heating	W	1060	1535	2040
Rated Running Current	Cooling	A	5.2	7.6	9.7
	Heating	A	4.8	6.9	9.1
Max. Input Power		W	1700	2300	3200
Max. Input Current		A	9	12.7	14.2
EER Cooling		W/W	3.0	3.0	3.0
COP Heating		W/W	3.5	3.45	3.3
Power Supply Source		V/Ph/Hz	220-240V/1Ph/50Hz	220-240V/1Ph/50Hz	220-240V/1Ph/50Hz
Voltage Range		V	130-270V	130-270V	130-270V
Refrigerant			R32	R32	R32
Refrigerant Charged		g	610	990	1100
Max. Discharge Pressure		MPa	4.15	4.15	4.15
Max. Suction Pressure		MPa	1.6	1.6	1.6
Air Flow Volume		m ³ /h	660	1000	1150
Moisture Removal		L/h	1.2	1.8	2
Indoor Noise Level		dB	37/35/30	44/41/36	46/43/40
Outdoor Noise Level		dB	50	55	55
Indoor Unit Weight (Net Weight/ Gross Weight)		Kg	9/11	12/14.5	15/17.5
Outdoor Unit Weight (Net Weight/ Gross Weight)		Kg	24/26.5	29/31	39/42.5

Inverter (Variable Speed) Air Conditioners

1.1 Basic Series



Aero Extend Basic Inverter



- EER 3.2
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- 2 Row Condenser (18K only)
- R32 Refrigerant

Capacity	IDU Size (mm)
12K	805
18K	957



Superior Performance & Energy Saving



Strong & Durable Copper Coils for Best Cooling



Automatically Cleans the AC & Eliminates Moisture



Efficient cooling performance even at 50 degrees



High Capacity Rotary Compressor



Heat & Cool

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Double Row Condenser (18k Only)
- LVS @ 140 Volts
- Gold Fin Evaporator and Condenser

*Testing under controlled conditions

**Available in 18K and 24K only

Inverter (Variable Speed) Air Conditioners

1.1 Basic Series Specification:

Model			12K Aero Extend	18K Aero Extend
Power supply		V-Ph-Hz	220_240V_1Ph,FREQUENCY_50HZ	220_240V_1Ph,FREQUENCY_50HZ
Cooling (Standard conditions)	Capacity	Btu/h	12500	18000
	Range		1300~3750	2500~5700
	Input	W	1140	1640
	Range		270~1260	550~1960
	Current	A	5.0	7.2
	Range		1.85~5.52	3.80~8.60
	EER	W/W	3.21	3.22
Heating (Standard conditions)	Capacity	Btu/h	12500	18500
	Range		1500~3950	2600~5520
	Input	W	980	1500
	Range		450~1100	500~1585
	Current	A	4.3	6.5
	Range		3.15~4.81	3.50~6.95
	COP	W/W	3.74	3.61
Rated Power Input		W	1800	2650
Rated Current		A	9.0	13.5
Starting current		A	/	0
Indoor air flow (Hi/Mi/Lo)		m3/h	600/445/355	840/670/550
Indoor noise level (Hi/Mi/Lo)		dB(A)	42/35.5/30	43.5/38/32
Outdoor noise level		dB(A)	55	56

Inverter (Variable Speed) Air Conditioners

2. Mid Series



SUBLIME T3 Basic+ Inverter



- EER 3.3
- Full DC (18K only)
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- 2 Row Condenser (18K only)
- R32 Refrigerant

Capacity	IDU Size (mm)
12K	790
18K	920



*Powerful Cooling up to 50°C Ambient Temperature



*Operates on as low as 2.5A



*Minimal Start-Up Voltage at 155 Volts



*Flexible Electricity Consumption



For Homogeneous Cooling Across the Room



Automatically Cleans the AC & Eliminates Moisture



Strong & Durable Copper Coils for Best Cooling



Superior Performance & Energy Saving

*Testing under controlled conditions

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Double Row Condenser (18K SUBLIME Only)
- LVS @ 155 Volts
- Gold Fin Evaporator and Condenser

Inverter (Variable Speed) Air Conditioners

2. Mid Series Specification:

Model No.			12K Sublime	18K Sublime	24K Sublime
Rated cooling capacity(T1)		Btu/h	12000 (3412~12863)	18000(4094~20472)	24000 (5118~24908)
Rated cooling capacity(T1)		W	3517(1000~3770)	5275(1200~6000)	7034 (1500~7300)
Rated heating capacity		Btu/h	12000 (3412~13000)	19500(5000~21500)	24500 (5118~25590)
EER for cooling(T1)		Btu/h-W;	10.95	11.60	11.60
EER for cooling(T1)		W/W	3.21	3.40	3.40
COP for heating		Btu/h-W;	12.32	12.10	12.64
SEER			14.85	14.00	14.00
Moisture removal		Liters/h	1.20	1.80	2.40
Pressure	High(DP)	MPa	3.70	3.70	3.70
	Low(SP)	MPa	1.20	1.20	1.20
Indoor noise level at cooling	Super	dB(A)	40.00	45.00	50.00
	High	dB(A)	37.00	41.00	45.00
	Med.	dB(A)	33.00	37.00	41.00
	Low	dB(A)	29.00	32.00	36.00
	Quite	dB(A)	27.00	29.00	32.00
Outdoor noise level		dB(A)	50.00	55.00	57.00
Climate type			T1	T1	T1
Power supply			220-240V~/50Hz/1P	220-240V~/50Hz/1P	220-240V~/50Hz/1P
Current	Cooling(T1)	A	4.6(1.3~9.0)	6.8 (1.4~11)	9.18 (1.5~13.0)
	Heating	A	4.5(1.3~9.0)	7.08 (1.4~11)	8.6 (1.5~13.0)
Power input	Cooling(T1)	W	1049(290~1500)	1552 (300~2050)	2068 (200~2600)
	Heating	W	1049(290~1500)	1612 (220~2130)	1938 (230~2600)
Rated Current	Cooling(T1)	A	10.00	12.00	14.70
	Heating	A	10.00	12.00	14.70
Rated Power input	Cooling(T1)	W	1730	2550	3310
	Heating	W	1730	2550	3310
Refrigerant		kg	R32/0.49kg	R32/0.88kg	R32/1.06kg
Indoor air volume(Cooling/Heating)		m ³ /h	560/560	850 / 850	1200/1200
Connecting Pipe	Gas	Inches	Φ9.52(3/8")	Φ9.52(3/8")	Φ12(1/2")
	Liquid	Inches	Φ6(1/4")	Φ6(1/4")	Φ6(1/4")

Inverter (Variable Speed) Air Conditioners

2.1 Mid Series



JUMBO BLACK
Mid Inverter

- Black panel
- EER 3.5
- Gold Fin Evaporator & Condenser
- R410a Refrigerant

Capacity	IDU Size (mm)
12K	805
18K	1025
24K	1025





*Operates on as Low as 2.5A

*Minimal Start-Up Voltage at 165 Volts

3 Stage Air Filtration





Smart Connection For Easy Control

Automatically Cleans the AC & Eliminates Moisture

Strong & Durable Copper Coils for Best Cooling

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Bigger IDU 18k & 24K JUMBO BLACK 1025mm / 1025mm
- Double Row Condenser (18K SUBLIME Only)
- LVS @ 155 Volts
- Gold Fin Evaporator and Condenser

Inverter (Variable Speed) Air Conditioners

2.1 Mid Series Specification:

PEL Model			12K Jumbo Black	18K Jumbo Black	24K Jumbo Black
Power supply		Ph-V-Hz	220V/50HZ	220V/50HZ	220V/50HZ
Type			Split	Split	Split
Power Supply Mode			Indoor	Indoor	Indoor
Cooling	Capacity	BTU	12500	18500	24000
	Power Input	W	1038	1545	2004
	Current	A	4.7	7.0	9.1
	EER	W/W	3.53	3.51	3.51
Heating	Capacity	BTU	12500	18500	24000
	Power Input	W	1015	1502	1948
	Current	A	4.6	6.8	8.9
	COP	W/W	3.61	3.61	3.61
Moisture removal		L/h	1.2	1.8	2.4
Max. input consumption		W	1557	2317	3006
Max. current		A	7.3	10.9	14.1
Indoor air flow	Indoor air flow	m3/h	550/500/450/400	1150/1080/1000/850	1150/1080/1000/850
	Noise level	dB(A)	42/40/38/36	46/43/40/37	46/43/40/37
	Unit dimension (W*H*D)	mm	805×270×200	1025×320×226	1025×320×226
	Packing (W*H*D)	mm	864×331×265	1102×395×305	1102×395×305
	Noise level	dB(A)	54	54	55
	Unit dimension (W*H*D)	mm	780×560×270	780×560×270	820×635×310
	Packing (W*H*D)	mm	889×612×359	889×612×359	969×688×402
	Refrigerant type		R410A	R410A	R410A
	Refrigerant charge	Kg	0.8	1.35	1.42
	Refrigerant pipe				
Liquid side	Liquid side	inch	1/4	1/4	1/4
	Gas side	inch	3/8	1/2	5/8

Inverter (Variable Speed) Air Conditioners

3. High End Inverter - Series

Inverteron
TECHNOLOGY



TURBO DC - ULTIMATE
High End Inverter

Capacity	IDU Size (mm)
12K	802
18K	971
24K	1083

- Full DC Inverter
- EER 3.4
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- 2 Row Condenser (18K only)
- R32 Refrigerant
- T3

100% DC INVERTER
Outstanding Efficiency & Electricity Savings

A++ Inverter Technology
*Operates on as Low as 2.5A

140V Low Voltage Start
*Minimal Start-Up Voltage at 140 Volts

TE
*Powerful Cooling up to 50°C Ambient Temperature

WIFI SMART
Smart Connection For Easy Control

eco gear
*Flexible Electricity Consumption

MULTI HEALTH FILTERS
3 Stage Air Filtration

SELF-CLEAN EVAPORATOR
Automatically Cleans the AC & Eliminates Moisture

*Testing under controlled conditions

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Double Row Condenser (18k Only)
- Gold Fin Evaporator and Condenser
- LVS @ 140V AC

- Gen Mode Function
- Wi-Fi Separate USB Dongle provided with Unit.

Inverter (Variable Speed) Air Conditioners

3. High End Inverter - Series

Model			12K Turbo DC Ultimate	18K Turbo DC Ultimate	24K Turbo DC Ultimate
Power supply		V-Ph-Hz	220_240V_1Ph,FREQUENCY_50H Z	220_240V_1Ph,FREQUENCY_50H Z	220_240V_1Ph,FREQUENCY_50H Z
Cooling (Standard conditions)	Capacity	Btu/h	12000	18000	24000
	Range		3900~13500	5100~20100	10300~30000
	Input	W	1004	1507	2420
	Range		100~1182	400~1883	340~3450
	Current	A	4.5	6.8	10.5
	Range		1.1~5.4	1.9~8.3	1.4~15
	EER	W/W	3.6	3.6	3.0
Heating (Standard conditions)	Capacity	Btu/h	12000	18000	25000
	Range		4200~12800	6100~18500	5200~32300
	Input	W	925	1388	2130
	Range		240~1083	420~1400	300~3150
	Current	A	4.2	6.0	9.3
	Range		2.5~4.8	2.0~6.2	1.3~13.7
	COP	W/W	3.80	3.80	3.44
Rated Power Input		W	1800	2700	3700
Rated Current		A	9.0	13.5	19
Indoor air flow (Hi/Mi/Lo)		m3/h	560/400/330	810/630/530	1090/770/610
Indoor noise level (Hi/Mi/Lo)		dB(A)	39.5/32/29	44.5/37.5/33.5	46/37/34.5
Outdoor noise level		dB(A)	56	56.5	60
Refrigerant type		kg	R32/0.6	R32/1.02	R32/1.45
Design pressure		MPa	4.3/1.7	4.3/1.7	4.3/1.7
Refrigerant piping	Liquid side/ Gas side	mm(inch)	6.35mm(1/4in)/12.7mm(1/2in)	6.35mm(1/4in)/12.7mm(1/2in)	9.52mm(3/8in)/15.9mm(5/8in)

Inverter (Variable Speed) Air Conditioners

3.1 High End Inverter - Series

Inverteron
TECHNOLOGY



JUMBO DC - CLASSIC High End Inverter

Inverteron
TECHNOLOGY

- Full DC Inverter
- EER 3.4
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- 2 Row Condenser (18K Only)
- R32 Refrigerant

Capacity	IDU Size (mm)
12K	788
18K	1121

100%
DC
INVERTER
Outstanding Efficiency & Electricity Savings

A++
Inverter
Technology
*Operates on as Low as 2.5A

140V
Low Voltage
Start
*Minimal Start-Up Voltage at 140 Volts

R32
Superior Performance & Energy Saving

4D
Air Flow
For Homogeneous Cooling Across the Room

SELF-CLEAN
EVAPORATOR
Automatically Cleans the AC & Eliminates Moisture

PURE COPPER
COILS
Strong & Durable Copper Coils for Best Cooling

*Testing under controlled conditions

Key Features

- 05 Year PCB Warranty
- R32 Gas
- Double Row Condenser (18k Only)
- Gold Fin Evaporator and Condenser
- LVS @ 140V AC



PEL Range – Air Conditioner



Inverter (Variable Speed) Air Conditioners

3.1 High End Inverter - Series



MODEL & SPECIFICATIONS			12K Jumbo DC Classic	18K Jumbo DC Classic	24K Jumbo DC Classic
Rated Capacity	Cooling	W	3520(800-4000)	5300 (1300-5800)	6400 (1800-7200)
	Heating	W	3550(800-4200)	5600 (1300-6000)	6800 (1800-8000)
	Dehumidifying	Kg/h	1.0	1.6	2.5
Rated Power Consumption	Cooling	W	1100(300-1350)	1560(300-2400)	2000 (230-2760)
	Heating	W	970(300-1350)	1450(300-2200)	1840 (230-2530)
Rated Running Current	Cooling	A	5.3(1.5-6.0)	7.3(1.4-10.6)	8.96 (1-12)
	Heating	A	4.6(1.5-6.0)	6.9(1.4-10)	8.24 (1-12)
Max. Input Power		W	1500	2400	2760
Max. Input Current		A	8.5	12	16
EER Cooling		W/W	3.20	3.40	3.2
COP Heating		W/W	3.66	3.86	3.6
Power supply source		V/Ph /Hz	220-240V~/50Hz	220~240V/1/50Hz	220-240/50
Refrigerant		/	R32	R32	R32
Refrigerant Charged		g	490	900	1070
Max. Discharge Pressure		Mpa	4.3	4.3	4.3
Max. Suction Pressure		Mpa	2.5	2.5	2.5
Air Flow Volume		m3/h	600	1200	1250
Noise level		dB (A)	42	49	50
Noise level		dB (A)	52	56	58
Indoor unit weight (Net)		Kg	8.5	14	14
Outdoor unit weight (Net)		Kg	22.5	27	42.5

Inverter (Variable Speed) Air Conditioners

3.2 High End Inverter - Series

Inverter
TECHNOLOGY



JUMBO DC – PRIME T3 WiFi
High End Inverter

Inverter
TECHNOLOGY

- Full DC Inverter
- EER 3.3
- Hidden LED Display
- Gold Fin Evaporator and Condenser
- 2 Row Condenser (18K & 24K Only)
- R410a Refrigerant

Capacity	IDU Size (mm)
12K	811
18K	1132
24K	1132

100%
DC
INVERTER
Outstanding Efficiency & Electricity Savings

A++
Inverter
technology
*Operates on as Low as 2.5A

165V
Low Voltage
Start
*Minimal Start-Up Voltage at 165 Volts

MODE
*Flexible Electricity Consumption

TE
*Powerful Cooling up to 50°C Ambient Temperature

4D
Air Flow
For Homogeneous Cooling Across the Room

SELF-CLEAN
EVAPORATOR
Automatically Cleans the AC & Eliminates Moisture

PURE COPPER
COILS
Strong & Durable Copper Coils for Best Cooling

*Testing under controlled conditions

Key Features

- 05 Year PCB Warranty
- R410a Gas
- Bigger IDU 18k 1132mm
- Double Row Condenser (18k & 24K Only)
- Gold Fin Evaporator and Condenser
- LVS @ 165V AC

- Gen Mode Function
- Wi-Fi Module Built-In No separate USB need to Install

Inverter (Variable Speed) Air Conditioners

3.2 High End Inverter - Series

SPECIFICATIONS			12K	18K	24K
Model Name			Jumbo DC Prime T3	Jumbo DC Prime T3	Jumbo DC Prime T3
Rated Capacity	Cooling	W	3400(1000~3770)	5275(1200~6200)	7034 (1200~7400)
	Heating	W	3420(1000~3810)	5422(1200~6406)	7181 (1500~7800)
Rated Power Consumption	Cooling	W	1130(290~1500)	1579(300~2500)	2132 (280~3300)
	Heating	W	922(290~1720)	1309(300~2500)	1984 (370~3500)
Rated Running Current	Cooling	A	5.8(1.5~9.0)	7.2(1.4~12.0)	9.7 (1.2~12)
	Heating	A	4.7(1.5~10.0)	6.0(1.4~13.0)	9.0 (1.6~14.5)
Max. Input Power		W	1720.00	2500.00	3500.00
Max. Input Current		A	7.54	13.00	14.50
EER Cooling		W/W	3.01	3.34	3.30
COP Heating		W/W	3.71	4.14	3.62
Power Supply Source		V/Ph/Hz	220-240V~/50Hz/1P	220-240V~/50Hz/1P	220-240V~/60Hz/1P
Voltage Range		V	165~265	165~265	165~265
Refrigerant Charged		g	R32/570	R410A/1350	R410A/1850
Max. Discharge Pressure		MPa	4.50	4.50	4.50
Max. Suction Pressure		MPa	1.90	1.90	1.90
Air Flow Volume		m ³ /h	550/550	1200/1200	1250/1250
Moisture Removal		L/h	1.20	1.80	2.40
Indoor Noise Level		dB	22.00	49.00	52.00
Outdoor Noise Level		dB	50.00	55.00	59.00
Indoor Unit Weight (Net Weight/ Gross Weight)		Kg	8/10.5	14/17	14/17
Outdoor Unit Weight (Net Weight/ Gross Weight)		Kg	24/26(w/o pipe) 27(w/i pipe)	31/36	47/54

Inverter (Variable Speed) Air Conditioners

4. Floor Standing - Series

FLOOR STANDING Inverter

BOLD+



24K BTU
EER 2.80
Gold Fin
Easy Assembling
R410A
Hydrophilic Fins

SUPREME



24K BTU Round T
EER 2.70
Turbo Sirocco
Fans for Longer Air Flow
Quick Cooling & Heating

ULTIMATE



48K BTU
EER 2.80
R32
Gold Fin
03 Row
Hydrophilic Fins

NEW ARRIVAL

ALL SEASONS TECHNOLOGY

Inverter
TECHNOLOGY

Heat & Cool made for Summers & Winters

Inverter (Variable Speed) Air Conditioners

4. Floor Standing - Series

PEL Model Name			24K Bold Plus	24k Supreme
Function			H&C	H&C
Capacity			24000 Btu/h	24000 Btu/h
Climatic Type			T1	T1
Rated Capacity	Cooling	W	6500(1800-7000)	7000
	Heating	W	6700(1900-7500)	7000
Rated Power Consumption	Cooling	W	2320(600-2800)	2550 (300-3000)
	Heating	W	2320(550-2850)	2000 (650-3000)
Rated Running Current	Cooling	A	10.5(2.6-13.0)	11.8(1.0-13)
	Heating	A	9.5(2.5-12.6)	9.2(1.0-13)
Max. Input Power		W	3500	3400
Max. Input Current		A	16	18
EER Cooling		W/W	2.80	2.75
COP Heating		W/W	2.89	3.6
Power Supply Source		V/Ph/Hz	220-240V-1-50Hz	220-240/50
Refrigerant			R410A	R410A
Refrigerant Charged		g	1380	1500
Max. Discharge Pressure		MPa	4.15	4.15
Max. Suction Pressure		MPa	1.15	1.15
Air Flow Volume		m ³ /h	1200	1250
Moisture Removal		Kg/h	2.5	2.56
Indoor Noise Level		dB	51	46
Outdoor Noise Level		dB	58	57
	Liquid tube Diameter × Thickness	mm	3*0.5	0.46*6
	Gas tube Diameter × Thickness	mm	6*0.7	0.7*12

Inverter (Variable Speed) Air Conditioners

4. Floor Standing - Series

Model				48K Ultimate
Electric parameter	Power supply	Indoor	V/ph/Hz	220~240-1-50/60
	Power supply	Outdoor	V/ph/Hz	380~415-3-50/60
	Input consumption	Rated	W	6000
	Input current	Rated	A	10.5
Cooling	Capacity	Nominal	W	14000
		Range(Min~Max)	W	3300~14500
	Input	Nominal	W	5000
	Current	Nominal	A	9.0
	EER	Nominal	W/W	2.80
Heating	Capacity	Nominal	W	14500
		Range(Min~Max)	W	3000~15000
	Input	Nominal	W	4100
	Current	Nominal	A	7.6
	COP	Nominal	W/W	3.54
Indoor unit	Dimension	WxHxD	mm	580×1870×380
	Packing	WxHxD	mm	690×2000×480
	Weight	Net/Gross	kg	50/61
	Air Volume	Hi/Med/Lo	m ³ /h	1750/1500/1300
	Sound Level (SPL)	Hi/Med/Lo	dB(A)	52/46/44
	Sound Level (PWL)	Hi	dB(A)	64
	Moisture Removal		L/H.r	5.0
Outdoor unit	Dimension	WxHxD	mm	975×835×360
	Packing	WxHxD	mm	1045×460×960
	Weight	Net/Gross	kg	75.5/84.5
	Air Volume	Hi	m ³ /h	5300
	Sound Level (SPL)	Hi	dB(A)	61
	Sound Level (PWL)	Hi	dB(A)	73
	Refrigerant	Type	-	R32
		Amount	kg	2.70
Refrigerant piping	Liquid/ Gas		mm(inch)	Φ9.52/Φ15.88 (3/8'/5/8')

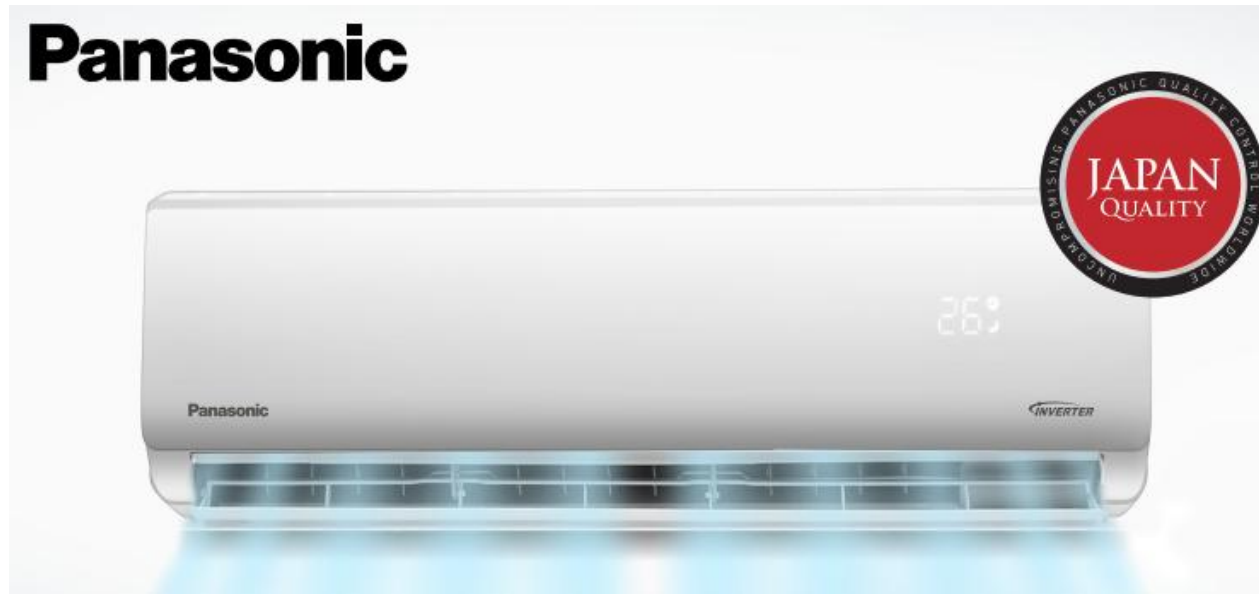
Inverter (Variable Speed) Air Conditioners

5. Panasonic Inverter

Inverteron
TECHNOLOGY

Key Features

- 05 Years Compressor Warranty
- LVS @ 150V AC
- R410a Gas
- SEER 3.30
- Double Row Condenser
- Gold Fin Evaporator and Condenser
- High Ambient Running T1 (50 °C)



Hot/Cold



Turbo Cooling

23°C

Hidden Display



R410A Eco Friendly Gas

Inverter (Variable Speed) Air Conditioners

5. Panasonic Inverter (Cont..)

Comfortable and Convenient Cooling with Inverter Technology

Inverter Technology

Panasonic is one of the pioneers of Inverter Technology, providing exceptional energy saving and allowing for a super quick and consistent cooling experience inside your home.

Gold Fin Condenser



The condenser fins are specially gold-coated to enhance its durability. These specially coated layers serve as protection against corrosion from air, water and other elements.

High Quality Flame Retardant PCB Box

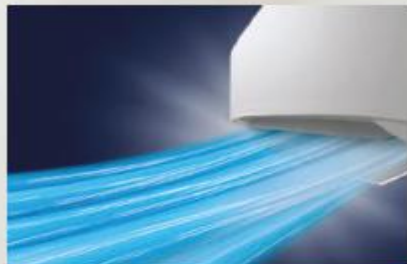


The outdoor is designed with High Quality Flame Retardant PCB Box provides more safety

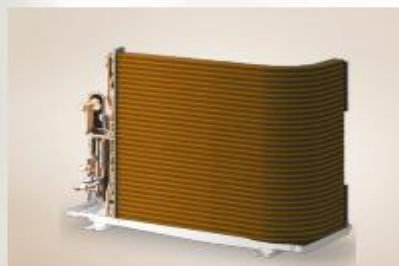
4-Way Swing



The 360 Air Flow function enables quick and even cooling at every part of the room by blowing cool air in multiple directions. This allows for comfort and convenience.



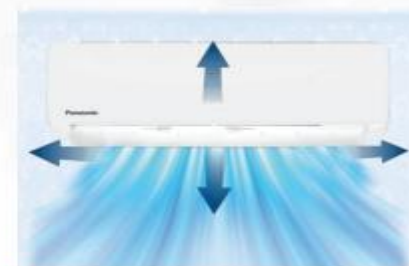
POWERFUL COOLING



STRONG & DURABLE



SAFE AND RELIABLE



SMART COOLING

Inverter (Variable Speed) Air Conditioners

5. Panasonic Inverter (Cont....)

SPECIFICATIONS

MODEL			CS/CU-UE12XKF-9	CS/CU-UE18XKF-9	CS/CU-UE24XKF-9
Type			Cooling & Heating	Cooling & Heating	Cooling & Heating
Power supply		Ph-V-Hz	1Ph-220-240V~50Hz	1Ph-220-240V~50Hz	1Ph-220-240V~50Hz
Voltage Range		V	150~265	150~265	150~265
Cooling	Capacity	Btu/h	12500	18000	24000
	Rated power Input	W	1110	1599	2132
	Rated current	A	5	7.30	9.70
	EER	W/W	3.30	3.30	3.30
Heating	Capacity	Btu/h	13500	18500	26000
	Rated power Input	W	1099	1506	2117
	Rated current	A	5.00	6.80	9.60
	COP	W/W	3.60	3.60	3.60
Indoor Air Flow	(Hi)	m3/h	550	1100	1000
Indoor Unit	Dimension(W*D*H)	mm	811×292×203	1010×315×220	1010×315×220
	Net weight	kg	8.50	13.00	13.00
Outdoor Unit	Dimension(W*D*H)	mm	777×290×498	853×349×602	920×380×699
	Net weight	kg	22.50	28.00	28.00
Refrigerant			R410A	R410A	R410A
Ambient Temperature		°C	15-60	15-60	15-60

PEL Range – Air Conditioner



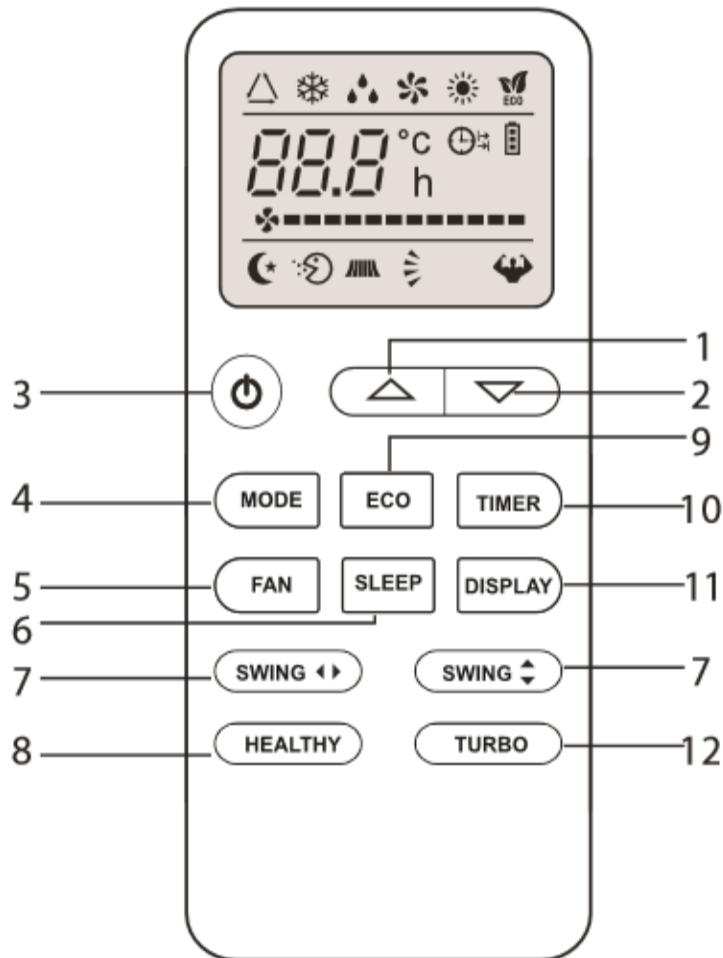
Inverter (Variable Speed) Air Conditioners

5. Panasonic Inverter (Cont....)

Technical Specifications			12K	18K	24K
Power supply		Ph-V-Hz	1Ph-220-240V~50Hz	1Ph-220-240V~50Hz	1Ph-220-240V~50Hz
Voltage Range		V	150~265	150~265	150~265
Cooling	Capacity	Btu/h	12500(3412~12966)	18000(4094~19107)	24000(4947~25590)
	Capacity	W	3664(1000~3800)	5275(1200~5600)	7035(1450~7500)
	Capacity	kcal/h	3150.00	4536.00	6048.00
	Rated power Input	W	1110(300~1700)	1599(300~2550)	2132(350~2500)
	Rated current	A	5.0(1.4~9.0)	7.3(1.4~12.0)	9.7(1.7~12.0)
	EER	W/W	3.30	3.30	3.30
Heating	Capacity	Btu/h	13500(3412~14330)	18500(4094~19790)	26000(4947~27296)
	Capacity	W	3957(1000~4200)	5422(1200~5800)	7620(1450~8000)
	Capacity	kcal/h	3402.00	4662.00	6552.00
	Rated power Input	W	1099(300~1500)	1506(300~2550)	2117(350~2500)
	Rated current	A	5.0(1.4~9.0)	6.8(1.4~12.0)	9.6(1.7~13.0)
	COP	W/W	3.60	3.60	3.60
Moisture removal		Liters/h	1.00	1.80	2.60
Indoor noise level		dB(A)	41.00	NA	50.00
Outdoor noise level		dB(A)	54.00	55.00	59.00
Indoor air flow (Cool/Heat)		m3/h	550/590	1100/1100	1000/1100
Indoor unit	Dimension(W*D*H)	mm	811×292×203	1010×315×220	1010×315×220
	Packing (W*D*H)	mm	885×366×275	1096×390×297	1096×390×297
	Net/Gross weight	Kg	8.5/10.5	13/16	13/16
Outdoor unit	Dimension(W*D*H)	Mm	777×290×498	853×349×602	920×380×699
	Packing (W*D*H)	Mm	818×325×520	890×385×628	960×430×732
	Net/Gross weight	Kg	22.5/27/5(with pipe)	28/33	37/45(with pipe)
Refrigerant piping	Liquid side/ Gas side	mm(inch)	Φ6(1/4")/Φ9.52(3/8")	Φ6(1/4")/Φ9.52(3/8")	Φ6(1/4")/Φ12(1/2")
Pressure	High(DP)	Mpa	4.50	4.50	4.50
	Low(SP)	Mpa	1.90	1.90	1.90
Refrigerant			R410A/0.68	R410A/1.19kg	R410A/1360g

Panasonic Inverter (Variable Speed) Air Conditioners

5.1 Remote Control Function:



No.	Button	Function
1	Temp Up	Press to increase temperature / time setting
2	TEMP Dn	press to decrease temperature / time setting
3		Press it to start or stop operation
4	MODE	To select mode of the operation
5	FAN	To select the fan speed of auto/low/mid/high
6	SLEEP	To activate the function "SLEEP"
7	SWING	To activate or deactivate the movement of the horizontal deflectors
8	SWING	To activate or deactivate the movement of the vertical deflectors
9	HEALTH	To switch - on/off "HEALTH" function, it is the button which controls the ionizer or plazma generator (if applicable)
10	ECO	Press this button to activate or deactivate the eco function which enables the unit automatically to sets the operation to achieve energy savings.
11	TIMER	Press it to set auto-off timer.
12	DISPLAY	To switch on/off the display
13	TURBO	Press the button to activate / deactivate the super function which enables the unit to reach the preset temperature in shortest time.

Panasonic Inverter (Variable Speed) Air Conditioners

5.1 Remote Control Function:

Inverter
TECHNOLOGY



No.	Symbols	Meaning
1		FEEL mode indicator
2		COOLING indicator
3		DEHUMIDIFYING indicator
4		FAN ONLY OPERATION indicator
5		HEATING indicator(For cooling only type, fanning function is activated.)
6		ECO indicator
7		CLOCK indicator
8		TIMER ON & TIMER OFF indicator
9		BATTERY indicator
10		LOW/MIDDLE/HIGH FAN SPEED indicator
11		SLEEP indicator
12		HEALTHY indicator
13		FLAP SWING indicator
14		TURBO indicator

Non-Inverter (Fixed Speed) Air Conditioners

1. Wall Mount



MAJESTIC T3 Fixed Speed – Cool Only

- EER 2.6
- Hidden LED Display
- Blue Fin Condenser
- R410a Refrigerant

Capacity	IDU Size (mm)
12K	750
18K	792
24K	1082



*Powerful Cooling up to 50°C
Ambient Temperature



High Capacity
Rotary Compressor



For Homogeneous Cooling
Across the Room



Efficient cooling performance
even at 50 degrees



Automatically Cleans the AC
& Eliminates Moisture



Strong & Durable Copper
Coils for Best Cooling

Non-Inverter (Fixed Speed) Air Conditioners

1.1 Wall Mount



MAJESTIC GLORY

Fixed Speed – Cool Only

- EER 3.0
- Hidden LED Display
- Gold fin evaporator
- R410a Refrigerant

Capacity	IDU Size (mm)
12K	805
18K	957
24K	1040



PEL Range – Air Conditioner

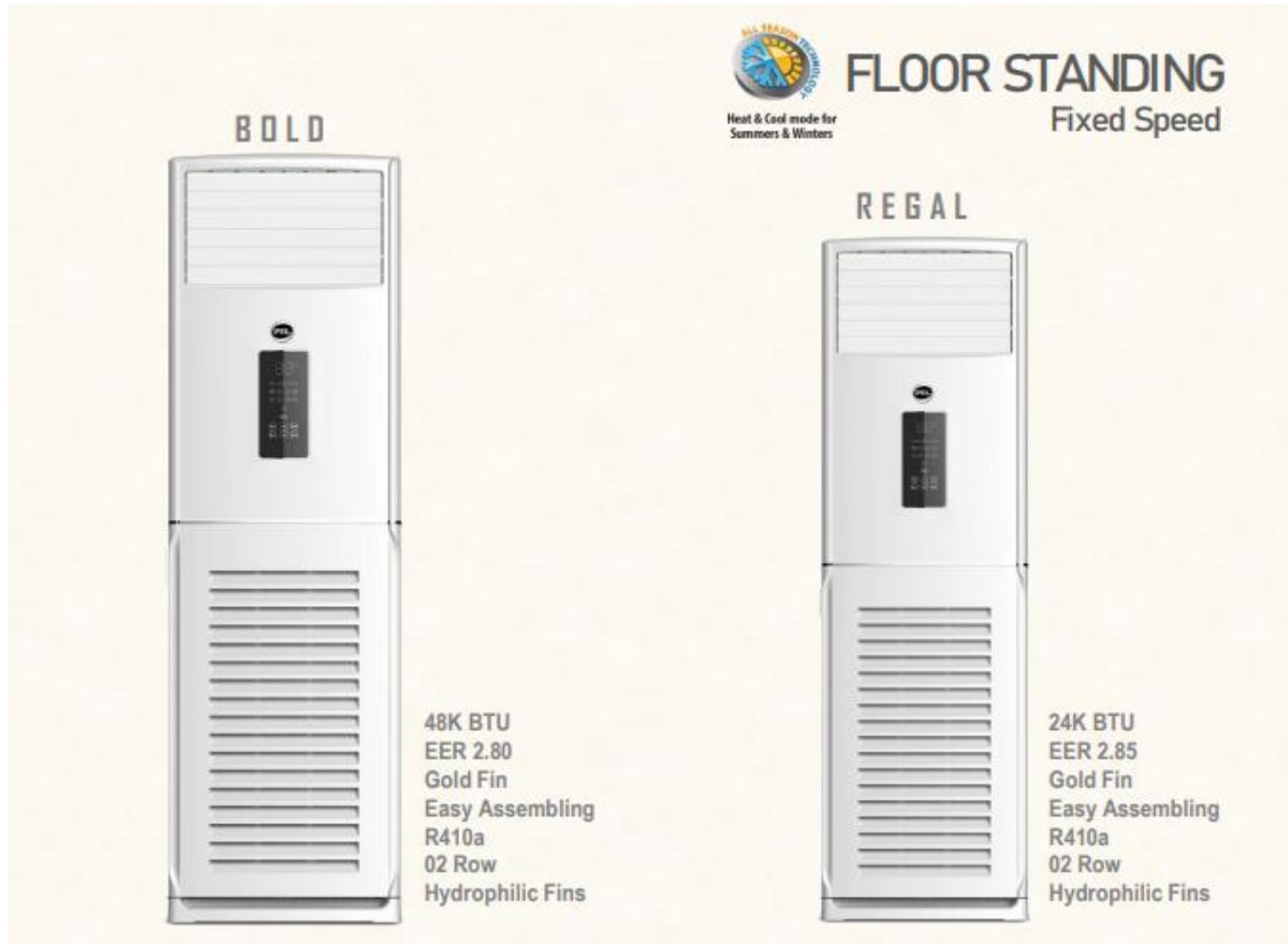


Non-Inverter (Fixed Speed) Air Conditioners Specification (Majestic Glory)

Model			12K Majestic Glory	18K Majestic Glory	24K Majestic Glory
Function			Cool Only	Cool Only	Cool Only
Power supply		V-Ph-Hz	220_240V_1Ph, 50Hz	220_240V_1Ph, 50Hz	220_240V_1Ph, 50Hz
Cooling (Standard conditions)	Capacity	Btu/h	12000	18000	22000
	Input	W	1252	1740	2015
	Current	A	5.5	7.7	9
	EER	W/W	2.81	3.03	3.20
Rated Power Input		W	1650	2200	3250
Rated Current		A	9.8	11.6	17
Starting current		A	25	38	42
Indoor air flow (Hi/Mi/Lo)		m3/h	566/440/349	760/630/520	1020/815/680
Indoor noise level (Hi/Mi/Lo)		dB(A)	37.5/32.5/29.5	42.5/37/31.5	47.5/42.5/38
Indoor unit	Dimension(W*D*H)	mm	805x194x285	957x213x302	1040x220x327
	Packing (W*D*H)	mm	870x270x365	1035x295x385	1120x405x315
	Net/Gross weight	kg	8.1/10.2	10.6/13.5	13.6/16.8
Outdoor noise level		dB(A)	51.5	55.5	57.5
Outdoor unit	Dimension(W*D*H)	mm	720x270x495	765x303x555	805x330x554
	Packing (W*D*H)	mm	835x300x540	887x337x610	915x370x615
	Net/Gross weight	kg	24.9/26.7	31.6/34	40.5/42.7
Refrigerant type		kg	R410A/0.49	R410A/0.65	R410A/1.1
Design pressure		MPa	4.2/1.5	4.2/1.5	4.2/1.5
Refrigerant piping	Liquid side/ Gas side	mm(inch)	6.35mm(1/4in)/12.7mm(1/2in)	6.35mm(1/4in)/12.7mm(1/2in)	9.52mm(3/8in)/15.9mm(5/8in)
	Max. refrigerant pipe length	m	20	25	25
	Max. difference in level	m	8	10	10

Non-Inverter (Fixed Speed) Air Conditioners

2. Floor Standing



The advertisement features two tall, silver floor-standing air conditioners against a light beige background. The unit on the left is labeled 'BOLD' and the unit on the right is labeled 'REGAL'. Above the units, a circular logo with a sun and moon is accompanied by the text 'ALL SEASONS COMFORT' and 'Heat & Cool inside for Summers & Winters'. To the right of this logo, the text 'FLOOR STANDING Fixed Speed' is displayed. Below each unit, a list of specifications is provided.

Model	Capacity	EER	Coil	Assembly	Refrigerant	Row	Feature
BOLD	48K BTU	2.80	Gold	Easy	R410a	02	Hydrophilic Fins
REGAL	24K BTU	2.85	Gold	Easy	R410a	02	Hydrophilic Fins

PEL Range – Air Conditioner



Non-Inverter (Fixed Speed) Air Conditioners Specifications (Regal & Bold)

PEL Model Name				24K Regal	48K BOLD PLUS
Category				FS	FS
Function				H&C	H&C
Nameplate Parameters	Rated Capacity	Cooling	W	7050	14000(6500-15500)
		Heating	W	7600	14000(5000-16000)
	Rated Power Consumption	Cooling	W	2470	5350(1300-6000)
		Heating	W	2530	4500(1000-5800)
	Rated Running Current	Cooling	A	11.5	9(2.5-10)
		Heating	A	11.5	8(2.0-9.5)
	Max. Input Power		W	3800	9000
	Max. Input Current		A	18	14
	EER Cooling		W/W	2.85	6.1
	COP Heating		W/W	3.0	4.0
	Power Supply Source		V/Ph/Hz	220-240V~/50Hz	380-415V 3N~/50Hz
	Refrigerant			R410A	R32
	Refrigerant Charged		g	1400	3000
	Max. Discharge Pressure		MPa	4.3	4.3
	Max. Suction Pressure		MPa	1.15	2.5
	Air Flow Volume		m ³ /h	1200	2100
	Moisture Removal		Kg/h	2.57	5.2
	Indoor Noise Level		dB	53	66
	Outdoor Noise Level		dB	59	69
	Indoor Unit Weight (Net Weight/ Gross Weight)		Kg	38	52
	Outdoor Unit Weight (Net Weight/ Gross Weight)		Kg	51	94
Indoor	Dimensions	Unit Dimension (WxDxH)	mm	506×1780×315	580×1925×400
		Packing Dimension (WxDxH)	mm	620×1970×460	690×2100×565
Connection	Connecting pipe	Material		copper	copper
		Length	m	3	3
		Liquid tube Diameter × Thickness	mm	0.45*3000	0.59*9
		Gas tube Diameter × Thickness	mm	0.75*3000	1*19
Outdoor	Dimensions	Unit Dimension (WxDxH)	mm	900×700×350	940×1320×340
		Packing Dimension (WxDxH)	mm	1020×770×430	1070×1435×450

❑ Microwave Oven



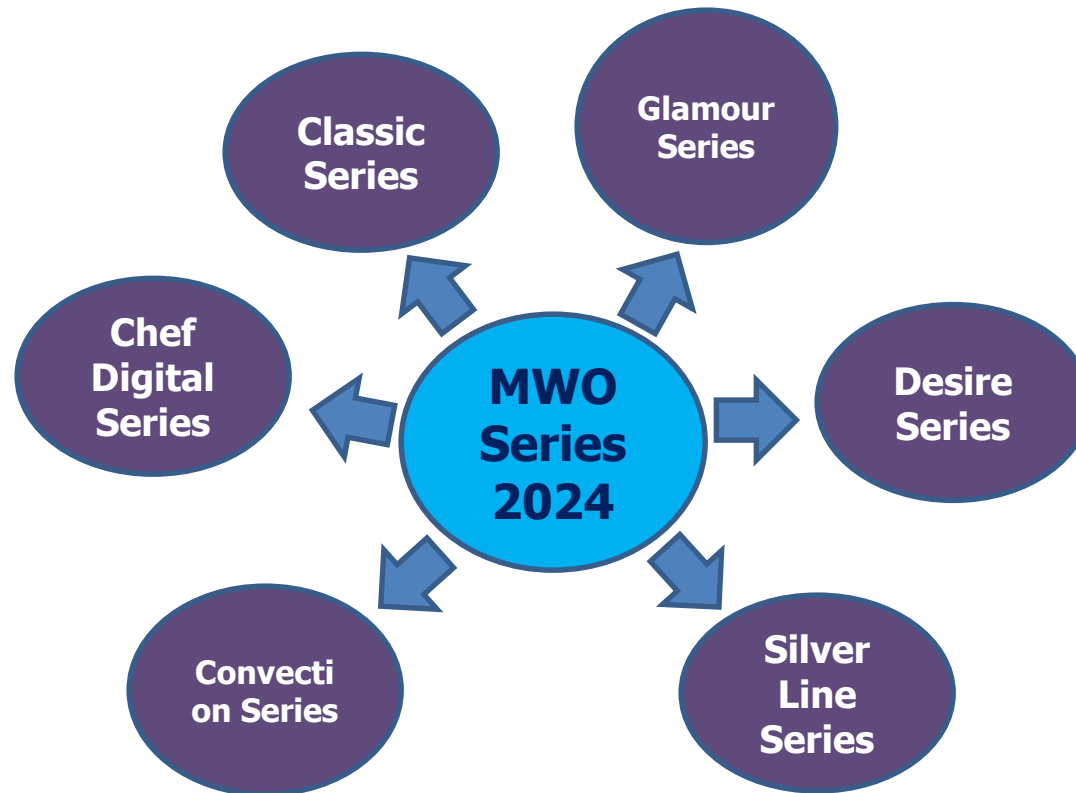
PEL
Change your life

PEL MICROWAVE OVEN
Hot hai!



Microwave Range 2024 : Subcategories

1. Classic Series : 20 LTRS B&W, 20 LTRS B&W Classic Plus, **PMO 20 WGM**
2. Glamour Series : 23LTRS baking, 30ltr Grill, 38 LTRS
3. Desire Series : 23, 26, 30 LTRS (Touch Screen)
4. Silver Line Series : 23 LTRS Black (Manual), 23 LTRS Black Digital
5. Convection Series : 25 & 30 LTRS (Manual)
6. Chef Digital Series : 23 & 26 LTRS Black Digital



1. Classic Series

CLASSIC SERIES



A durable crown with tactile response that makes navigation easier



Gives convenience of setting power levels for variety of cooking



Selection of cooking type and heat time with additional convenience of clock and kitchen timer



Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal



Automatically adjusts weights and power level for speedy defrosting



The only microwave in Pakistan which offers 2 year magnetron warranty

PMO-20



PMO-20



PMO-20 WGM



PMO-20 CP



PMO-20 CP



2. Glamour Series



Capacitive buttons that ensure convenience and ease of use



Gives convenience of setting power levels for variety of cooking



Enables you to grill meat and vegetables which works just like any other rotisserie giving you perfect cooking in less time



Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal



Speedy Defrost

Automatically adjusts weights and power level for speedy defrosting



The only microwave in Pakistan which offers 2 year magnetron warranty

PMO-23 Baking

PMO-30 Grill

PMO-38 Grill



3. Desire Series

 DIGITAL OPERATION Capacitive buttons that ensure convenience and ease of use	 MULTIPLE POWER LEVELS Gives convenience of setting power levels for variety of cooking	 GRILL FUNCTION Enables you to grill meat and vegetables which works just like any other rotisserie giving you perfect cooking in less time
 EXPRESS COOKING Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal	 Speedy Defrost Automatically adjusts weights and power level for speedy defrosting	 2 YEAR MAGNETRON WARRANTY The only microwave in Pakistan which offers 2 year magnetron warranty

PMO-23 Desire

PMO-26 Desire

PMO-30 Desire



4. Silver Line Series



Capacitive buttons that ensure convenience and ease of use



Gives convenience of setting power levels for variety of cooking



Achieves target temperature uniformly throughout the food, assuring microbiological safety



Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal



Speedy Defrost

Automatically adjusts weights and power level for speedy defrosting



The only microwave in Pakistan which offers 2 year magnetron warranty

PMO-23 SLM



PMO-23 SLD





5. Convection Series



Enables you to grill meat and vegetables which works just like any other rotisserie giving you perfect cooking in less time



Gives convenience of setting power levels for variety of cooking



Achieves target temperature uniformly throughout the food, assuring microbiological safety



Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal



A technology allowing uniform heating that maintains the nutritional value and assures microbiological safety



A portable pizza cooking surface that makes preparing pizza in the microwave easy and absolutely hassle free

PMO-25

PMO-30





6. Chef Digital Series



Enables you to grill meat and vegetables which works just like any other rotisserie giving you perfect cooking in less time



Gives convenience of setting power levels for variety of cooking



Achieves target temperature uniformly throughout the food, assuring microbiological safety



Unique sensor technology allows you to select appropriate weight, temperature settings and time to cook your required meal



A technology allowing uniform heating that maintains the nutritional value and assures microbiological safety



A portable pizza cooking surface that makes preparing pizza in the microwave easy and absolutely hassle free

PMO-23

PMO-26



Microwave Oven With 3 Technologies

Convection Series



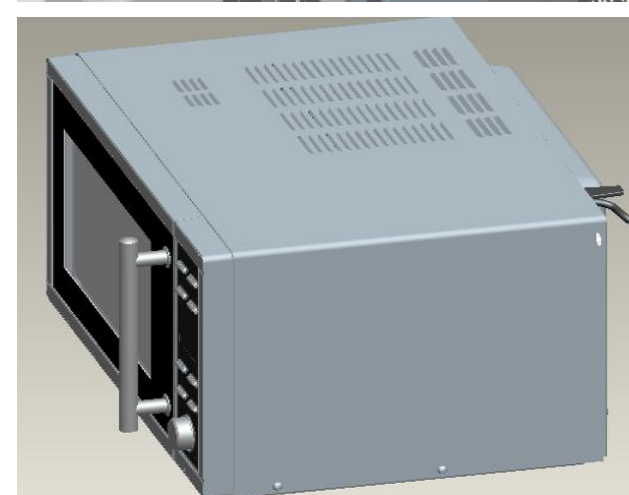
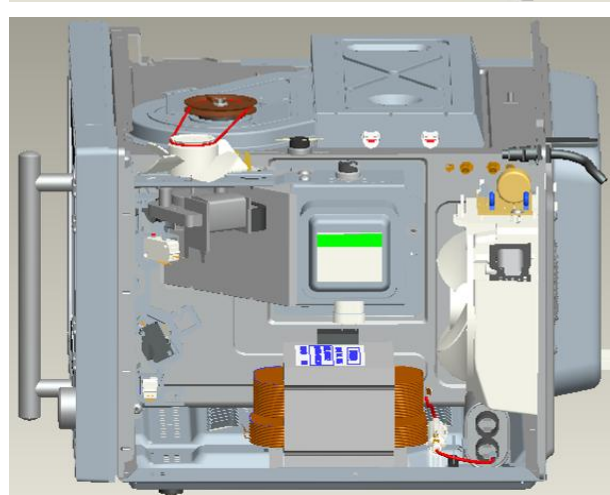
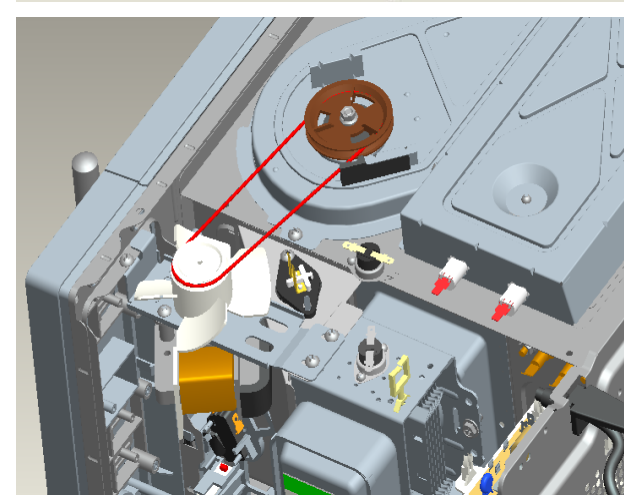
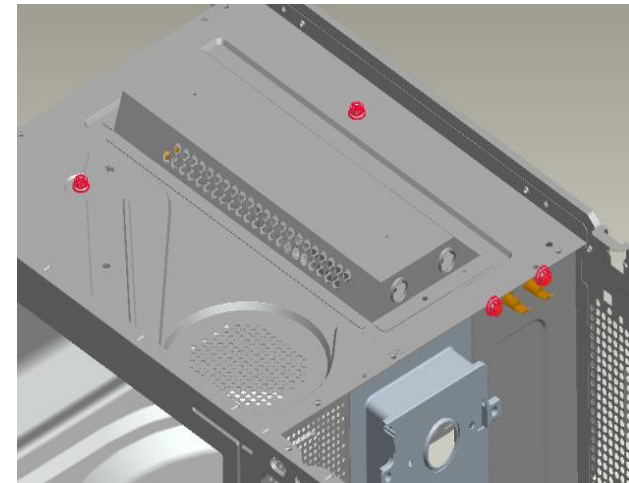
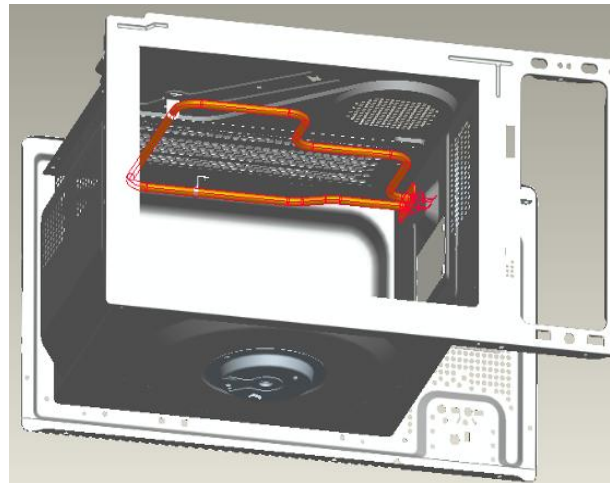
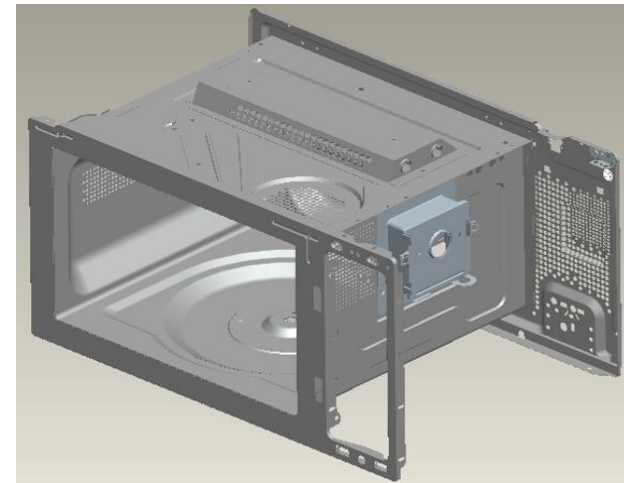
PEL Model		PMO 25L Convection
Supplier		GALANZ
Capacity (Liter)		25
Power Consumption (W)	Microwave Input	1,400
	Microwave Output	900
	Grill	1,000
	Convection	1,950





Microwave Oven With 3 Technologies

Convection Series



Microwave Oven With 3 Technologies

Convection Series

S No	Range	Once	Twice	3 Times	4 Times	5 Times	6 Times	7 Times	8 Times	9 Times	10 Times	11 Times
1	Micro	100% (P-100)	90% (P-90)	80% (P-80)	70% (P-70)	60% (P-60)	50% (P-50)	40% (P-40)	30% (P-30)	20% (P-20)	10% (P-10)	0% (P-00)
2	Combination	Micro 30% Grill 70%	Micro 55% Grill 45%									
3	Convection	110 ºC	120 ºC	130 ºC	140 ºC	150 ºC	160 ºC	170 ºC	180 ºC	190 ºC	200 ºC	
4	Micro + Convection	Combi.1 110 ºC	Comb 2 140 ºC	Comb 3 170 ºC	Comb 4 200 ºC							

MULTI-STAGE COOKING:

1. Input Microwave Oven program. Do not touch start button.
2. Input Grill Program. Do not touch start button.
3. Input Convection program.
4. Press the start/quick start button.



Microwave Oven With 3 Technologies

Convection Series

S No	Error	Fault	Symptoms
1	E01	Super High Temp.	Press Stop/Cancel Button
2	E02	Super Low Temp.	Press Stop/Cancel Button
3	E03	Sensor Short Circuit	Check/Replace the Sensor

Cooling Feature:

➤ Microwave Oven Fan will work for 3 minutes to cool the system. In case of cooking time more than 02 minutes.

Water Dispenser

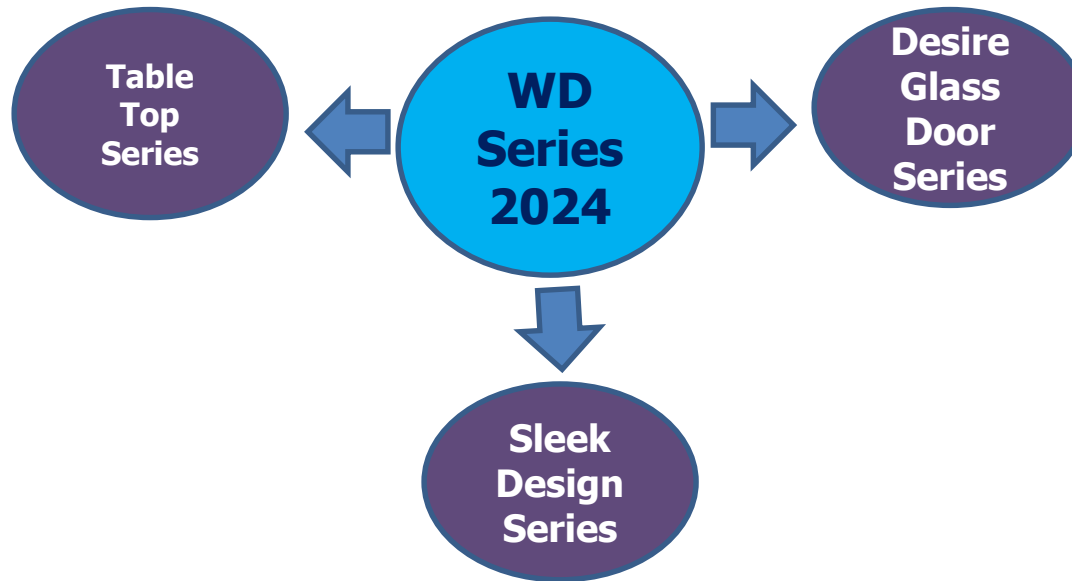


PEL Range – Water Dispenser



Water Dispenser 2024 : Subcategories

1. Desire Glass Door Series : Flat Glass Door (PWD), Curved Glass Door (PCWD)
2. Sleek Design Series : Smarter, Premier
3. Table Top Series : Simple (Without Door White), Glass Door (RBS/LMI)



PEL Range – Water Dispenser



1. Desire Glass Door Series (Flat Glass Door)

			
No Gas Leakage Less Noise & Low Power Consumption	Coolest Cabinet to Store Food & Drinks up to 3°C	25 Glasses of Consistently Chilled Water	304 SS Food Grade Rust Free & Antibacterial

Red Blaze Purple Blaze Maroon Blaze PMI



Our latest technology provides chilled water in no time.



This unique feature gives you less noise and low power consumption.



Cool cabinet that can store food and drinks up to 3° C.



Our latest technology 304 SS food grade that gives you rust-free and antibacterial feature.

PEL Range – Water Dispenser



1. Desire Glass Door Series (Curved Glass Door)



Red Blaze

Purple Blaze

Grey Blaze

PMI



Our latest technology provides chilled water in no time.



This unique feature gives you less noise and low power consumption.



Cool cabinet that can store food and drinks up to 3° C.



Our latest technology 304 SS food grade that gives you rust-free and antibacterial feature.

1. Desire Glass Door Series Specifications



**STAINLESS
STEEL WATER
TANKS**



**LOW VOLTAGE
START**
ABOUT 187 V



**REFRIGERATOR
FUNCTIONS**



The temperature
inside the fridge stays
in the correct range.

PEL Model		PWD-425 (Flat GD)	PWD-525 (Curved GD)
Color		RB/PB/MZ/PMI	RB/PB/PMI/GB
Capability of Heating		5 L/h $\geq 90^{\circ}\text{C}$	5 L/h $\geq 90^{\circ}\text{C}$
Capability of Cooling		2 L/h $\leq 10^{\circ}\text{C}$	2 L/h $\leq 10^{\circ}\text{C}$
Power Consumption	Heating Power (W)	500	500
	Cooling Power (W)	100	100
	Rated Voltage (V)	220~240	220~240
	Cooling Rate (Amp)	0.4 – 0.6	0.4 – 0.6
Refrigerant	Type	R134a	R134a
	Charge (g)	40	40

PEL Range – Water Dispenser



2. Sleek Design Series



No Gas Leakage
Less Noise & Low Power Consumption



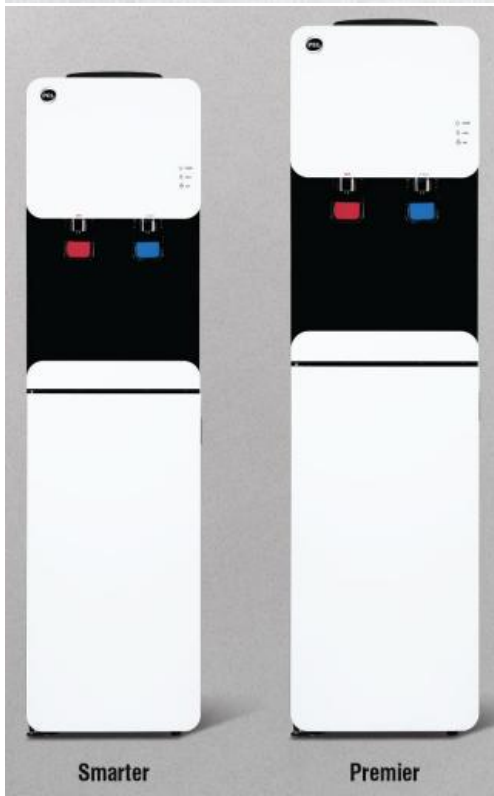
Coollest Cabinet
to Store Food & Drinks up to **3°C**



25 Glasses
of Consistently Chilled Water



304 SS Food Grade
Rust Free & Antibacterial



Our latest technology provides chilled water in no time.



This unique feature gives you less noise and low power consumption.



Cool cabinet that can store food and drinks up to 3° C.



Our latest technology 304 SS food grade that gives you rust-free and antibacterial feature.

2. Sleek Design Series Specifications

Description	Units	PWD-316-PRM	PWD-315-SMT
Type	-	Sleek Design	Sleek Design
Colour	-	White	White
Net Weight	kg	17	17.5
Gross Weight	kg	19	19.5
Dimensions (W x D x H)	mm	310 x 375 x 1190	310 x 375 x 1115
Voltage/Frequency	Volts/Hz	220/50	220/50
Cooling Current	Amperes	0.6	0.6
Heating Current	Amperes	2.2	2.2
Cooling Power	Watt	100	100
Heating Power	Watt	500	500
Cold Tank Capacity	Liters	3.7	3.7
Hot Tank Capacity	Liters	1.2	1.2
Cooling Temperature	°C	2 L/Hr ≤ 10 °C	2 L/Hr ≤ 10 °C
Heating Temperature	°C	5 L/Hr ≥ 90 °C	5 L/Hr ≥ 90 °C
Refrigerant (R134a)	Grams	40	40
Refrigerator Volume	Liters	20	18

3. Table Top Series



**STAINLESS
STEEL WATER
TANKS**



**LOW VOLTAGE
START**
ABOUT 187 V



The temperature
inside the fridge stays
in the correct range.

PEL Model		PWD-115 TT RBS/ LMI	PWD-115 TTW
Color		Red / Black	White
Cold Tank Capacity		3.7 Liters	3.7 Liters
Hot Tank Capacity		1.2 Liters	1.2 Liters
Power Consumption	Heating Power (W)	500	500
	Cooling Power (W)	100	100
	Current Rate (Amp)	0.4~2.2	0.4~2.2
Refrigerant	Type	R134a	R134a
	Charge (g)	24	24



❑ Washing Machine



PEL Range – Washing Machine



Fully Automatic



Semi Automatic





Fully Automatic Washing Machine

-- MODELS AVAILABLE:

1. Soft Panel
 1. PAWM – 900
 2. PAWM - 1100
2. Smart Panel
 1. PAWM – 900i
 2. PAWM – 1100i

-- COLORS AVAILABLE:

1. Metallic Grey
2. Golden



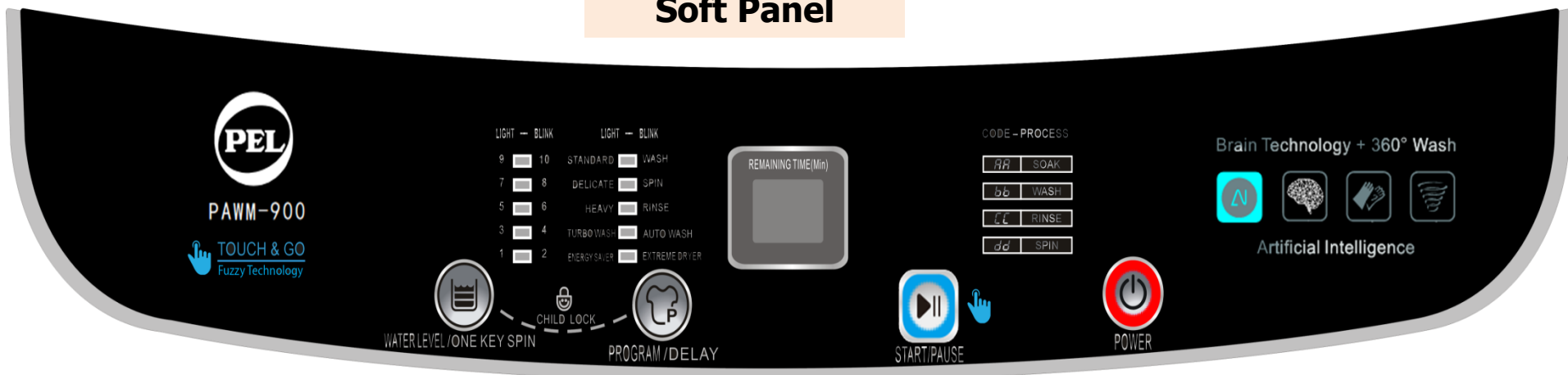
Fully Automatic Washing Machine Models Specification

MODEL	PAWM-900	PAWM-1100	PAWM-900i (Smart Touch)	PAWM-1100i (Smart Touch)
Rated Voltage/ Frequency	220V~ / 50Hz	220V~ / 50Hz	220V~ / 50Hz	220V~ / 50Hz
Water Levels	10	10	10	10
Water Input Pressure	0.03 MPa ~ 0.78 MPa	0.03 MPa ~ 0.78 MPa	0.03 MPa ~ 0.78 MPa	0.03 MPa ~ 0.78 MPa
Rated Power Input	Wash: 380W Spin: 260W	Wash: 420W Spin: 280W	Wash: 380W Spin: 260W	Wash: 420W Spin: 280W
Water Proof Degree	IPX4	IPX4	IPX4	IPX4
Net Weight	About: 34kg	About: 41kg	About: 34kg	About: 41kg
Machine Size	555mm x 560mm x 960mm	595mm x 620mm x 1000mm	555mm x 560mm x 960mm	595mm x 620mm x 1000mm
No. of Programs	10	10	10	10
Water Inlet	Automatic	Automatic	Automatic	Automatic
Touch & Go	Yes	Yes	Yes	Yes
Delay Start	Yes	Yes	Yes	Yes
Child Lock	Yes	Yes	Yes	Yes
Memory Backup	Yes	Yes	Yes	Yes
Door Cover	Tempered Glass with Damper	Tempered Glass with Damper	Tempered Glass with Damper	Tempered Glass with Damper
Control Panel	Soft Touch	Soft Touch	Smart Touch	Smart Touch

Fully Automatic Washing Machine

a. Panel Types:

Soft Panel



Smart Panel





Fully Automatic Washing Machine

b. Features

☐ Artificial Intelligence

Machine has the ability to sense what kind of clothes are placed and what specific time of each function is required to wash those clothes

☐ Brain Technology

When the machine is in working state, cycle is in process and light trips off.

When the light comes back, machine starts from the same time it stopped

☐ Touch & Go

Just press the start button and machine will automatically sense the amount of water required start the washing cycle

☐ Delay Start

You can set a specific time on the machine after which it will automatically turn on to a preset cycle

☐ Fuzzy Logic

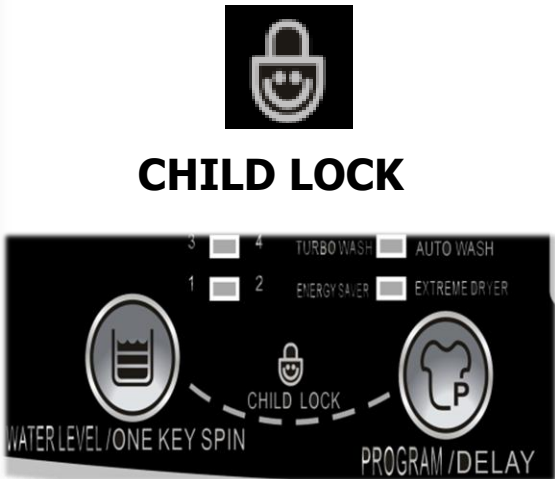
Inbuilt sensors monitor the washing process and make corrections to produce the best washing results

☐ Rust Resistant Body Gives More durability to machine



Fully Automatic Washing Machine

c. Operating Procedure:





Fully Automatic Washing Machine

d. Checks:

1. C1

Press "Program" & "Power" keys simultaneously to check LED's

2. C2

Press "Program" & "Power" keys simultaneously & then "Program" again to check Wash Cycle & Water Inlet Valve

3. C3

Press "Program" & "Power" keys simultaneously & then "Program" again twice to check Spin & Drain Motor

4. C6

Press "Water Level" & "Power" keys simultaneously & then to Reset, Fuzzy Reset and to check water level



PEL Range – Washing Machine



Fully Automatic Washing Machine

e. Major Components:

1. Motors:

- 150 W – PAWM 900
- 180 W – PAWM 1100

2. Hangers:

- For Model 900 Upper /Lower Yellow Front side , upper / lower white backside,
- For 1100 Upper yellow lower white front, upper/lower white backside ,
- Difference b/w 900 & 1100 1100 model hangers small about 10mm in comparison 900 model

3. Water Inlet Valve (Work with Pipe and air pressure, 5VDC)

4. Drain Motor: Rectangular shape in Model 1100

5. Water Level Sensor

6. Gear Box: Single Gear in Model 1100 and Double Gear in Model 900

7. Spin Tub:

8. PCB: Smart and Soft

9. Capacitor: 11uF in Model 900, 15uF in Model 1100

10. Hinges: In Model 900 Plastic Shaft and on Model 1100 Aluminum cast Shaft

11. V-Belts: Size O-480E (C-22) for Fully Auto, Same size O-695E for Semi Auto)

12. Harness Wire (900—Additional wire yellow)



Fully Automatic Washing Machine (900/1100)

f. Errors Code List:

Code	Reason of Alarm	Solution
E0	Abnormal Memory	Cut-off the power, power on again.
E1	The water inlet more than 20 minutes, but still cannot reach the water level has been set.	Check the water supply.
E2	After draining for 7 minutes, there is still has water in the tub.	Open the lid check there is any obstruction in drainage through pulsator, also check drain pipe.
E3	The top lid is open in reservation, wash and the spin working.	Close the top lid
E4	Unbalanced spin	Open the top lid, put the clothes in order, then close it.
E5	Water level over MAX line	Machine stop washing and drain continue wash while water level return to normalcy.
E6	Water level sensor abnormal	Cut-off the power and sent to repair.
E7	“CHILD LOCK” function has been set but open the top lid in working.	Close the top lid, or release the “CHILD LOCK”.

Twin Tub Semi-Automatic Washing Machine

WHITE COLOR



GREEN COLOR



Twin Tub Semi-Automatic Washing Machine

Models Availability:

PWM - 1050T

Top Lid Colors:

1. Green - White
2. Fully White

Specification:

1. Spin Motor is Direct driven
2. Wash Motor is Belt Driven via Gear
3. Belt is V-type # O-480
4. Both Motors are Bush Type

5. PWM – 1050T

Spinner Motor = 70 W

Wash Motor = 200 W

Combine Capacitor = 10uF+4uF

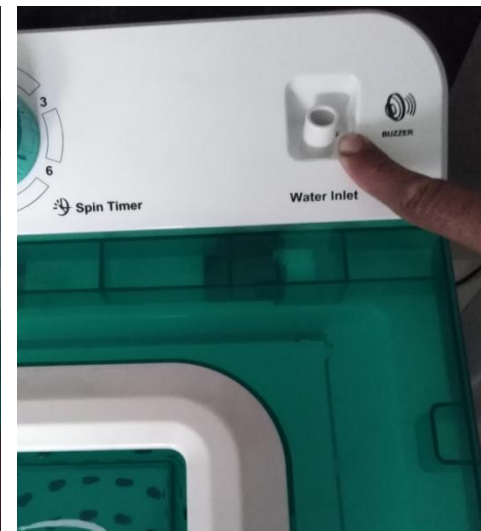




Twin Tub Semi-Automatic Washing Machine

Features:

- ❖ Twin tub for greater convenience
- ❖ Two way washing mechanism
- ❖ Powerful pulsation gives strong washing effect
- ❖ Jumbo size washing tub and spinner
- ❖ Strong overflow rinse
- ❖ Noise free operation



PEL Range – Washing Machine



Single Tub Semi-Automatic Washing Machine



Single Tub Semi-Automatic Washing Machine

-- MODELS AVAILABLE:

1. Washer

1. PWMS – 8050
2. PWMS – 1250

-- COLORS AVAILABLE:

1. GREEN
2. GREY
3. WHITE

(*Above Colors LID/COVER Only*)

SEMI AUTOMATIC SERIES

			
Wider pulsator removes stains more efficiently	Rust proof body gives strength to machine	Reduces the direct load on motor, eventually increasing its life	Prevents rats from entering, which ensures wiring is not damaged





Single Tub Semi-Automatic Washing Machine



Single Tub Semi-Automatic Washing Machine

Feature:

- ☐ Big Pulsator
- ☐ Gear Train Technology
- ☐ Rat Guard
- ☐ Biggest Load
- ☐ Heavy Duty
- ☐ Anti Rust Body



MODEL	PWMS - 8050	PWMS - 1250
Rated Voltage / Frequency	220V~ / 50Hz	220V~ / 50Hz
Rated Power Input	350 W	450 W
Water Proof Degree	IPX4	IPX4
Net Weight	About: 16 Kg	About: 17 Kg
Machine Size	570mm x 540mm x 970mm	570mm x 540mm x 1100mm
Water Inlet	Manual	Manual

PEL Range – LED TV



PEL Range – LED Television



Difference b/w LCD/CRT & LED

	CRT & LCD TV	LED TV
Full form	Cathode Ray Tube & Liquid Crystal Display Television	Light Emitting Diode Television
Definition	Ordinary, LCD TV's make use of the cold cathode fluorescent lamps for backlighting.	It is a type of LCD TV. Instead of using fluorescent lamps, LED TV's use an array of tiny and efficient LED to get the screen illuminated
Slim Structure	Comparatively less	Comparatively more Slim
Picture Quality	Normal	Excellent
Power Consumption	High	Low
Colors	Comparatively it offers less color than LED's with RGB backlighting.	Offers more colors
Light Output (brightness)	Comparatively less	Brighter than the LCD
Resolution	Low	High
Weight	Comparatively bulky	Comparatively light



PEL Range – LED Television



Resolution Detail:

Name	Alt. Names	Columns	Lines	Total Pixels	Common Media
		Width in pixels	Height in pixels		
480p	Standard	720	480	345,600	DVD Standard Channels
720p	HD (HD Ready)	1280	720	921,600	HD channels (some are 1080i)
1080p	FHD (Full HD)	1920	1080	2,073,600	Blu-ray Game consoles
4k	UHD (Ultra High Definition)	3840	2160	8,294,400	Streaming Blu-ray 4k gaming

DIFFERENT RESOLUTIONS STANDARD

2K

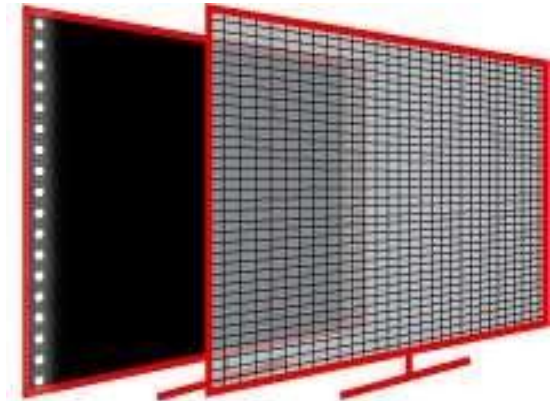
2048 x 1080 - this refers to displays that have a horizontal resolution of approximately 2000 pixels. Although it is close to 1080p, it is considered as a different resolution standard.

1440p

2560 x 1440 - commonly known as QHD or Quad HD resolution, and typically seen on gaming monitors and on high-end smart phones. 1440p is four times the resolution of 720p HD or "HD ready."



Edge Lit



Full Array



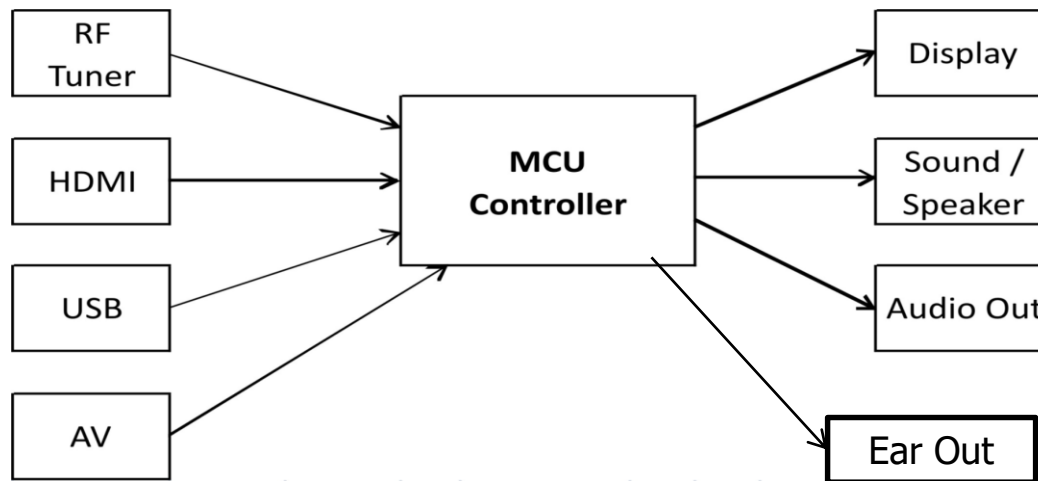
PEL Range – LED Television



Unique Selling Points :



Block Diagram



1. **MCU** (Microcontroller Unit)
2. **RF Tuner** (Radio Frequency)
3. **Ear Out**
4. **Audio Out**
5. **Audio-Video Component** (Y,Pb,Pr)

PEL Range – LED TV



Model No.	Ram	Internal Memory	Resolution
PLD-65US	2GB	16GB	3840 x 2160

PEL Range – LED TV



Model No.	Ram	Internal Memory	Resolution
PLD-55US	2GB	16GB	3840 x 2160

PEL Range – LED TV



Model No.	Ram	Internal Memory	Resolution
PLD-50US	2GB	16GB	3840 x 2160

PEL Range – LED TV



Model No	Ram	Internal Memory	Resolution
PLD-43FS	1.5 GB	8GB	2560 x 1440

PEL Range – LED TV



Model No.	Ram	Internal Memory	Resolution
PLD-40FS	1.5GB	8GB	1920 x 1080

PEL Range – LED TV



Model No.	Ram	Internal Memory	Resolution
PLD-32HS	1.5 GB	8GB	1280 x 720

PEL Range – LED TV



32" HD READY BEZEL-LESS DESIGN



MEDIA TV 1 x USB 2.0 3x HDMI

Model No.	Resolution
PLD-32HR	1280 x 720



Warranty Duration

"PEL Products"



Product	Compressor (Years)	Spare Parts (Year)	PCB Kit (Years)	Gas (Year)	Magnetron (Years)	Panel (Years)	Motor (Years)	Sensors (Years)
Refrigerator (Conventional)	03 ~ 05 (DOP < 01-Jan- 18)	1	NA	1	NA	NA	NA	NA
Refrigerator (Conventional)	10 (DOP ≥ 01-Jan- 18)	1	NA	1	NA	NA	NA	NA
Refrigerator (Inverter)	10	1	NA	1	NA	NA	NA	NA
Deep Freezer (Conventional)	2	1	NA	1	NA	NA	NA	NA
Deep Freezer (Inverter)	10	1	NA	1	NA	NA	NA	NA
Split Air Conditioner (Inverter)	10	1	04 & 05 (21 & 22)	1	NA	NA	NA	NA
Split Air Conditioner (Conventional)	3	1	NA	1	NA	NA	NA	NA
FS Air Conditioner (Conventional/Inv)	3	1	NA	1	NA	NA	NA	NA
Water Dispenser	1	1	NA	1	NA	NA	NA	NA
Microwave Oven	NA	1	NA	NA	2	NA	NA	NA
LED Television	NA	1 Year < DOM 2024	NA	NA	NA	1 Year < DOM 2024	NA	NA
LED Television	NA	2 Years DOM ≥ 2024	NA	NA	NA	2 Years DOM ≥ 2024	NA	NA
Washing Machine (Semi-Automatic)	NA	1	NA	NA	NA	NA	1 Year < DOM 2021	NA
Washing Machine (Semi-Automatic)	NA	1	NA	NA	NA	NA	3 Years (DOM ≥ 01-01-21)	NA
Washing Machine (Fully Automatic)	NA	1	NA	NA	NA	NA	05 < 2021 & 10 ≥ DOM 2021	NA
Air Purifier	NA	NA	NA	NA	NA	NA	2	2
Vertical Freezer	2	1	NA	1	NA	NA	NA	NA



Warranty Duration

"Corporate Cabinets"



1. ICE BERG (HICO) Warranty Duration:

Category	Duration
Manufacturing Year	Product code and Manufacturing year provided by QC
Visit Charges	Free 01 year from the DOM.
Gas Charges	
Compressor	3 Years
Cabinet	5 Years
Other Parts	1 Year

2. Unilever (Walls) Warranty Duration:

Category	Duration
Manufacturing Year	Product code and Manufacturing year provided by QC
Visit Charges	Free 01 year from the DOM.
Gas Charges	
Compressor	3 Years
Cabinet	5 Years
Other Parts	1 Year

3. EFL (Omore) Warranty Duration:

Category	Duration
Manufacturing Year	Product code and Manufacturing year provided by QC
Visit Charges	Free 01 year from the DOM.
Gas Charges	
Compressor	02 ½ Years
Cabinet	5 Years
Other Parts	1 Year

4. Error Code & Troubleshooting



4. Error Code & Troubleshooting



Sensor Faults

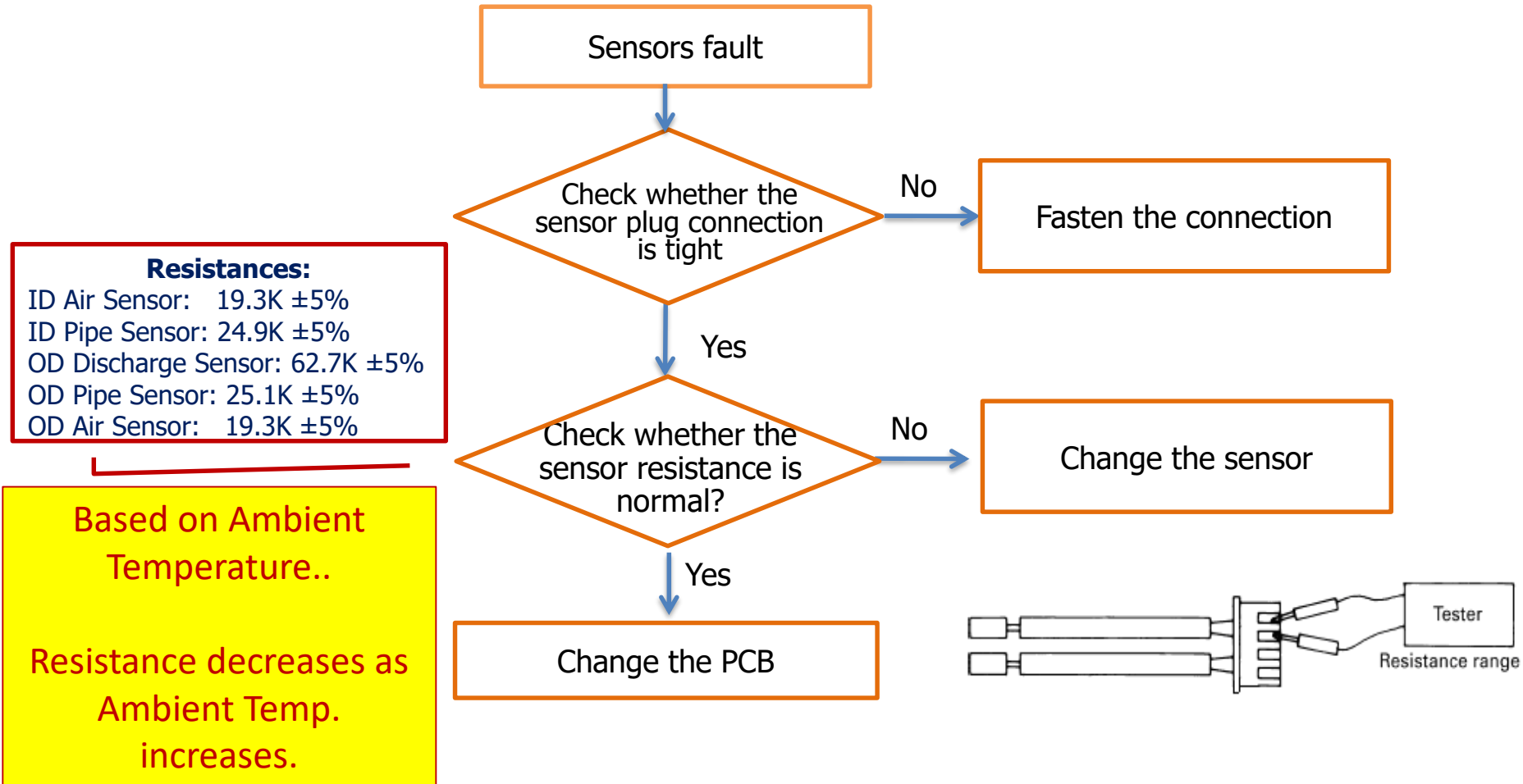
Indoor display: Error Codes as follows for different PEL Models

Error Code	Error Description	PEL AC Models
E1	Indoor Room temperature sensor failure	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME/ FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
E3	Indoor Coil/pipe sensor failure	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
E2	Indoor Coil/pipe sensor failure	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
E2	Outdoor Coil/Pipe Sensor failure	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
F6	Outdoor ambient temperature Sensor failure	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
E3	Outdoor Coil/Pipe Sensor failure	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
E7	Outdoor ambient temperature Sensor failure	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
Error Code	Error Description	PEL AC Models
F1	Indoor Room temperature sensor failure	Jumbo DC Black
F3	Indoor Coil/pipe sensor failure	Jumbo DC Black
F4	Outdoor Coil/Pipe Sensor failure	Jumbo DC Black
F2	Outdoor ambient temperature Sensor failure	Jumbo DC Black
Error Code	Error Description	PEL AC Models
EH 60	Indoor Room temperature sensor failure	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
EH 61	Indoor Coil/pipe sensor failure	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
EC 52	Outdoor Coil/Pipe Sensor failure	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
EC 53	Outdoor ambient temperature Sensor failure	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
Error Code	Error Description	PEL AC Models
33	Indoor Room temperature sensor failure	Alpha/ O-Glory
34	Indoor Coil/pipe sensor failure	Alpha/ O-Glory
1	Outdoor Coil/Pipe Sensor failure	Alpha/ O-Glory
14	Outdoor ambient temperature Sensor failure	Alpha/ O-Glory
Error Code	Error Description	PEL AC Models
81	Indoor Room temperature sensor failure	48K Ultimate
83	Indoor Coil/pipe sensor failure	48K Ultimate
2	Outdoor Coil/Pipe Sensor failure	48K Ultimate
1	Outdoor ambient temperature Sensor failure	48K Ultimate

4. Error Code & Troubleshooting



Sensor Fault



4. Error Code & Troubleshooting



Compressor Malfunction / Failure

Indoor display :

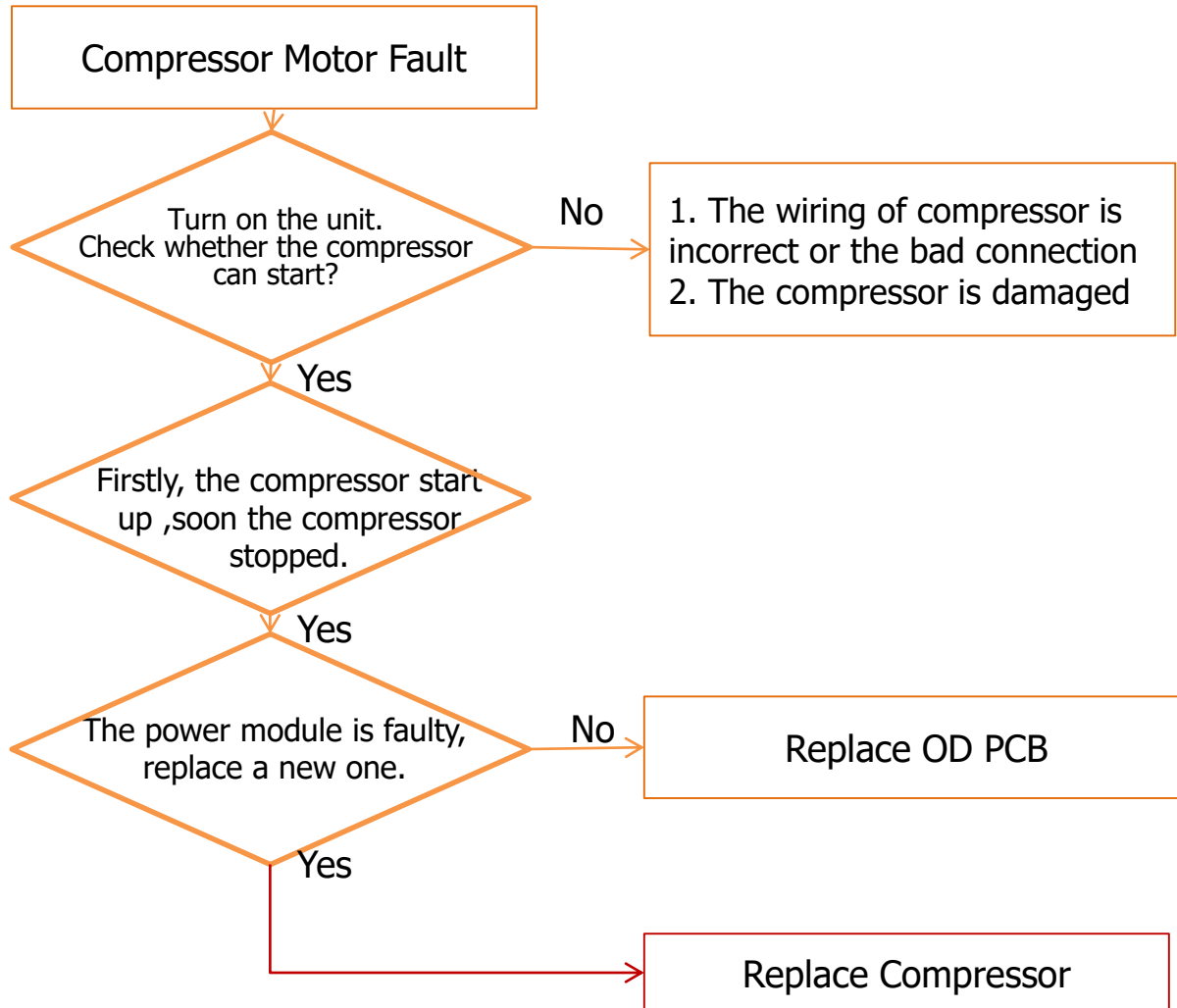
Error Codes as follows for different PEL Models

Error Code	Error Description	PEL AC Models
F3	The compressor fails to start or run out of step	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
Error Code	Error Description	PEL AC Models
E9	IPM / Compressor driving control abnormal.	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
Error Code	Error Description	PEL AC Models
E4	Compressor Starting abnormal	Jumbo DC Black
Error Code	Error Description	PEL AC Models
PC 04	Compressor drive error	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
Error Code	Error Description	PEL AC Models
18	Outdoor Compressor cannot start	Alpha/ O-Glory

4. Error Code & Troubleshooting



Compressor Malfunction / Failure



4. Error Code & Troubleshooting



Indoor Blower/Fan Motor Malfunction

Indoor display :

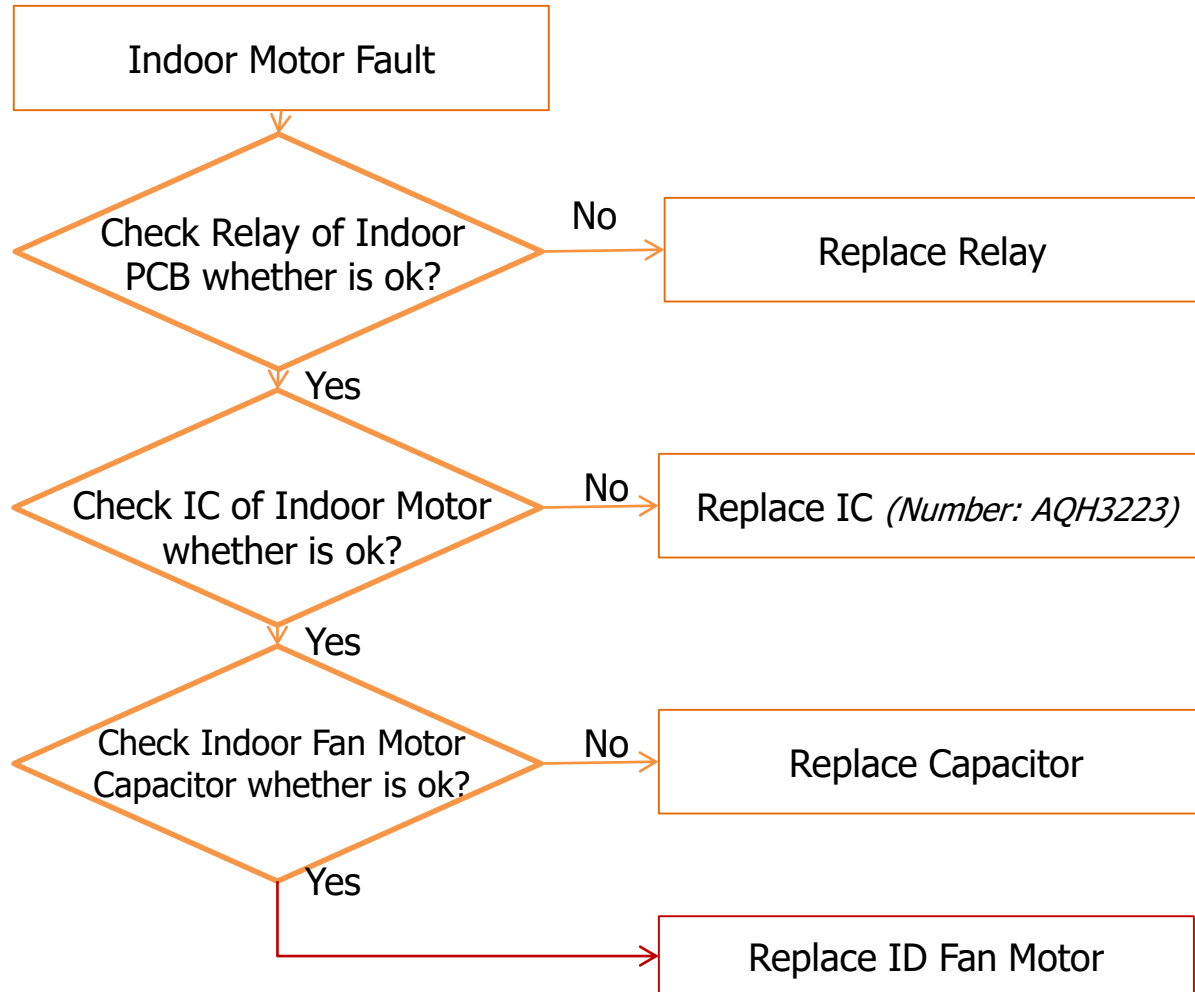
Error Codes as follows for different PEL Models

Error Code	Error Description	PEL AC Models
E4	Indoor Fan Motor Fault or DC Motor feedback fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
E4/Fb	Indoor Fan Loop Is Faulty	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME (DOM2024 and Onwards)
E6	IDU PG Fan motor / DC fan motor works abnormal	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
F0	Indoor Fan Motor Fault or DC Motor feedback fault	Jumbo DC Black
39	Indoor Fan Motor Abnormal Operation	Alpha/ O-Glory
EH 03	Indoor Fan speed out of control	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
72	Indoor Fan Motor Fault	48K Ultimate

4. Error Code & Troubleshooting



Indoor Blower/Fan Motor Malfunction



4. Error Code & Troubleshooting



Communication Error (Case – 01) Outdoor Kit Dead

Indoor display :

Following any Error Model wise listed can be seen on Indoor Display.

Indoor Unit on but outdoor unit complete shut-off.

1. Outdoor Kit completely dead

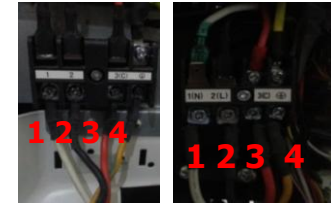
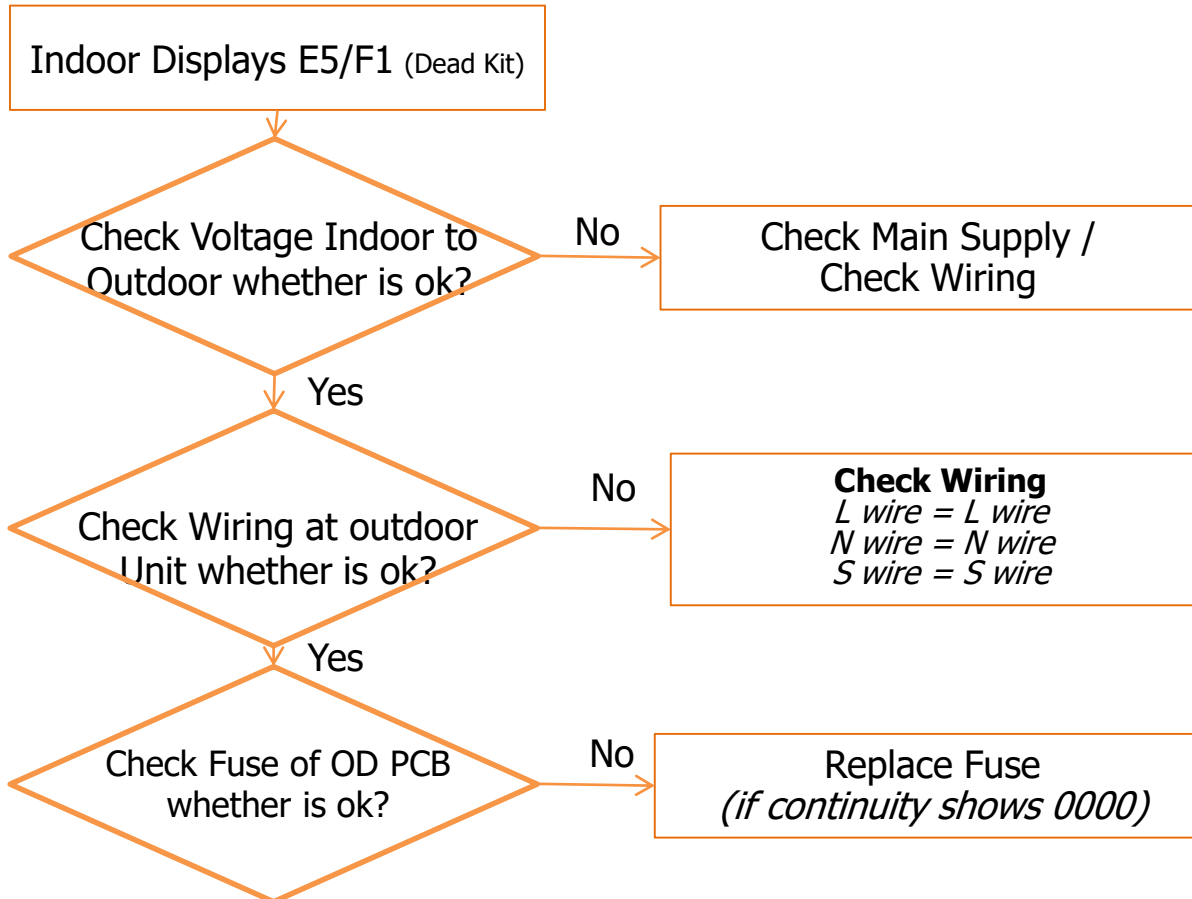
2. Outdoor Kit On

Error Code	Error Description	PEL AC Models
E5	Indoor & Outdoor communication fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
E5 (5E)	Indoor & Outdoor communication fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME (DOM2024 and Onwards)
E0	Indoor & Outdoor communication fault	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
F6	Indoor & Outdoor communication fault	Jumbo DC Black
36	Indoor & Outdoor communication fault	Alpha/ O-Glory
EL 01	Indoor & Outdoor communication fault	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
64	Indoor & Outdoor communication fault	48K Ultimate

4. Error Code & Troubleshooting



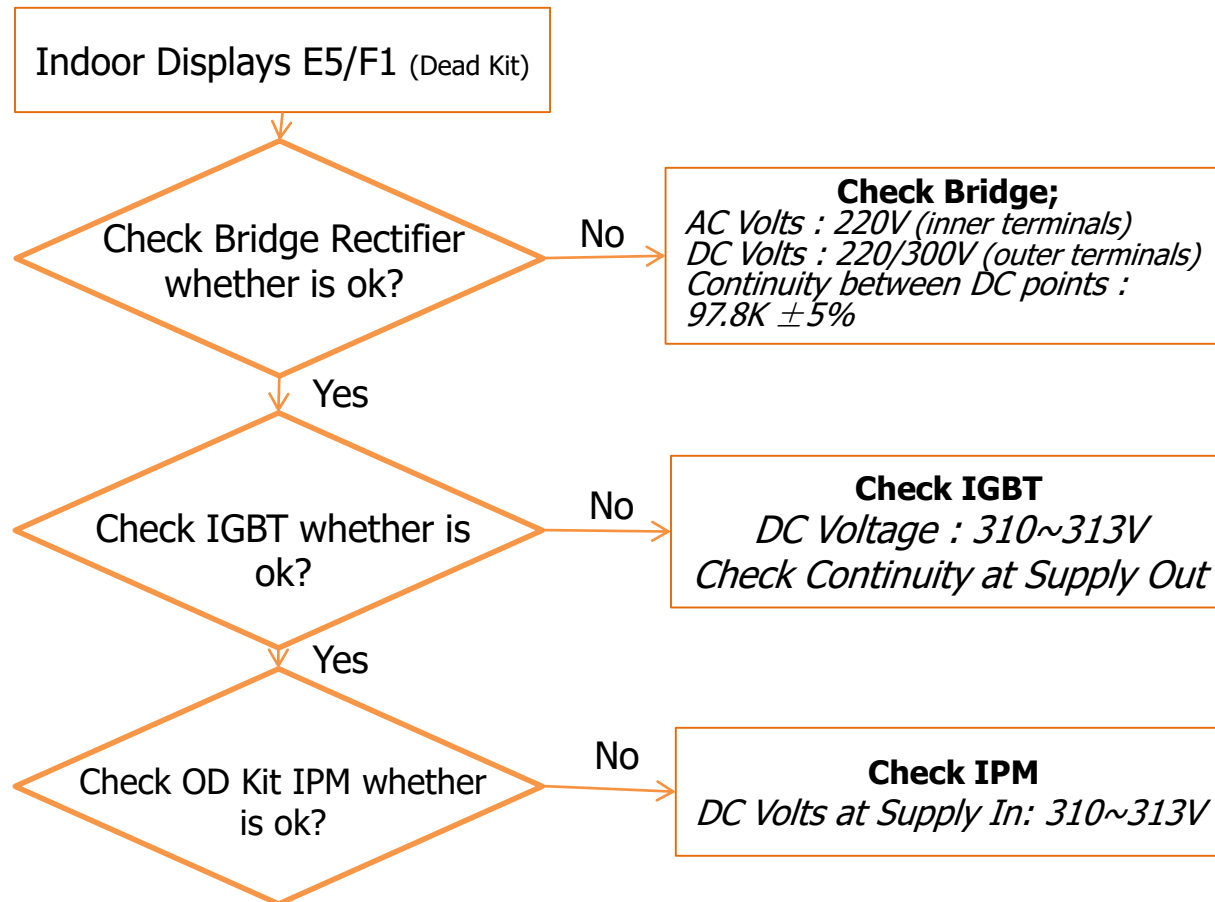
Communication Error (Case – 01) Outdoor Kit Dead



4. Error Code & Troubleshooting



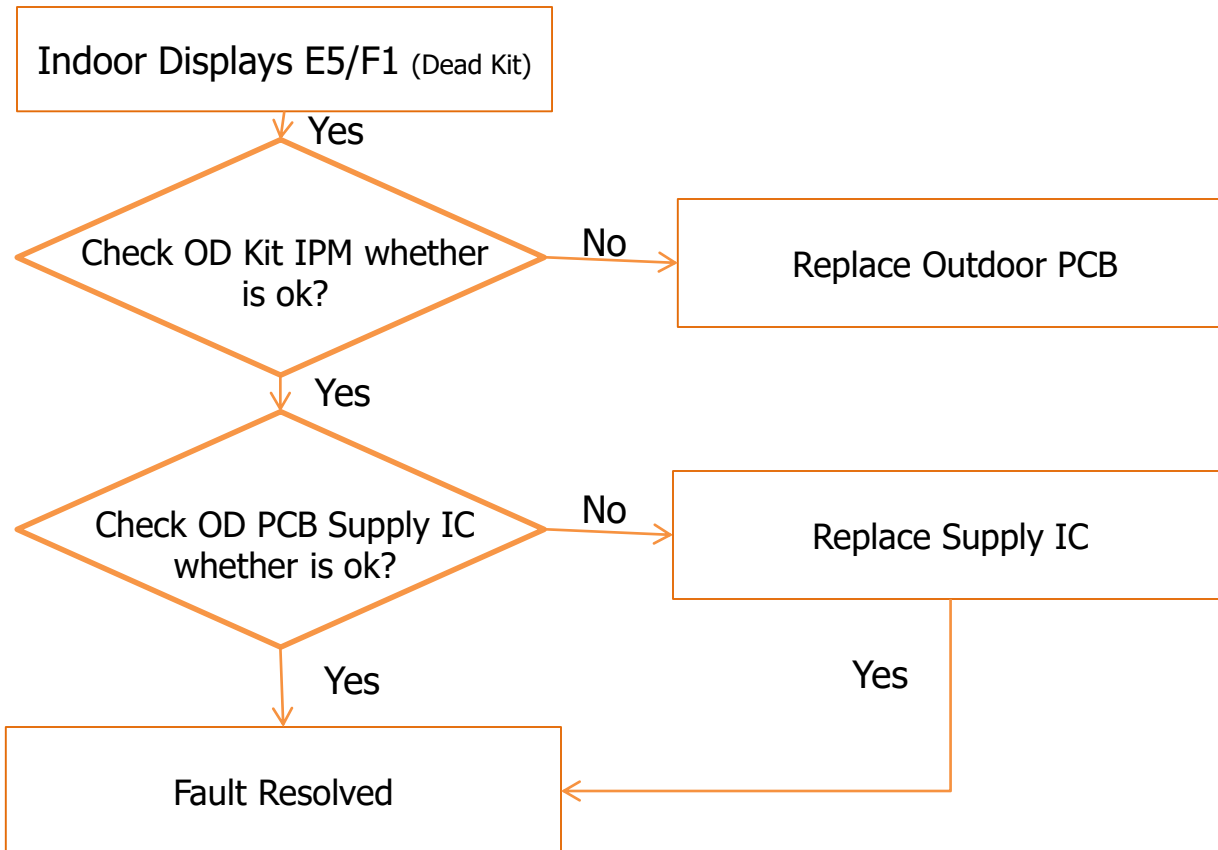
Communication Error (Case – 01) Outdoor Kit Dead



4. Error Code & Troubleshooting



Communication Error (Case – 01) Outdoor Kit Dead



4. Error Code & Troubleshooting



Communication Error (Case – 02) Outdoor Kit On

Indoor display :

Following any Error Model wise listed can be seen on Indoor Display.

Indoor Unit on but outdoor unit trip again & again.

1. Outdoor Kit completely dead

2. **Outdoor Kit On**

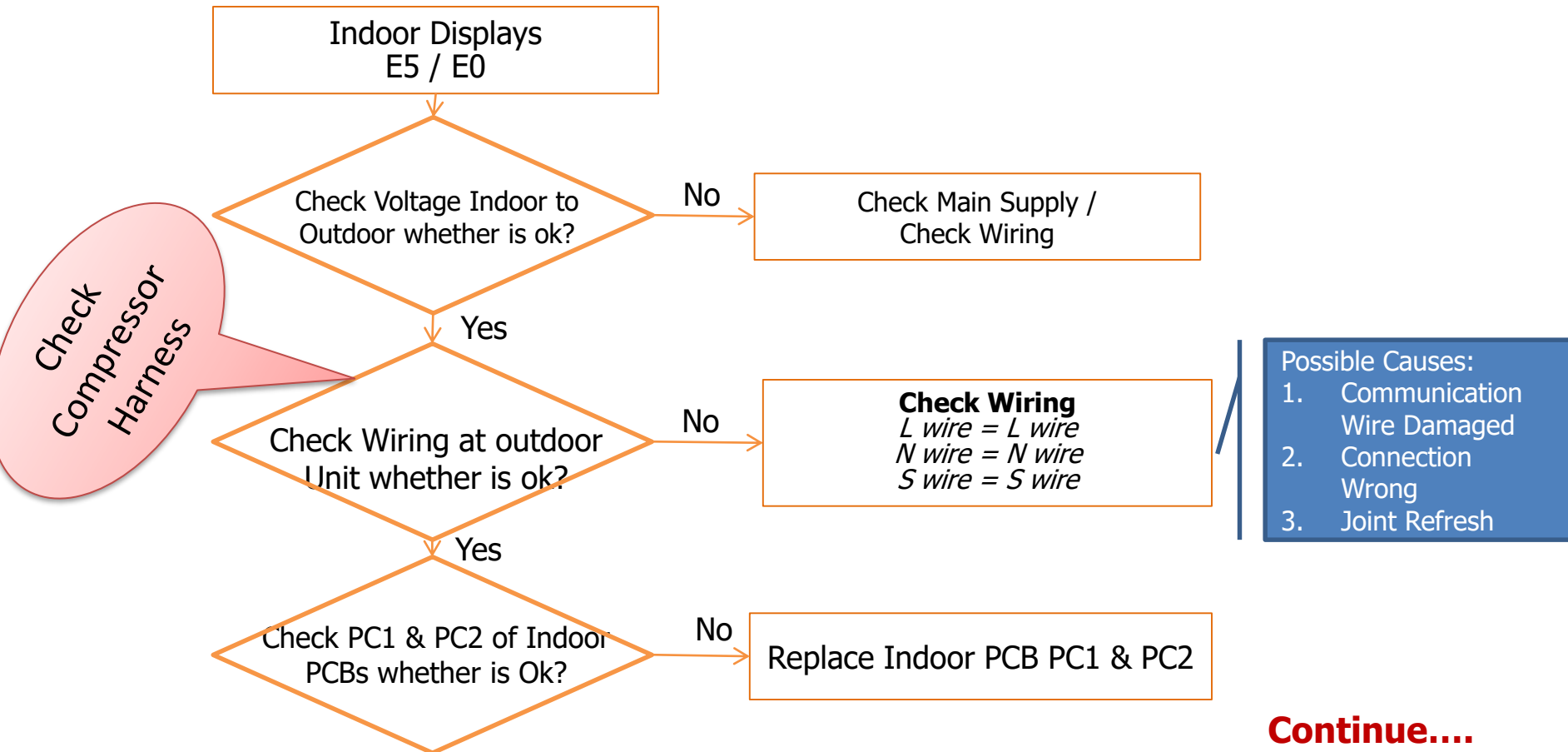
Error Code	Error Description	PEL AC Models
E5	Indoor & Outdoor communication fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME
E5 (5E)	Indoor & Outdoor communication fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME (DOM2024 and Onwards)
E0	Indoor & Outdoor communication fault	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
F6	Indoor & Outdoor communication fault	Jumbo DC Black
36	Indoor & Outdoor communication fault	Alpha/ O-Glory
EL 01	Indoor & Outdoor communication fault	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
64	Indoor & Outdoor communication fault	48K Ultimate

Continue....

4. Error Code & Troubleshooting



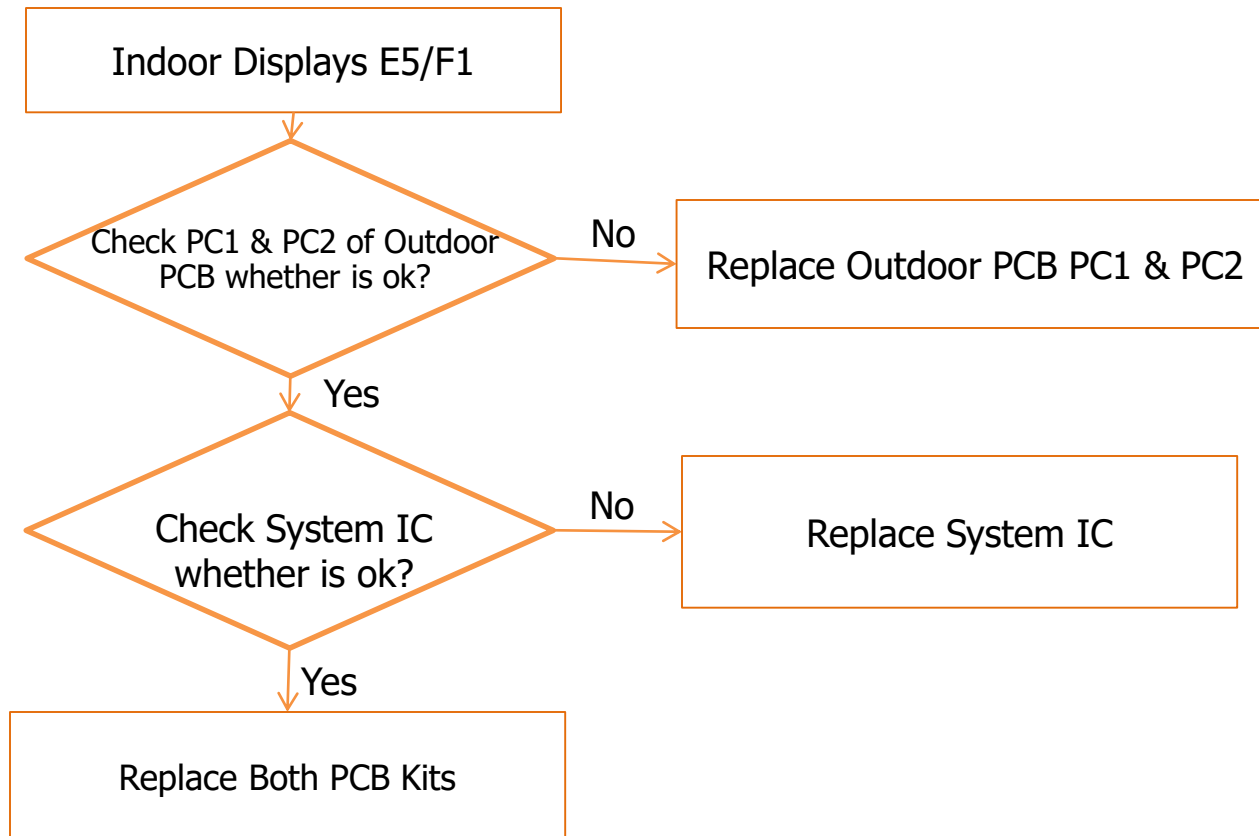
Communication Error (Case – 02) Outdoor Kit On



4. Error Code & Troubleshooting



Communication Error (Case – 02) Outdoor Kit On



4. Error Code & Troubleshooting



Outdoor Fan Motor Malfunction

Indoor display :

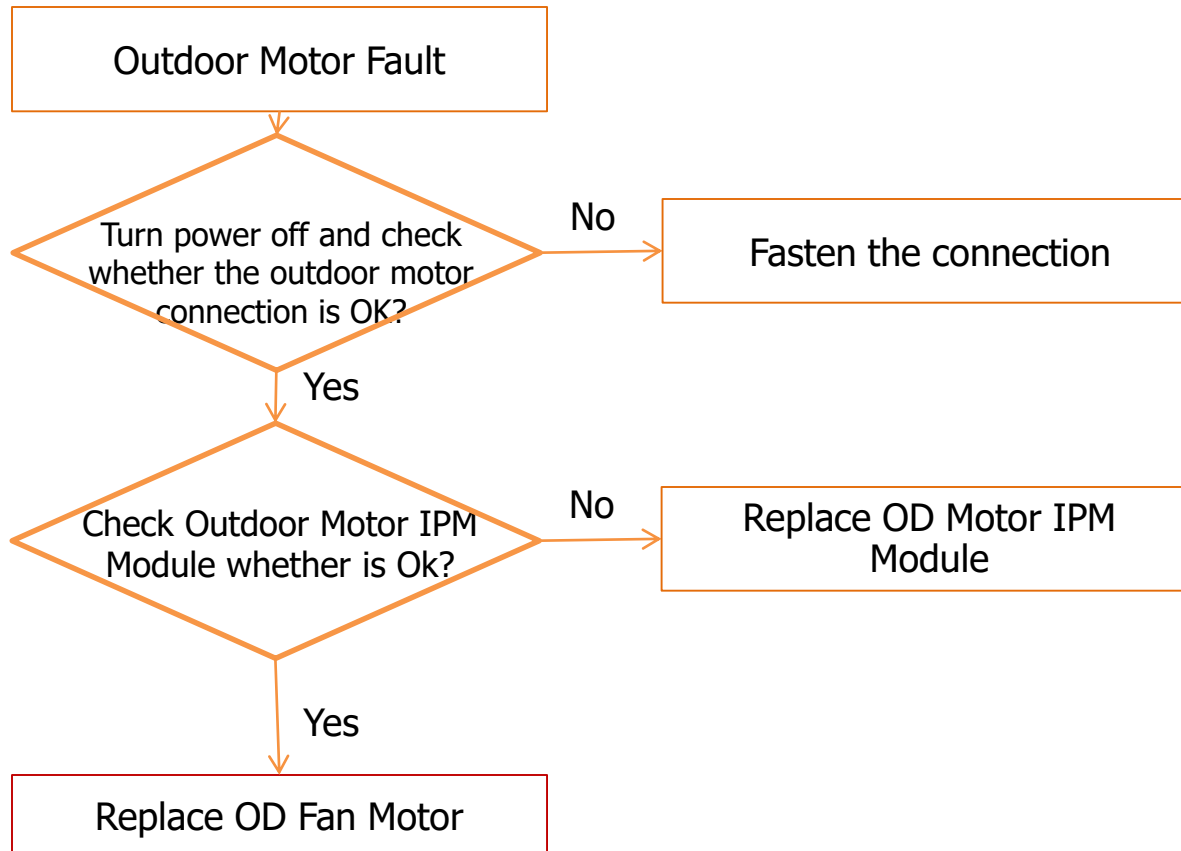
Following any Error Model wise listed can be seen on Indoor Display.

Error Code	Error Description	PEL AC Models
F0	Outdoor DC Fan Motor Malfunction Fault	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME (OLD 2023 DOM AND DOM 2024 AND ONWARDS)
EF	Outdoor DC Fan Motor Malfunction Fault	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
E2	Outdoor DC Fan Motor Malfunction Fault	Jumbo DC Black
20	Outdoor DC Fan Motor Malfunction Fault	Alpha/ O-Glory
EC 07	Outdoor DC Fan Motor Malfunction Fault	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
48	Outdoor DC Fan Motor Malfunction Fault	48K Ultimate

4. Error Code & Troubleshooting



Outdoor Fan Motor Malfunction



4. Error Code & Troubleshooting



Power Module Protection

Indoor display :

Following any Error Model wise listed can be seen on Indoor Display.

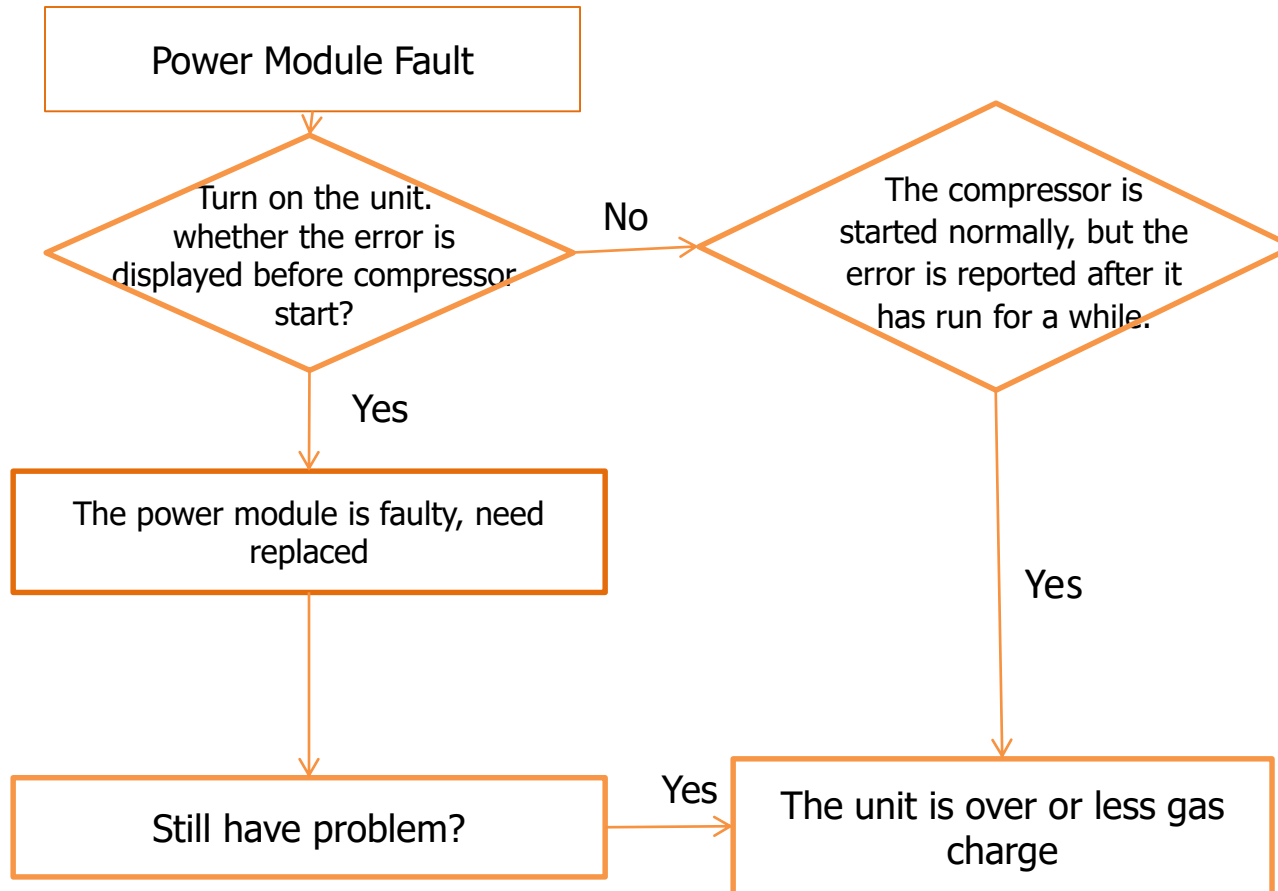
Outdoor PCB Power Module(IPM) Failure.

Error Code	Error Description	PEL AC Models
F1 / L4	IPM Module Failure	ACE / APEX /ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME (OLD 2023 DOM “F1” AND DOM 2024 “L4”)
P0 / E9	IPM module protection / IPM abnormal	FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC
F9	IPM Module Failure	Jumbo DC Black
19	Outdoor Main PCB Board Drive Failure	Alpha/ O-Glory
PC 00	IPM malfunction or IGBT over-strong current protection	Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine
45	IPM Fault	48K Ultimate

4. Error Code & Troubleshooting



Power Module Protection



4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(ACE / APEX / ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME):

Failure	The Reason of Fault and Solution
E1	Indoor ambient temperature sensor is faulty
E2	Outdoor coil temperature sensor fault
E3	Indoor coil temperature sensor fault
E4 / Fb	Indoor fan motor or DC motor feedback fault
E5 / 5E	Indoor PCB & outdoor PCB communication fault
F0	Outdoor DC Fan motor fault
F1 / L4	IPM modular fault / IPM of the compressor driver module is faulty
F2	PFC modular fault

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(ACE / APEX / ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME):

Failure	The Reason of Fault and Solution
F3	Compressor operation fault / Compressor fails to start or run out of step.
F4	Compressor Exhaust Temperature sensor fault
F5	Compressor top cover protection / Compressor press housing sensor is faulty
F6	Outdoor ambient temp sensor fault / Outdoor environment sensor is faulty
F7	Over and under voltage protection
F8	Communication between the outdoor module is faulty
F9	Outdoor EEPROM fault
FA	Suction temperature sensor fault / Return air sensor is faulty
FC	Four-way valve is faulty
Eb	Internal/Indoor EEPROM is faulty

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(ACE / APEX / ALLURE/ JUMBO DC/ JUMBO DC CLASSIC/ MAJESTIC/ REGAL/ SUPREME):

No.	Error Code	Error Name	Probable Trouble Location
1	E1	Internal room temperature sensor error	Indoor room temperature sensor, main indoor PCB Kit
2	E2	External coil sensor error	External coil sensor error, main external control panel
3	E3	Internal coil sensor error	Internal coil sensor error, main internal control panel
4	E4 / Fb	Indoor fan error	Mechanical jam of internal fan blade, internal fan, main internal control panel
5	E5(5E)	Indoor and outdoor unit communication error	Bridge cable, main internal control panel, main external control panel, module panel
6	F0	Outdoor DC fan error	Mechanical jam of external fan, external DC fan, main external control panel
7	F1 / L4	Module protection error	Power voltage, compressor cable, reactor, module panel, main external control panel, compressor
8	F2	PFC protection error	Power voltage, reactor, module panel, main external control panel
9	F3	Compressor out-of-step error	System pressure, compressor cable, module panel, main external control panel, compressor
10	F4	Exhaust air sensor error	Exhaust air sensor, main external control panel
11	F5	Compressor cap sensor error	System pressure, compressor cap sensor (protection switch), main external control panel
12	F6	External room temperature sensor error	External room temperature sensor, main external control panel
13	F7	OVP or UVP error	Power voltage, reactor, module panel, main external control panel
14	F8	Main external control panel and module panel communication error	Connection wire of module and main control data, module panel, main external control panel
15	F9	Outdoor EE error	Main external control panel
16	FA	Suction/Return Air sensor error	Return sensor, four-way valve, main external control panel

4. Error Code & Troubleshooting



The fault code table has
three LED status display,
respectively,

★: light ○: blink ×: off.
Order: LED1 LED2 LED3

c. Error Codes for Inverter Air Conditioners

Alpha/ O-Glory Inverter Air Conditioners (○: Blink; ×: Off; ★: Light)

IDU display	ODU LED lamp			Description
	LED1	LED2	LED3	
1	★	×	★	Outdoor coil temperature sensor fault
2	★	×	×	Outdoor compressor discharge temperature sensor fault
14	★	★	×	outdoor air temperature failure
15	×	★	○	Outdoor compressor shell temperature protection
19	○	×	○	Outdoor Main PCB Board drive failure
20	○	○	★	Outdoor fan motor was stuck, failure
21	×	★	×	Outdoor coil over heat/cold protection
33	NA	NA	NA	Indoor air sensor failure
34	NA	NA	NA	Indoor coil sensor failure
EA	NA	NA	NA	Indoor Display communication failure
36	×	×	○	Indoor and outdoor communication failure
39	NA	NA	NA	Indoor fan motor abnormal operation
6	★	★	○	unit Power Over-voltage protection
8	★	○	×	unit current overload (6s)
9	★	○	★	unit Maximum current protection (millisecond)
10	×	★	★	Outdoor controller and drive module communication failure (ODU main PCB failure)
11	★	★	★	Outdoor E2PROM fault (ODU main PCB failure)
17	○	×	×	Outdoor PFC failure (Reactor Failure/ ODU main PCB failure)
16	×	○	○	indoor coil over heat or over cold protection
18	○	○	×	Outdoor compressor cannot start
38	NA	NA	NA	Indoor E2PROM fault (Indoor PCB failure)
41	NA	NA	NA	Indoor zero-crossing fault (Indoor PCB failure)

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC):

Code	Reason	Remark
E0	IDU & ODU Communication failure	The IDU & ODU wiring connection, correct?
E1	IDU Room Temperature sensor failure. (IDU RT failure)	IDU sensor and PCB.
E2	IDU coil temperature sensor Failure. (IDU IPT Failure)	IDU sensor and PCB.
E3	ODU Coil temperature sensor failure. (OPT)	ODU coil sensor and ODU PCB
E4	AC Cooling system abnormal	Gas leakage? or 4-way valve blocked etc.
E5	IDU/ODU mismatched failure (specially performance test on the production line)	
E6	IDU PG Fan motor / DC fan motor works abnormal (IDU failure)	Fan motor, fan blade and PCB.
E7	ODU Ambient Temperature sensor Failure	ODU ambient sensor and ODU PCB.
E8	ODU Discharge Temperature sensor Failure.	ODU discharge sensor and ODU PCB.
E9	IPM / Compressor driving control abnormal.	ODU PCB , compressor, etc.
EA	ODU Current Test circuit failure	ODU PCB broken?
Eb	The Communication abnormal of Main PCB and Display board (IDU failure)	Display board and main PCB.
EE	ODU EEPROM failure.	1. ODU PCB broken? 2. Try to re-power on AC unit.
EF	ODU DC fan motor failure.	Fan motor, ODU PCB.
EU	ODU Voltage test circuit abnormal.	ODU PCB.
P0	IPM module protection.	ODU PCB

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(FIT/ SAVER/ SUPER/ JUMBO DC Prime/ Sublime/ Turbo DC):

Code	Reason	Remarks
P1	Over / under voltage protection.	1. ODU PCB broken? 2. Power supply abnormal?
P2	Over current protection.	1. ODU PCB broken? 2. Power supply abnormal?
F0	Infrared Customer feeling test sensor failure. (IDU failure)	Querying by press remote controller
F1	Electric Power test module failure. (IDU failure)	Querying by press remote controller
F9	The module temperature test circuit failure.	ODU PCB
FA	The compressor Phase-current test circuit failure.	ODU PCB
F2	Discharge temperature sensor failure PROTECTION.	
F3	ODU coil temperature Failure PROTECTION.	
P8	Outdoor Over temperature/Under temperature protection.	
P9	Compressor driving protection (Load abnormal).	

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(Turbo DC Ultra/ Aero Plus/ Turbo DC Ultimate/ AERO Extend/ Alpine):

Operation lamp	Timer lamp	Code	Meaning of Troubles
☆ 1 time	X	EH 00	Indoor unit EEPROM parameter error
☆ 2 times	X	EL 01	Indoor / outdoor units communication error
☆ 3 times	X	EH 02	Zero-crossing signal detection error (Not available for DC indoor fan motor)
☆ 4 times	X	EH 03	Indoor fan speed has been out of control
☆ 6 times	X	EH 60	Indoor room temperature sensor T1 open circuit or short circuit
☆ 6 times	X	EH 61	Evaporator coil temperature sensor T2 open circuit or short circuit
☆ 8 times	X	EL 0C	Refrigerant leakage detection
☆ 9 times	X	EH 0b	Communication error between indoor PCB and display PCB
☆ 1 times	☆	PC 08	Over-current protection
☆ 5 times	X	EC 53	Outdoor ambient temperature sensor T4 open circuit or short circuit
☆ 5 times	X	EC 52	Condenser coil temperature sensor T3 open circuit or short circuit
☆ 5 times	X	EC 54	Compressor discharge temperature sensor T5 open circuit or short circuit
☆ 5 times	X	EC 51	Outdoor unit EEPROM parameter error
☆ 12 times	X	EC 07	Outdoor fan speed has been out of control (DC fan motor only)
☆ 7 times	☆	PC 00	IPM malfunction or IGBT over-strong current protection
☆ 2 times	☆	PC 01	DC voltage between P&N out of range
☆ 3 times	☆	PC 02	Hi temperature protection of compressor top or IPM
☆ 5 times	☆	PC 04	Inverter compressor drive error
		FH 0P	AP mode is active but there is no WIFI kit installed
X: OFF			O: On
			☆: Flashes in 2Hz

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(Jumbo DC Black):

变频显示故障/状态代码显示方式

中文			English		
室外机指示灯 闪烁次数	数码管显示	故障/状态名称	ODU LIGHT FLASH	IDU DISPLAY	FAULT DETAILS
1	EF	室外EE故障	1	EF	Outdoor unit EEPROM fault
2	F6	室内、外机通讯故障	2	F6	Indoor and outdoor communication fault
3	F8	主板与驱动板通信故障	3	F8	Main board and driver board communication fault
4	E4	压缩机启动异常(缺相、反转)	4	E4	Compressor starting abnormal (phase failure, reverse)
5	E3	压缩机失步故障	5	E3	Compressor out-of-step fault
6	F9	IPM模块故障	6	F9	IPM module fault
7	E0	压缩机壳顶故障/保护	7	E0	Compressor shell roof fault/protection
8	F5	排气温度传感器故障	8	F5	Exhaust temperature sensor fault
9	E5	吸气温度传感器故障(预留)	9	E5	Suction temperature sensor fault
10	F4	室外盘管温度传感器故障	10	F4	Outdoor coil temperature sensor fault
11	F2	室外环境温度传感器故障	11	F2	Outdoor ambient temperature sensor fault
12	E2	室外直流风机故障	12	E2	Outdoor DC fan motor fault
13	P2	室外机AC电流保护停机	13	P2	Outdoor AC current protection
14	P0	压缩机相电流保护停机	14	P0	Compressor phase current protection

4. Error Code & Troubleshooting



c. Error Codes for Inverter Air Conditioners

(Jumbo DC Black):

15	P7	室外机AC电压过高、过低保护	15	P7	Outdoor unit over-high/over-low AC voltage protection
16	P8	直流母线电压过高、过低保护	16	P8	Outdoor unit over-high/over-low DC voltage protection
17	P9	IPM温度过高保护停机	17	P9	IPM over-high temperature protection
18	P1	排气温度过热保护停机	18	P1	Exhaust temperature overheat protection
19	P5	制冷内盘防冻结保护停机	19	P5	Cooling indoor coil anti-freezing protection
20	P6	制冷外盘管过热保护停机	20	P6	Cooling outdoor coil overheat protection shutdown
21	P4	制热内盘管过热保护停机	21	P4	Heating indoor coil overheat protection
22	PA	制冷外环境温度过低保护停机	22	PA	Cooling outdoor ambient temperature over-low protection
23	PE	制热外环境温度过高保护停机	23	PE	Heating outdoor ambient temperature over-high protection
25	EE	室内机 EE 故障	25	EE	Indoor unit EEPROM fault
26	F0	室内风机故障	26	F0	Indoor fan motor fault
27	E1	室内风机过零检测异常	27	E1	Indoor PCB zero-crossing fault
28	F3	室内盘管传感器故障	28	F3	Indoor coil fault and sensor fault
29	F1	室内环境温度传感器故障	29	F1	Indoor room temperature sensor fault
/	E8	室外系统异常	/	E8	Outdoor system exception
/	P3	制热除霜提示	/	P3	Heating and removing cream tips
/	E9	wifi模块异常	/	E9	Exception of <u>wifi</u> module

Error codes will be Display on Display PCB.

7. Installation of Split AC



7. Installation of Split AC



Contents:

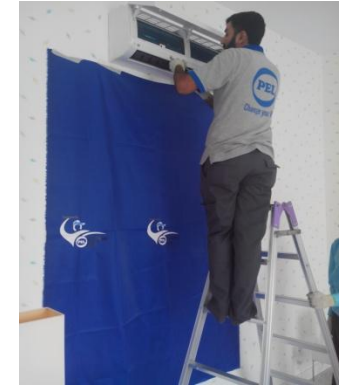
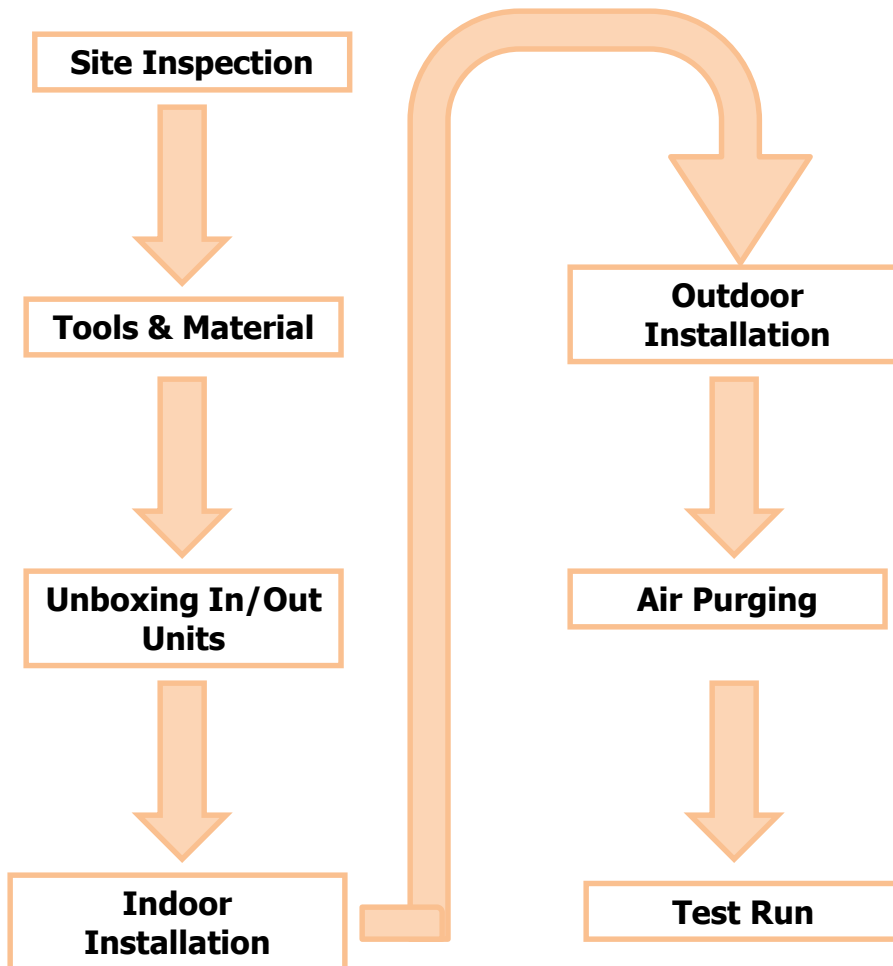
- 1) Flow Process
- 2) Appropriate position for installation.
- 3) Preparation of installation tools.
- 4) Preparation before installation
- 5) Installation of indoor unit
- 6) Installation of the outdoor unit
- 7) Vacuum
- 8) Leak detection
- 9) Trial Operation



7. Installation of Split AC



1). Flow Process:



7. Installation of Split AC



2). Appropriate position for installation – Indoor / Outdoor Unit

Recommended

Room Size

Room size	AC capacity
Up to 100 sq. ft	0.8 ton
Up to 150 sq. ft	1.0 ton
Up to 250 sq. ft	1.5 ton
Up to 400 sq. ft	2.0 ton

Standard Height

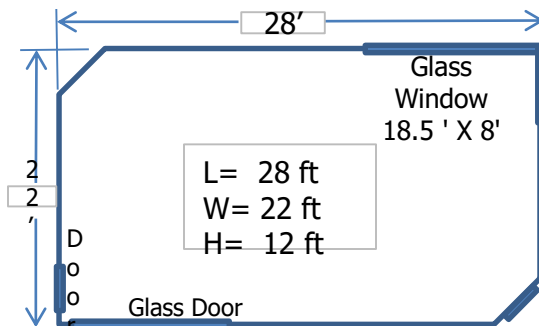
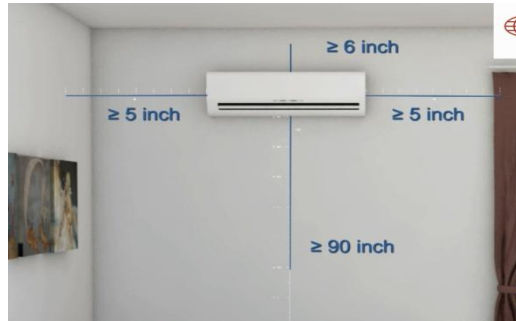
Draw Layout for approval

Main Supply
(Separate Breaker)

Earthing

Piping Route

Outdoor Location
(Bracket/Shed/Roof top)



7. Installation of Split AC



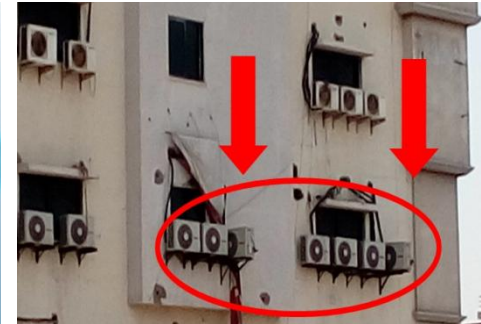
2). Appropriate position for installation – Indoor / Outdoor Unit

Not Recommended

Heat / Steam source near the installation position



Obstacles which blocks the air circulation near the installation position



Being too close to combustible objects as, curtain, clothing



Near or above the household appliances
(STD= distance about 40")



There's fume or droplet near the installation position



Obstacles, which may affect air flow



7. Installation of Split AC



3). Preparation of installation tools & equipment/material:

MATERIAL REQUIRED FOR INSTALLATION TOOLS REQUIRED FOR INSTALLATION

- i. IRON FRAME GUAGE (16 ~ 18)
- ii. RAWAL BOLTS (2 - 1/2")
- iii. SILICON TUBE (optional)
- iv. PLASTER OF PARIS
- v. MAIN SUPPLY WIRE (7/.036)
- vi. CONTROL WIRE 03-Core (76/110)
- vii. DUCT (Optional) (60*60)
- viii. DRAIN PIPE PVC
- ix. DRAIN PIPE G.I.
- x. SAMAD BOND (Optional)
- xi. AEROFLUX (1" DIA)
- xii. COTTON CLOTH
- xiii. MASKING , TEFLON & INSULATION TAPE

- 1) Hacksaw
- 2) Hole Core Drill
- 3) Hexagon Wrench
- 4) Gauge Manifold
- 5) Spanner Set
- 6) Adjustable Wrench
- 7) Tube Cutter
- 8) Flaring Tool Set
- 9) Swaging Tool Set
- 10) Hammer Set
- 11) Level
- 12) Reamer & Cleaning Brush
- 13) Flare / Union Nuts
- 14) Leak Detector
- 15) Measuring Tape
- 16) Gauge Manifold
- 17) Tongs Tester
- 18) Spring Bender
- 19) Screw Driver Set
- 20) Knife / Cutter
- 21) Pliers Set
- 22) Chisel Set
- 23) Temperature Meter
- 24) Lead Pencil
- 25) Gas Cylinder



7. Installation of Split AC



4). Preparation before installation:

Copper Pipe

Pipe cutting
(*optional*)

Cover up with PVC
tape

Sheathe the
insulation pipe

Sleeve the flare nut

Pipe expanding

Tools / Equipment:

1. Tube Cutter
2. PVC Tape
3. Insulation (Aeroflex)
4. Flare Nut
5. Flaring Tool Set

Safety/Caution:

1. Appropriate pressure to cut
2. Sleeve flare nut after sheathing insulation pipe



Cont.....

7. Installation of Split AC



4). Preparation before installation:

Wire stripping

Wire cutting

U-type plug

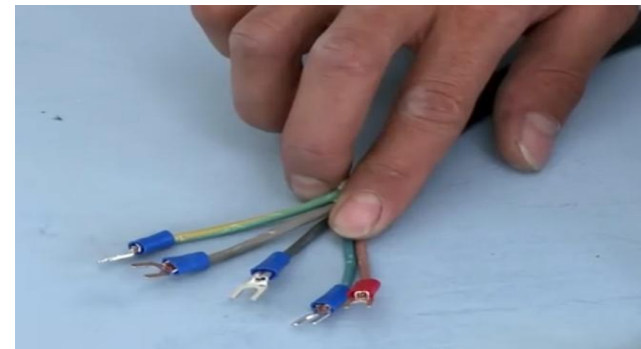
U-type plug fixing

Tools / Equipment:

1. Cutter Plier
2. Knife
3. Plier
4. U-Type plug

Safety/Caution:

1. Cut only about 6~8 inches
2. Do not punch with high pressure



Cont.....

7. Installation of Split AC



4). Preparation before installation:

Unit Unboxing

Check and match indoor and outdoor model of AC

Unbox Indoor unit

Check scratch/dent on unit

Accessories of indoor unit

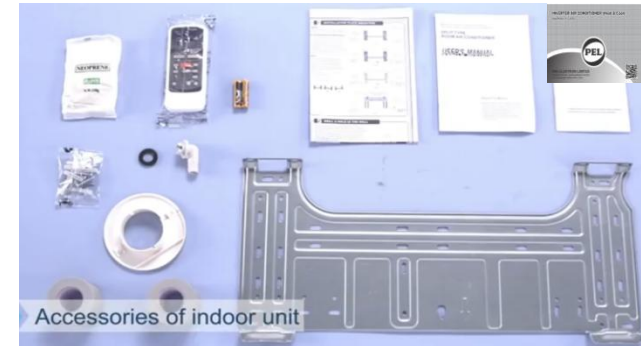
Selection of installation position of the indoor unit

Tools / Equipment:

Cutter/Knife

Safety/Caution:

1. Misplacement of W/C
2. Misplacement of Remote



7. Installation of Split AC



5). Installation of indoor unit:

Remove base plate
from indoor unit

Straighten pipes
safely

Wiring of Indoor
Unit

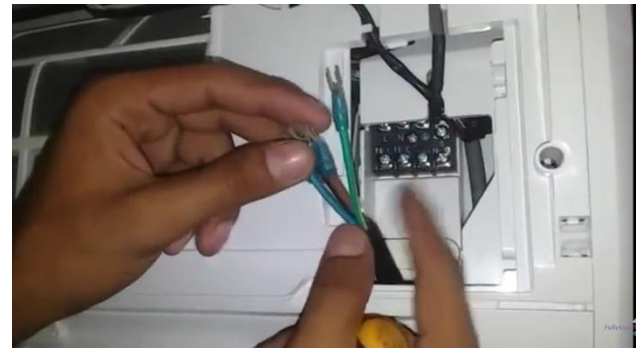
L = Brown
N = Blue
S = Black
E = Green/Yellow

Tools / Equipment:

Philips Screw Driver

Safety/Caution:

1. Pipe bending
2. Wiring connection



Cont.....

7. Installation of Split AC



5). Installation of indoor unit:

Mount / base plate fixing

Wrap up both pipes with Tape

Take size for hole

After marking center of the wall keep mount plate for hole opening

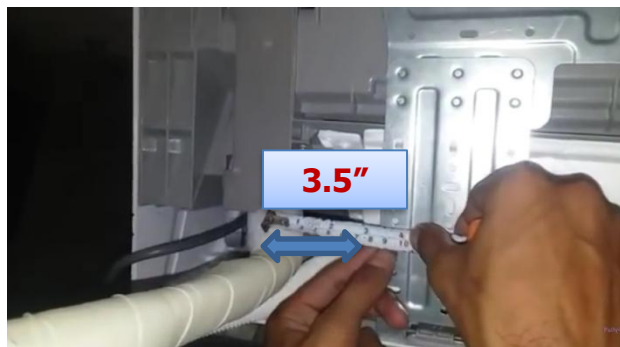
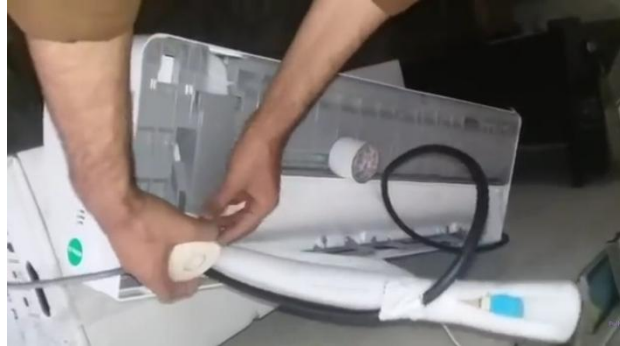
Mark point on wall for hole to join indoor pipes with outdoor

Tools / Equipment:

1. Weathering Tape
2. Measuring Tape
3. Lead Pencil

Safety / Caution:

1. Drain pipe loose
2. Pipe Bending



Cont.....

7. Installation of Split AC



5). Installation of indoor unit:

Mount / base plate fixing

Drill the wall to make hole

Take again size from hole to verify

Level the mounting plate with level meter

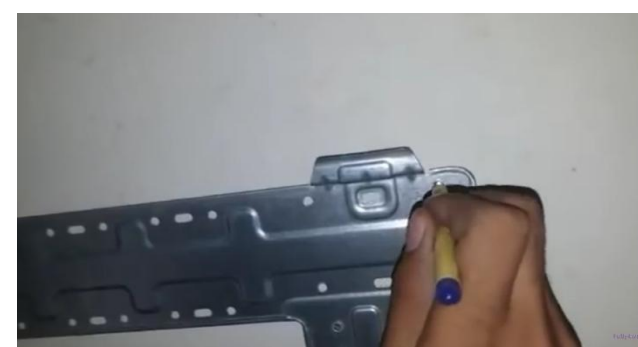
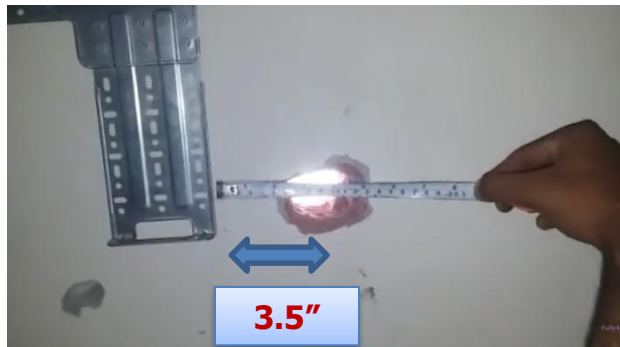
Mark and tight the screws to fix plate

Tools / Equipment:

1. Drill Machine
2. Hole saw (optional)
3. Paper/Vacuum Cleaner
4. Level Meter
5. Measuring Tape
6. Lead Pencil
7. Screw Driver (Flat)

Safety/ Caution:

Use a vacuum cleaner or paper to save wall with drilled powder



Cont.....

7. Installation of Split AC



5). Installation of indoor unit:

Indoor Unit Fixing

Cut of the hole for wires and drain

Wire and Drain pipe passes through the hole

Wrap up pipes pass through drilled hole

Insulating sleeve of In/Out pipes

Connect copper pipe and insulate

Reserve flare nuts open during wrapping for leak detection

Tools / Equipment:

1. Cutter / Knife, 2. Wrench

Safety/ Caution:

1. Make soft edge for wire
2. Check drain pipe
3. Keep 4" to 6" gap

1st Option



2nd Option



Cont.....

7. Installation of Split AC



5). Installation of indoor unit:

Pipe connecting

Bend pipe as per requirement

Tighten the flares, 1st by hand then with wrench

Reserve open for leak detection



Tools / Equipment:

Adjustable Wrench

Safety/Caution:

1. Bend softly
2. Check flare screw by hand tighten.

Cont.....

7. Installation of Split AC



5). Installation of indoor unit:

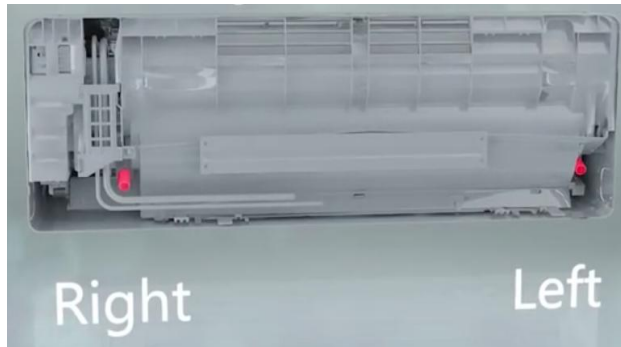
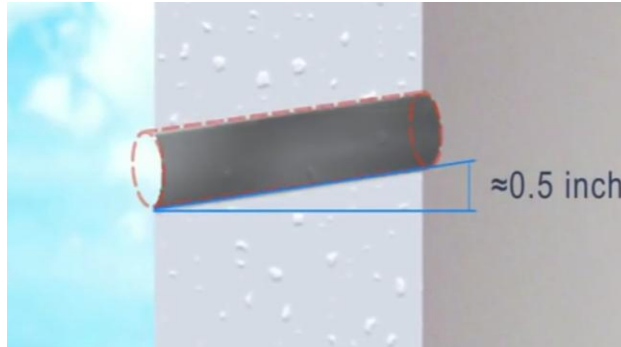
Drain pipe fixing

Slope in drilling
about 1/2 inch

Checking and
selection of drain
side

Cut the hole for
drain and wires

1. Use PVC pipe,
connecting to
drain pipe.
2. Make grooving
to better look

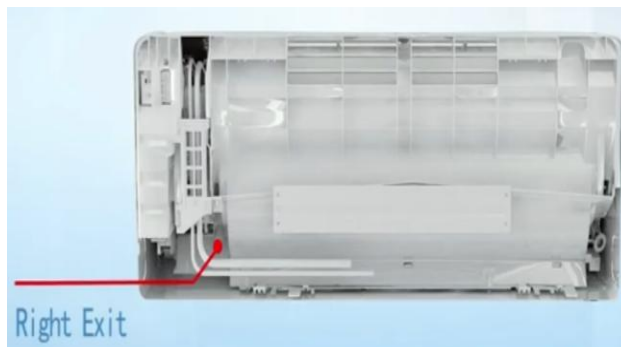


Tools / Equipment:

1. Cutter / Knife
2. PVC pipe & bend

Safety/Caution:

1. Make soft edge for wire
2. Check drain pipe
3. Unit hanged properly



Cont.....

7. Installation of Split AC

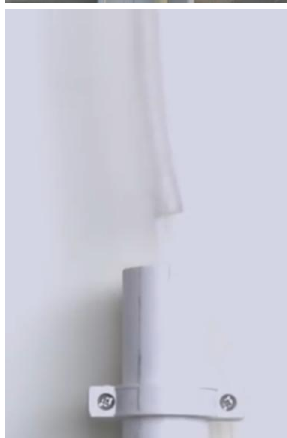


5). Installation of indoor unit:

Drain pipe fixing

Pros and Cons in Drain Pipe installation

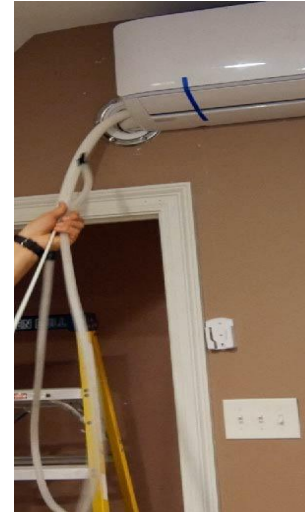
Wrong



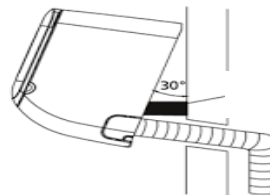
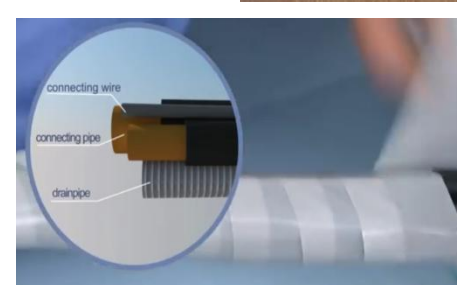
Right



Wrong



Right



7. Installation of Split AC



6). Installation of the outdoor unit:

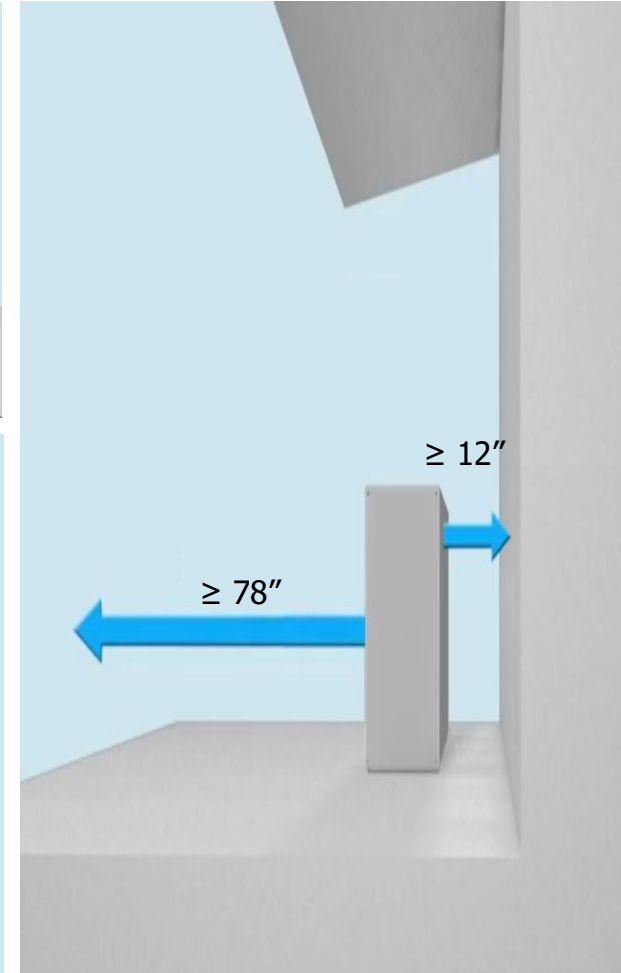
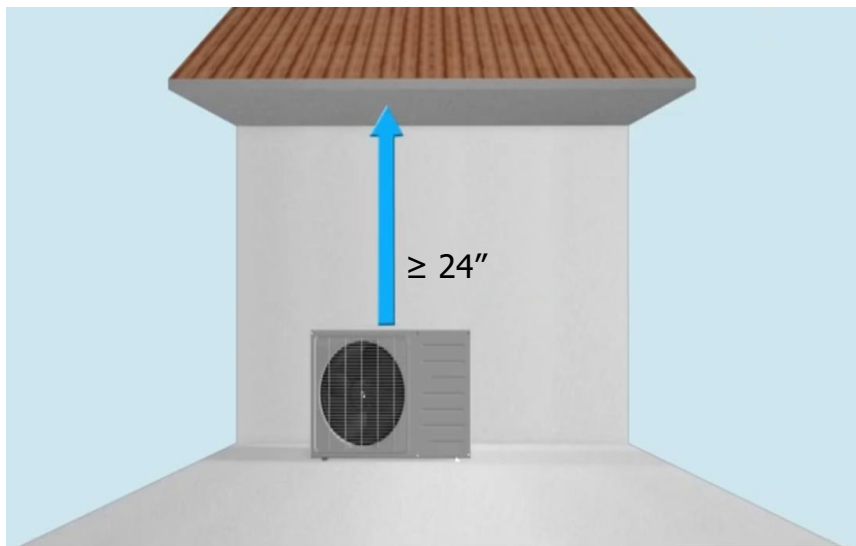
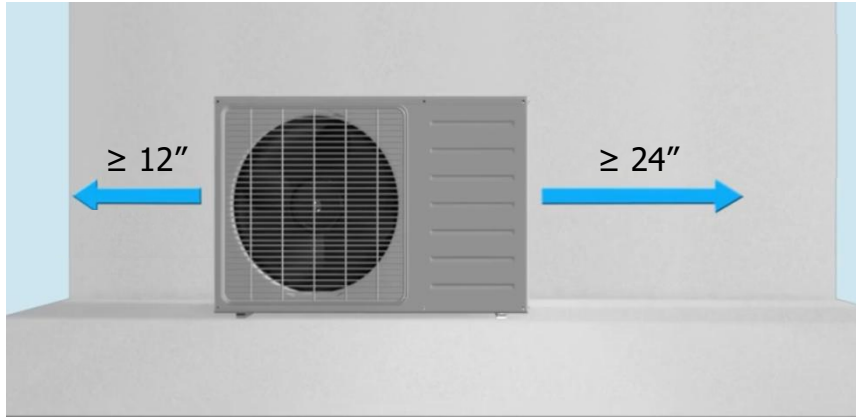
Space Requirement

Space left and right side

Space from top

Space from back

Space at front



Cont.....

7. Installation of Split AC



6). Installation of the outdoor unit:

Area Selection for Outdoor Installation

Check length to
make firmly

Mark the hole and
drill

Hammer rawal
bolts and fix the
unit with
spanner/wrench

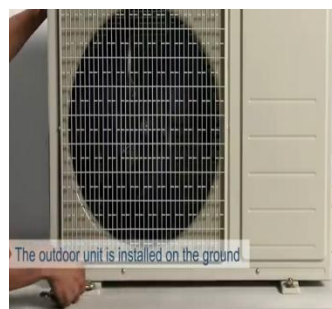
Tools / Equipment:

1. Measuring Tape
2. Lead pencil
3. Drill M/C
4. Level Indicator
5. Hammer
6. Spanners

Safety/Caution:

Selected surface should
be firm and strong

Installation on Ground



Installation on wall bracket



Level the bracket
and drill



Fix the unit and
tighten bolts



Cont.....

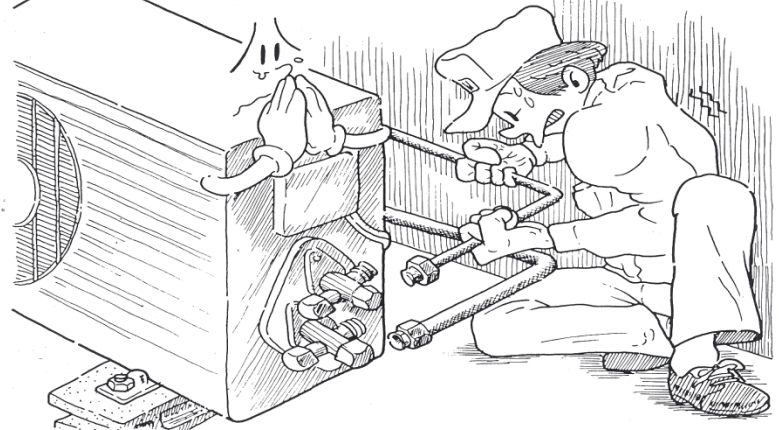
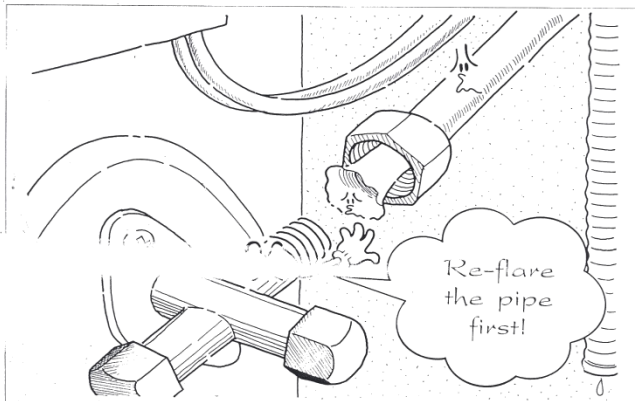
7. Installation of Split AC



6). Installation of the outdoor unit:

Connection of
Outdoor drain
pipe

**Safety
must
needed..**



Cont.....

7. Installation of Split AC



6). Installation of the outdoor unit:

Connection of Outdoor drain pipe

Heat pump units is required to connect to drain pipe for outdoor units



7. Installation of Split AC



6). Installation of the outdoor unit:

Connection of Outdoor connecting pipe

Unscrew the service valve cover

Connect flare properly

Connect both pipe with adjustable wrench

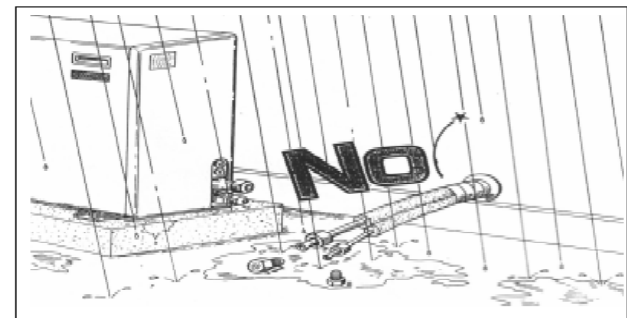
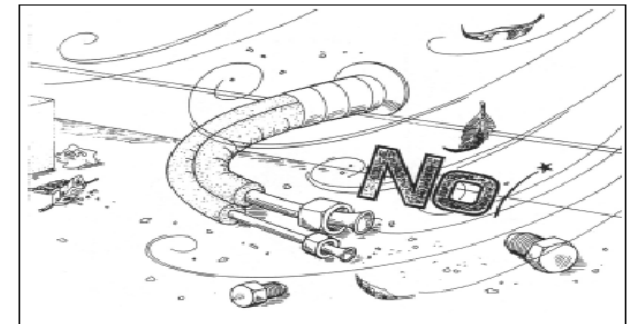
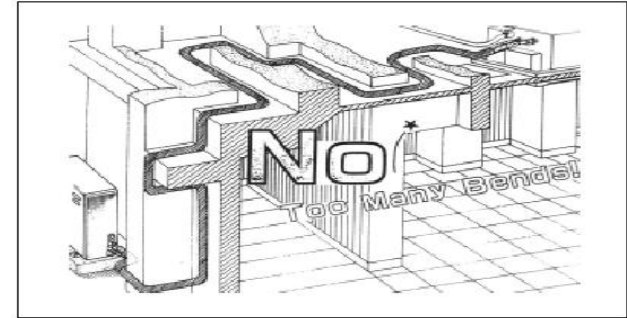


Tools / Equipment:

1. Screw Driver
2. Adjustable Wrench

Safety/Caution:

1. Too many bends
2. Pipe end should be covered with tape
3. Before outdoor installation check length of connecting pipe



Cont.....

7. Installation of Split AC



6). Installation of the outdoor unit:

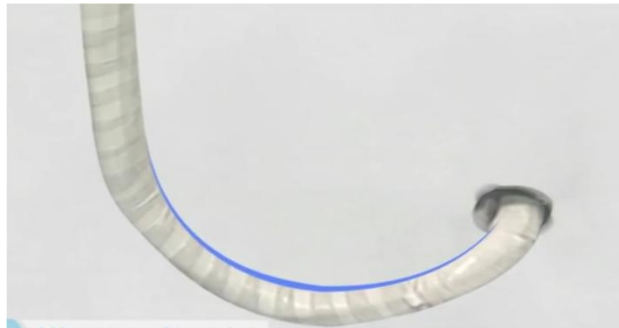
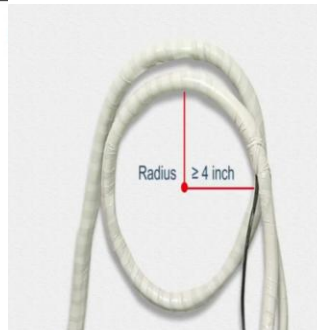
Connection of Outdoor pipe and wiring

Fixe safety pipe in hole for pipe

Pipes should not be bend less than 8" dia

Bend Connecting pipe to make water proof

Electrical connections for outdoor unit



Tools / Equipment:

Screw Driver

Safety/Caution:

1. Bend safely
2. Indoor unit wiring should be correspond with outdoor
3. Pull the wire back after fixing to make sure the connection is secure.

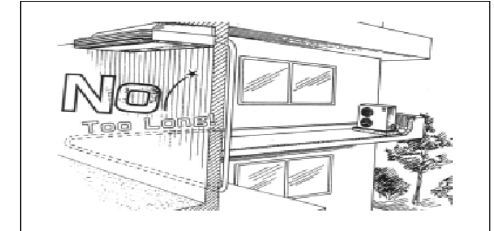
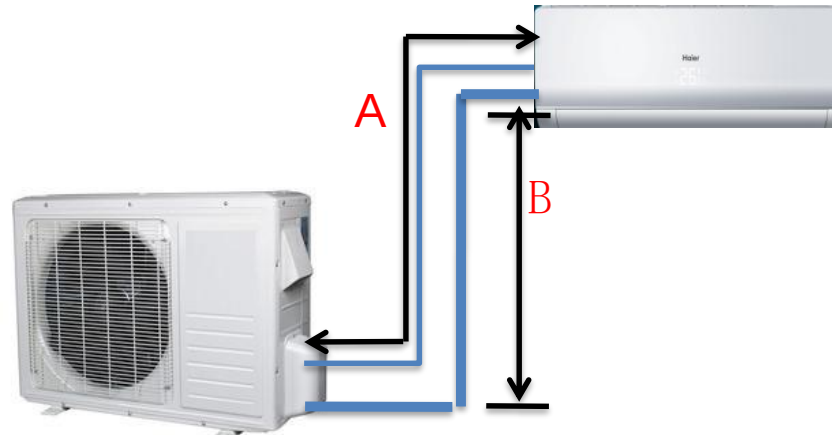
7. Installation of Split AC



6). Installation of the outdoor unit:

Connection of Outdoor connecting pipe

Standard maximum length of connecting pipe is 3m, but if more than 3m required then see table



Model	Diameter		Maximum length A (m)	Maximum height between indoor and outdoor B (m)	Additional charge g/m
	Discharge (Inch)	Suction (mm)			
12K	1/4	3/8 (1/2)	15	10	20
18K	1/4	1/2	20	15	20
24K	1/4 (3/8)	1/2(5/8)	20	15	50

Safety/Caution:

1. No too long
2. Pipe length should not be greater than 3m but if required the see table

7. Installation of Split AC



7). Vacuum:

Open low side charging valve and fix the hose

Power on vacuum pump and then open gauge valve

15 minutes later, when gauge reading is 29.92 in.Hg, then close the low pressure valve of gauge.

Now power off vacuum pump, after 5 minutes there should be no change

Open high side charging valve and fix the hose

Open high side charging valve and fix the hose, check the reading 147.65 in.Hg for 1 minute.

Tighten the hexagonal keys and flare nuts properly



7. Installation of Split AC



8). Leak Detection:

Apply soapy water of high concentration in to interfaces of nuts with brush

Repeat above same to indoor unit nuts

After satisfaction, wrap the pipe and fix indoor unit properly on mounting plate

Tools / Equipment:

Soapy water

Safety/Caution:

Use soapy water except washing powder



7. Installation of Split AC



9). Trial Operation:



Things to remember

a). Before Operation Test

1. Do not switch on power before installation is finished completely.
2. Electric wiring must be connected correctly and securely.
3. Cut-off valves (flare nuts) of the connection pipes should be opened.
4. All the impurities such scrap must be cleared from the unit.

b). Operation Test Mode

1. Switch on power and press , "ON/OFF" button on the wireless remote controller to start the operation.
2. Press MODE button to select the, "COOL, HEAT, FAN" to check whether the operation is normal or not.

7. Installation of Split AC



9). Trial Operation:

Drainage test

Current detection

Voltage detection

Block the wall hole



Tools / Equipment:

1. Water Bottle
2. Tongs Tester

Safety/Caution:

Area should be free of debris after all work.



7. Installation of Split AC



Installation – Can we install different brands IDU & ODU ?

It will work fine in the following cases;

- 1) The tonnage capacity is the same. You can't pair a 1tr idu to 1.5tr odu.
- 2) The technology is same. Avoid interchanging if you have an Inverter model.
- 3) Use the same Gas. An R22 IDU should preferably be paired with its counterparts and not an R410.
- 4) Check mode. Avoid using only cooling with heating cooling.



PEL Range – LED Television



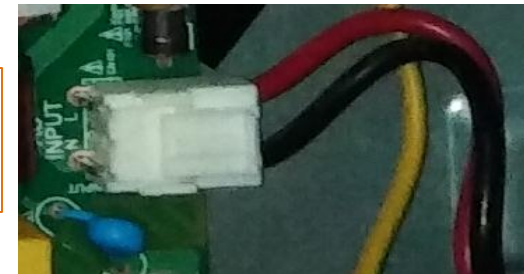
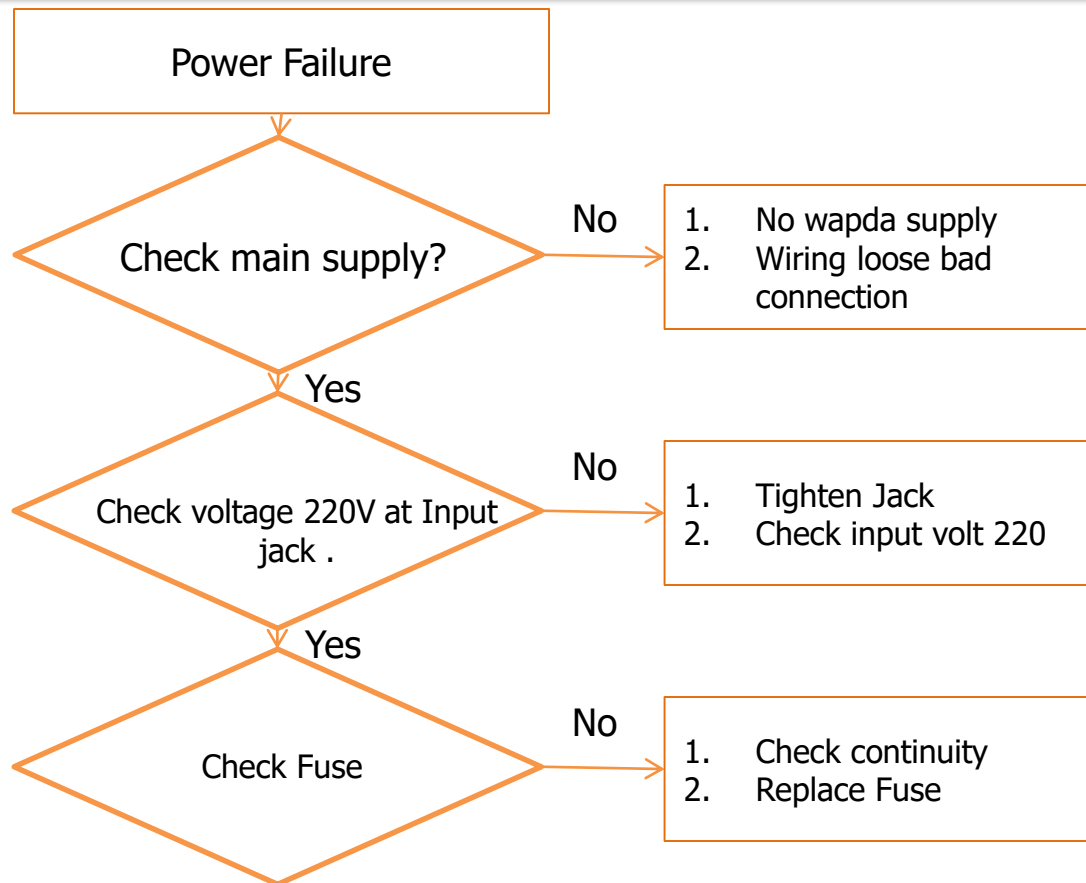
LED TV Troubleshooting



Power Failure

Display on Screen :

No blue light power on
No display



Continue....

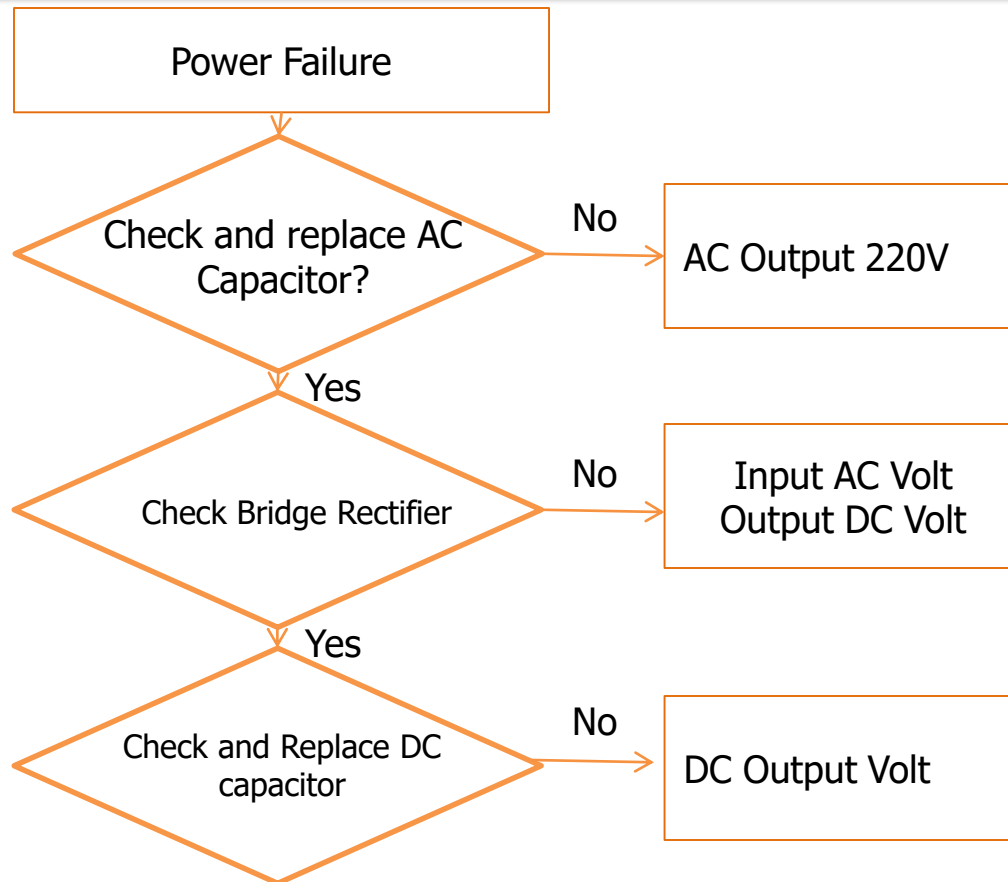
PEL Range – LED Television



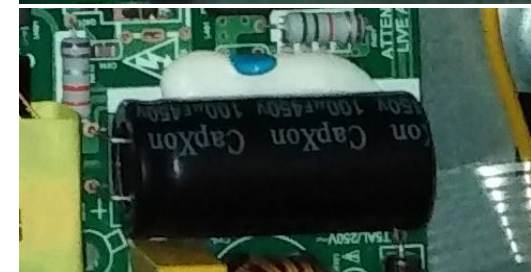
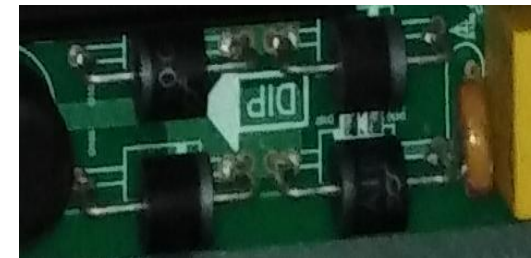
Power Failure

Display on Screen :

No blue light power on
No display



Continue....



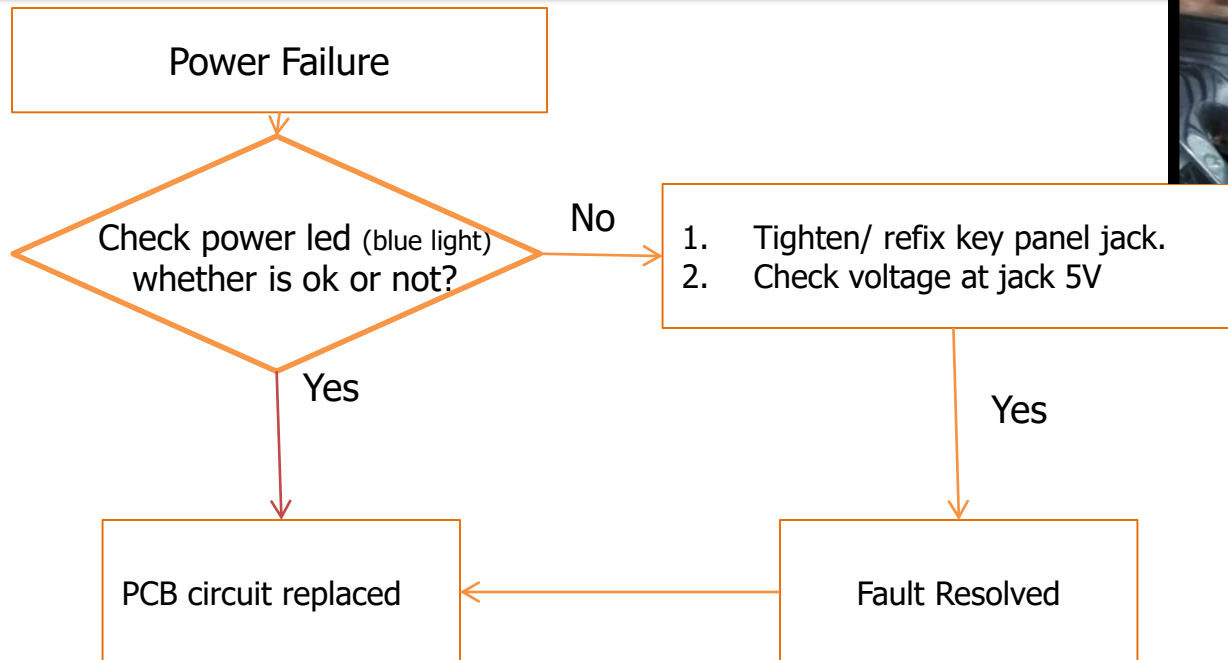
PEL Range – LED Television



Power Failure

Display on Screen :

No blue light power on
No display

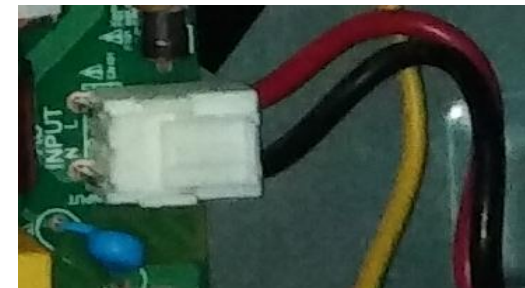
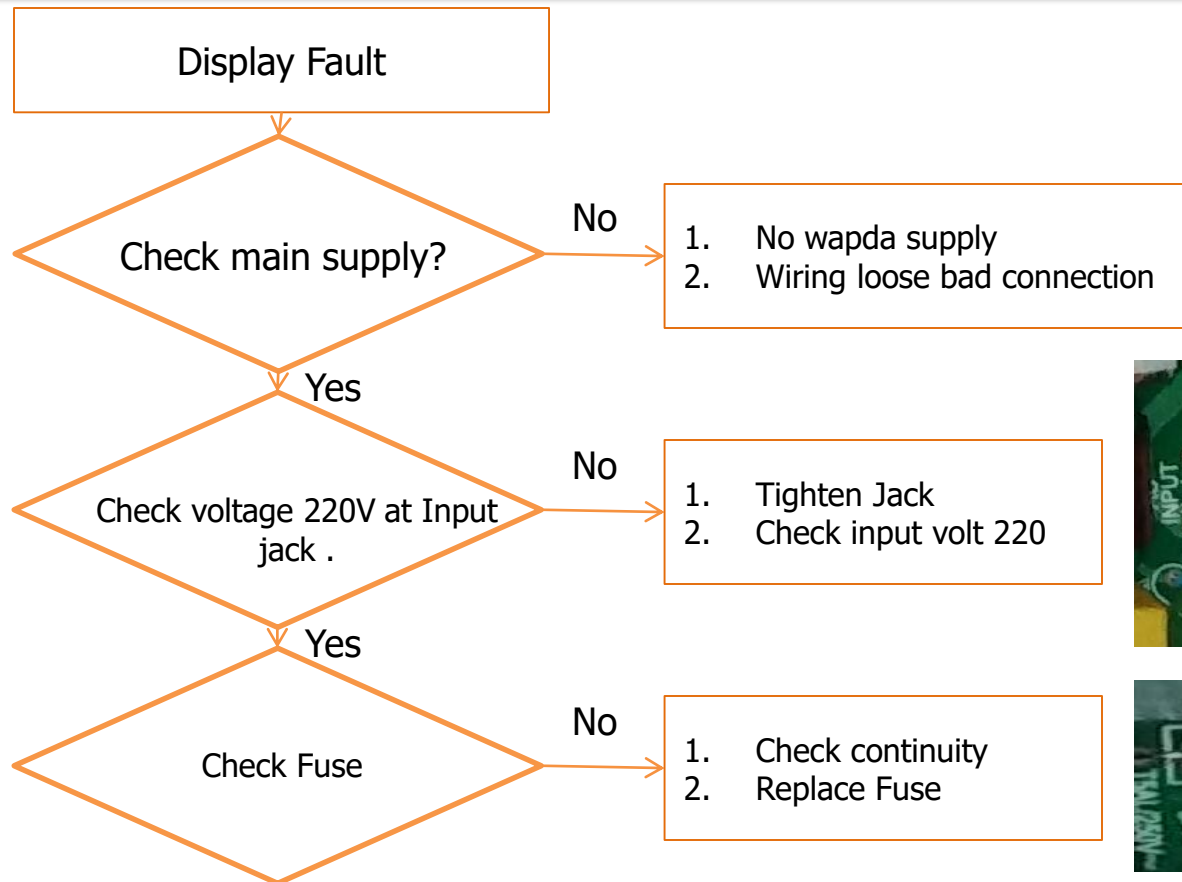
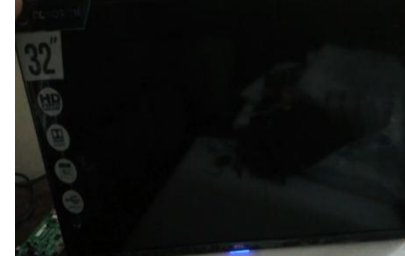




Display Fault

Display on Screen :

1. PEL logo displays for a second or screen shows lining
2. No Display



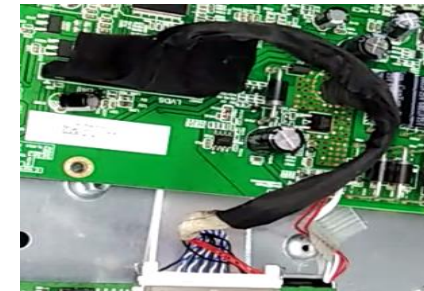
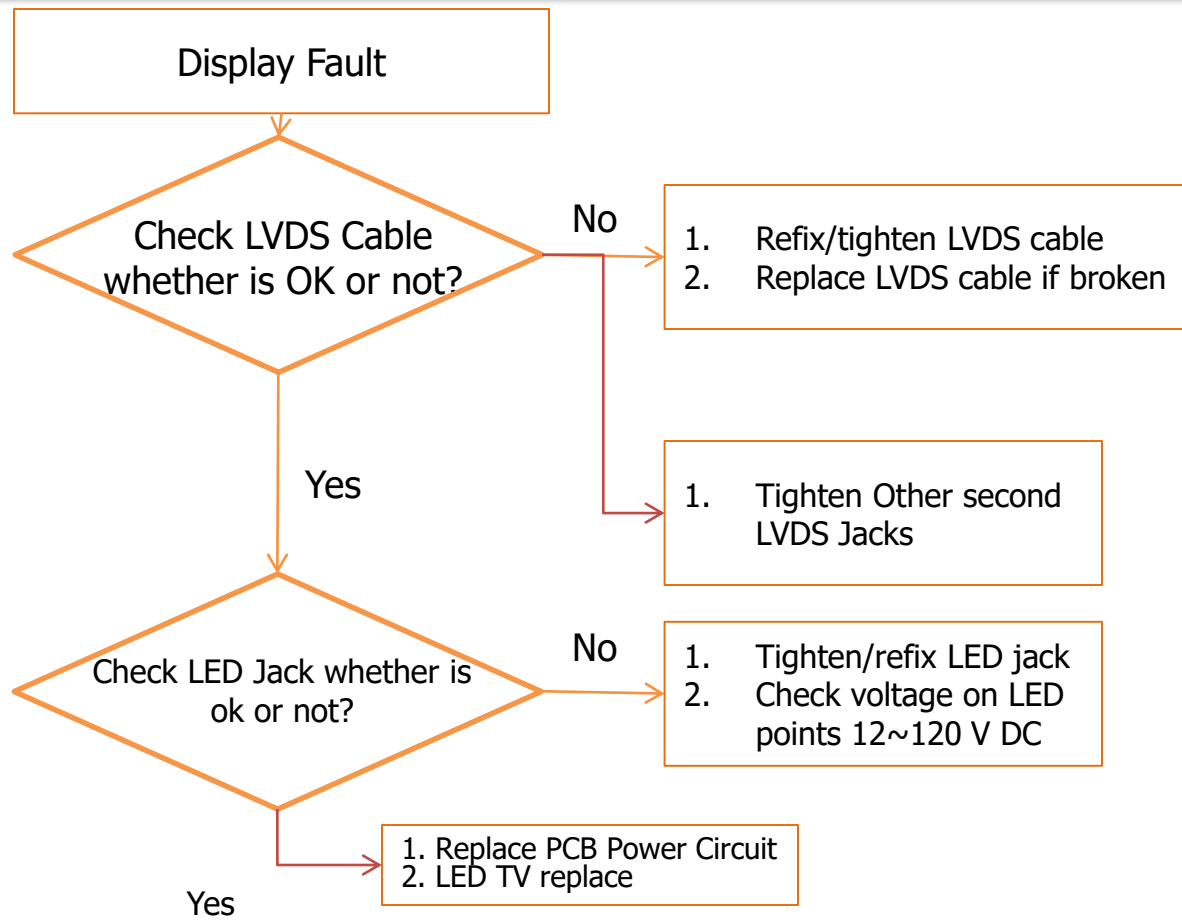
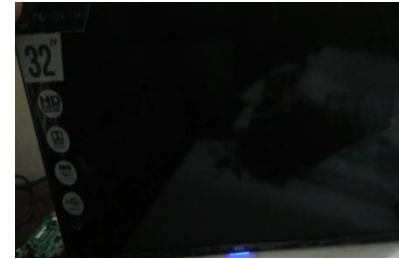
Continue....



Display Fault

Display on Screen :

1. PEL logo displays for a second or screen shows lining
2. No Display



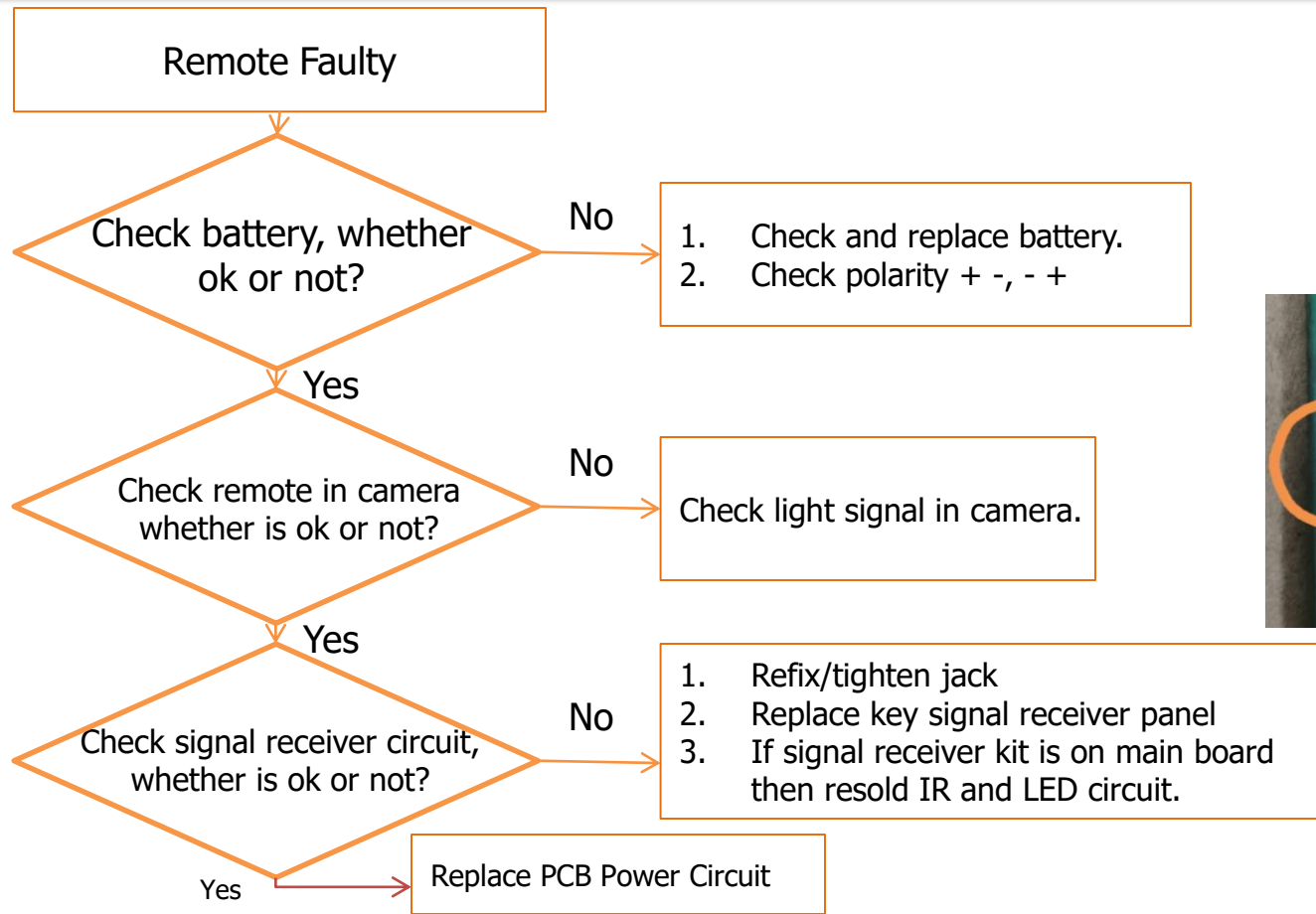
PEL Range – LED Television



Remote Faulty

Remote not Working :

1. Remote Defective
2. Signal Receiver Defective

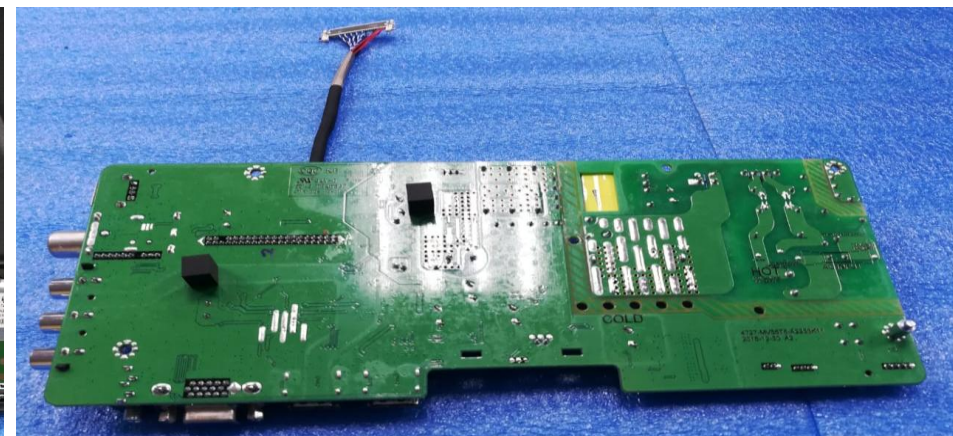


PEL Range – LED Television



1. LED OFF even on power supply:

1. Check Main power cable connected
2. Check Power LED ON (If not ON then move to step 4)
3. Press power Button or operate through remote Power Button
4. Remove back cover and check power cable properly connected on board
5. If above Checks do not work then Change the main Board ,Connect power cable and check



PEL Range – LED Television



2. No display on screen :

1. Check Power cord connected properly
2. Check Power LED ON
3. Remove back cover and check Display cable (LVDS) connected properly on both side
4. Change the cable and check again
5. Change main board and connect Display cable on both side
6. If no Display then TV set send to Company for further action



3. No sound display :

1. Adjust the Volume properly
2. Press Mute key on the remote control to turn on the sound
3. Enter main MENU to change the sound setting
4. Remove back cover and check the speaker connector
5. if no sound observed then Change the speaker

Note: There are 02 speakers in PEL LED TV.
i. Left side speaker connected with 03 wires
ii. Right side speaker connected with 02 wires



PEL Range – LED Television



4. Remote not working :

1. Check the remote-control battery
2. Move the object to avoid blocking the receiver WINDOW
3. Check the remote signal emitter bulb with Mobile phone cam
4. If remote not working then Change remote



5. No display on RF tuner :

1. Check whether cable antenna connector is loose
2. Check cable signal in good condition
3. Check cable/antenna type is VHF/UHF
4. Check RF tuner jack, resold or replace if defective



6. USB not detected :

1. Check whether USB is properly fixed in USB port
2. Connect USB in usb port, power off LED from remote and again power on through remote then detect it from source
3. Sometimes USB automatically connected, not through source
4. If USB not working then change the PCB

Note:

1. System supports USB flash drive and hard disk with standard USB 1.1 and 2.0 interface transmission protocol.
2. The file format for USB drive must FAT / fat 32 instead of NTFC.
3. Do not unplug the USB while playing or else, it will cause system or device error.



5. Others Troubleshooting

Problem	Fault	Symptoms
Slow YouTube	- Internet Issue - Cache Full - Old Version	- Check internet connection speed. - Clear RAM memory. - Remove main power and then start. - Install the new version of YouTube provided by TV TEAM. Never installed YouTube version through play store
LED shuts off during wapda off and again on UPS	UPS delay time	No fault in TV, its just happened due to UPS delay time
Mirroring issue	- No Wifi - Older version	- TV and mobile device must be connected through same WIFI router. - If TV shows registration error/machine code fail or other error then check the MAC address.
USB Connecting issue	USB Corrupted	- Format USB - Change USB
Cursor button not shown		Press F1 button for 3 second
Power off by Remote, but again automatically on after wapda on...	PCB board defective	Change main Board
How can we check MAC Address		First press "menu" and then press 2013
Screen shows nothing after power on	- LVDS Loose - LEDs Defective	- Check and tighten LVDS Cable - Replace LED strips



Hands On Session in Workshop